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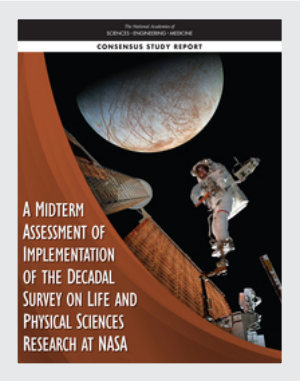

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A Midterm Assessment of Implementation of the Decadal Survey on Life and Physical Sciences Research at NASA

Committee on a Midterm Assessment of Implementation of the Decadal Survey on Life and Physical Sciences Research at NASA

Aeronautics and Space Engineering Board

Space Studies Board

Division on Engineering and Physical Sciences

A Consensus Study Report of

The National Academies of
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Preface

The Committee on a Midterm Assessment of Implementation of the Decadal Survey on Life and Physical Sciences Research at NASA was charged to review the progress of NASA's program in addressing the strategies, goals, and priorities outlined in the 2011 National Research Council (NRC) decadal survey report,[1] *Recapturing a Future for Space Exploration, Life and Physical Sciences Research for a New Era*.[2] The committee's complete statement of task is reprinted in Appendix A.

The tasks before the committee encompassed areas of science, science policy, and science implementation, which have been under extraordinary evolution since the publication of the *Recapturing a Future for Space Exploration*. To address both the tasks and the evolutions in the field, the committee held four in-person meetings and many teleconferences from January through December 2017. The committee heard extensively from the NASA Division of Space Life and Physical Sciences Research and Applications (SLPSRA), the NASA Human Exploration and Operations Mission Directorate (HEOMD), the NASA Office of the Chief Scientist, and the NASA Space Technology Mission Directorate (STMD). As part of the April 2017 meeting, a wide-ranging and interactive Community Input Symposium was held at a National Academy of Sciences building in Washington, D.C.

The committee witnessed the extensive work being done within SLPSRA and the broader International Space Station (ISS) research program, as well as within the Center for the Advancement of Science in Space (CASIS) and international programs that impact the portfolio. This multipronged approach to science is in line with the general priority of the decadal survey, which was restoring and then maintaining a broad portfolio for space life and physical sciences, led by NASA. This resurgence in growth was extensively discussed by the committee within the bounds of the statement of task, and the scope of the statement of task was thoroughly examined by the committee in the extensive context of the entities now involved.

The discussions of the committee were also guided by the now rapid evolution of an increasingly commercial or private space "ecosystem." This ecosystem already involves the use of commercial carriers to the ISS, and it also includes the potential further economic development of low Earth orbit (LEO) in a broader sense. This discussion is particularly relevant as NASA considers science in and around LEO during the period that will define NASA's operational transition away from the ISS as its focus. Thus, the potential transition of ISS capabilities

[1] Hereinafter also referred to as "the decadal survey" or "the 2011 decadal survey," or "the 2011 decadal survey on space life and physical sciences."

[2] National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C.

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toward the private sector and the focus of NASA more toward deep space had a dramatic impact on committee deliberations and recommendations.

The committee and the co-chairs thank the many very busy people at NASA, CASIS, in the larger U.S. government, and especially the members of the national and international community of space life and physical sciences, who helped the committee through presentations, written input, and discussions. Special thanks go to the staff of the Aeronautics and Space Engineering Board and the Space Studies Board. While the words of this report are those of the committee, the extraordinary expertise of the National Academies staff made the work of the committee possible, and their guidance consistently enabled the committee to pull informative data and reach enlightened conclusions. Without their support, this report would not have been possible.

Daniel L. Dumbacher and Robert J. Ferl, *Co-Chairs*
Committee on a Midterm Assessment of Implementation of the
Decadal Survey on Life and Physical Sciences Research at NASA

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Acknowledgment of Reviewers

This Consensus Study Report was reviewed in draft form by individuals chosen for their diverse perspectives and technical expertise. The purpose of this independent review is to provide candid and critical comments that will assist the National Academies of Sciences, Engineering, and Medicine in making each published report as sound as possible and to ensure that it meets the institutional standards for quality, objectivity, evidence, and responsiveness to the study charge. The review comments and draft manuscript remain confidential to protect the integrity of the deliberative process.

We thank the following individuals for their review of this report:

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Michael M. Weil, Colorado State University.

Although the reviewers listed above provided many constructive comments and suggestions, they were not asked to endorse the conclusions or recommendations of this report nor did they see the final draft before its release. The review of this report was overseen by Louis J. Lanzerotti, NAE, New Jersey Institute of Technology. He was responsible for making certain that an independent examination of this report was carried out in accordance with the standards of the National Academies and that all review comments were carefully considered. Responsibility for the final content rests entirely with the authoring committee and the National Academies.

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Summary

The 2011 National Research Council (NRC)[1] decadal survey on biological and physical sciences in space,[2] *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*,[3] was written during a critical period in the evolution of science in support of space exploration. The research agenda in space life and physical sciences had been significantly descoped during the programmatic adjustments of the Vision for Space Exploration in 2005,[4] and this occurred in the same era as the International Space Station (ISS) assembly was nearing completion in 2011. Out of that period of change, *Recapturing a Future for Space Exploration* presented a cogent argument for the critical need for space life and physical sciences, both for enabling and expanding the exploration capabilities of the National Aeronautics and Space Administration (NASA) as well as for contributing unique science in many fields that can be enabled by access to the spaceflight environment. The important areas of science that are enabled by microgravity and that enable the exploration of space were clearly and extensively articulated in *Recapturing a Future for Space Exploration*, producing a broad portfolio of science questions and approaches. Thus, *Recapturing a Future for Space Exploration* presented not only a broad portfolio but also a comprehensive guide for NASA to consider in its rebuilding of the space life and physical sciences program to address exploration imperatives.

As is standard practice following decadal surveys, NASA later requested that the National Academies of Science, Engineering, and Medicine perform a midterm review of NASA's progress in addressing the decadal survey recommendations. The details of the statement of task for this midterm review by the Committee on a Midterm Assessment of Implementation of the Decadal Survey on Life and Physical Sciences Research at NASA are shown in Appendix A. The statement of task asks for a typical midterm assessment of NASA's progress against the decadal survey recommendations, then goes on to request a focus on deep space exploration, thus framing much of the midterm assessment as a further ranking of those decadal survey recommendations that enable exploration, in particular, deep space exploration beyond low Earth orbit (LEO).

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[1] Effective July 1, 2015, the institution is called the National Academies of Sciences, Engineering, and Medicine. References in this report to the National Research Council are used in an historical context identifying programs prior to July 1, 2015.

[2] Hereinafter also referred to as "the decadal survey" or "the 2011 decadal survey," or "the 2011 decadal survey on space life and physical sciences."

[3] National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C., https://doi.org/10.17226/13048.

[4] NASA, 2004, *The Vision for Space Exploration*, https://www.nasa.gov/pdf/55583main_vision_space_exploration2.pdf.

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Since the 2011 publication of the decadal survey, NASA has seen tremendous change, including the retirement of the Space Shuttle Program and the maturation of the ISS. NASA formation of the Division of Space Life and Physical Sciences Research and Applications (SLPSRA) provided renewed focus on the research of the decadal survey. NASA has modestly regrown some of the budget of space life and physical sciences within the agency and engaged the U.S. science community outside NASA to join in this research. In addition, NASA has collaborated with the international space science community. It also has seeded the birth of privately funded research and development in space through the initiation, development, and coordination of the U.S. National Laboratory on the ISS, managed by the Center for the Advancement of Science in Space (CASIS). When considering the role of CASIS, the committee recognized that the U.S. ISS National Laboratory serves the broad need of developing commercial space science for terrestrial applications and need not necessarily address the decadal survey priorities.

The overall result is that the ISS is now running near its present capacity to perform research within current cargo and crew constraints. With NASA looking toward focusing its efforts on deep space exploration, NASA faces important prioritization decisions on the distribution of science research. Indeed, NASA faces a creative tension between developing its exploration goals while servicing the needs of the ISS National Laboratory. Maximizing the science that will enable deep space exploration consistent with the evolving NASA exploration strategy and ISS lifetime, while balancing other ISS science demands, is a challenge. This challenge is, in turn, balanced by the growth of human suborbital space platforms, increased commercial interest in developing LEO, and expansion of relevant ground research capabilities.

NASA's strategy for exploration has also evolved since the decadal survey. NASA is now focused on Mars as the horizon destination, and since the spring of 2017 has initiated planning for the Deep Space Gateway in cislunar space. This overarching exploration strategy establishes the context for research priorities and programmatic implementation. One key element that remains to be understood is the plan and strategy for the ISS and LEO research beyond 2024 and the current international partnership funding agreements. These unknowns affect how the research necessary for human space exploration will be accomplished.

This summary presents an overview of the mid-decadal assessment of the challenges and progress of the science portfolio since it was presented in *Recapturing a Future for Space Exploration*. That overview is extended with findings on the status of the research progress and program process within NASA and the extended science community that NASA engages to accomplish the goals of *Recapturing a Future for Space Exploration*. The overview, findings, and recommendations are organized in this summary in alignment with the statement of task that was developed for this report. The statement of task focuses the report toward the exploration of space given the NASA goal of moving beyond LEO. The statement of task specifically requested that, in addition to the assessment of progress, the committee develop a set of top-priority recommendations dedicated toward exploration. This report describes the assessment of individual science disciplines in more detail and articulates the highest-priority exploration-related science areas to be emphasized in the remaining years of the decade.

NASA'S APPROACH TO AND PROGRESS ON REALIZING DECADAL STRATEGIES, GOALS, AND PRIORITIES

The committee finds that NASA responded to *Recapturing a Future for Space Exploration* in a strong and positive manner at the programmatic level. As a major response to the programmatic recommendations of the decadal survey, the SLPSRA Division was established as part of NASA's Human Exploration and Operations Mission Directorate in 2011. "SLPSRA provides administrative oversight of NASA's Life and Physical Sciences Research, a programmatic home for an integrated research agenda, program leadership and execution under a single management structure and is housed in a NASA directorate organization that understands both the value of science and its potential application in future exploration missions."[5] In support of the relationship between SLPSRA and the decadal survey, the Committee on Biological and Physical Sciences in Space[6] was formed as

[5] See NASA, "SLPSRA Overview," updated August 3, 2017, https://www.nasa.gov/content/slpsra-overview.

[6] See National Academies of Sciences, Engineering, and Medicine, "Committee on Biological and Physical Sciences in Space," http://sites.nationalacademies.org/SSB/SSB_145312 accessed October 5, 2017.

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an activity of the Space Studies Board and the Aeronautics and Space Engineering Board of the National Academies of Sciences, Engineering, and Medicine. Data presented to the committee also demonstrated an increase in budgetary commitment to the science of the decadal survey. In the years following the formation of SLPSRA, the budgets for space life and physical sciences showed signs of stabilizing and improving at the program level and as measured by the levels of science supported and accomplished through engagement of the science community.

The findings and recommendations that follow are identified by the chapter in which they are discussed and their sequential order in that chapter.[7]

Finding 2-1: NASA addressed the overall strategy of the report *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era* in an appropriate and responsive programmatic manner through the formation of SLPSRA. NASA has enhanced community science input related to the decadal survey by developing a strong interaction with the Committee on Biological and Physical Sciences in Space.

Finding 2-2: NASA is supporting the science of the report *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era* by modestly increasing the budget and tasks for space life and physical sciences within overall funding constraints and the operational diversification of microgravity science across research platforms.

Recommendation 5-1: NASA should recognize the need for regular requests for research proposals, in order to keep an active external research community available to do exploration-related space life and physical sciences research.

Recommendation 5-9: In light of the resource constraints, NASA should raise the priority of Space Life and Physical Sciences Research and Applications Division research within the International Space Station (ISS) to address the risks and unknowns of human space exploration, particularly given the value of microgravity research for exploration and the urgency resulting from the potential transition of the ISS. These priorities should be directly traceable to the space exploration strategy, linked research priorities, and related technologies. Table 4.1 can be used to initiate this traceability.

Finding 4-1: Several exploration pathways have been proposed by NASA since the 2011 decadal survey, involving the Moon, asteroid missions, Mars, and cislunar space. NASA has also used a separate set of Design Reference Missions to develop its Technology Roadmaps. These frequent exploration strategy changes result in unclear traceability between research investigations and exploration needs.

Recommendation 5-5: NASA should establish and document traceability of the research priorities to its Technology Roadmaps, Design Reference Missions, and exploration strategy.

Recommendation 5-8 : In order to maximize the implementation of decadal survey priorities within its constrained resources, NASA should continue to be mindful of the full range of platforms (including drop towers, aircraft, balloons, suborbital vehicles, and free-flyers) and terrestrial analogs and ground-based laboratories available for decadal survey research.

In assessing the progress of implementation of the decadal survey portfolio, the committee found difficulty in navigating the research tracking within various parts of the Agency that report on research alignments with the decadal survey recommendations. While overall programmatic attention to space life and physical sciences was readily apparent in the many presentations from NASA to the committee, a cumulative alignment or mapping of agency research projects to specific decadal survey recommendations proved problematic.

[7] For example, Finding 2-1 is the first finding discussed in Chapter 2.

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Finding 2-3: NASA has a diverse approach to the reporting of investigation funding, published results, and experiments that is not fully coordinated among programs and offices, making overall tracking to decadal survey priorities difficult.

Recommendation 5-3: NASA should consider decadal survey priority tracking integration within Agency elements and utilize existing, commercially available, well-known research reporting, and open-science database tools that are in use across the academic research spectrum for accurate, timely, and sustainable information. NASA should also make a determined effort to build on the significant improvements in the International Space Station program for communicating the value of the investigations.

Recommendation 5-6: NASA should further balance communication and reporting efforts across the organization.

EXISTING AND EMERGING CHALLENGES AND OPPORTUNITIES RELATING TO PLANS FOR THE ISS

NASA is moving its exploration focus beyond LEO and into deep space, with cislunar space in the near term and Mars exploration on the horizon. In focusing its exploration agenda on deep space, NASA must contend with multiple and diverse concerns that all compete for resources within the agency. The nuances associated with the move of NASA to a deep space focus are presented in the 2014 NRC report *Pathways to Exploration: Rationales and Approaches for a U.S. Program of Human Space Exploration*.[8] NASA remains very cognizant of the need for space life and physical sciences research to enable deep space exploration and that at least some of that research requires long-term access to microgravity.

Finding 2-4: LEO science research needed for exploration will be required beyond 2024. Extended durations in microgravity, measured in years, will continue to be required to best meet deep space exploration research needs.

Crew time and cargo transfer, particularly downmass, to and from the ISS are critical constraints in the conduct of science on the ISS, and experiment durations need to be pushed toward longer durations in microgravity in order to prepare for deep space transits.

Recommendation 5-2: NASA should continue and increase its efforts to maximize International Space Station (ISS) resource synergy across the ISS National Laboratory, international partners, and the Division of Space Life and Physical Sciences Research and Applications, particularly with regards to crew time availability and research priority on the ISS. Continued efforts to increase cargo and crew transport to and from the ISS should be expedited as much as possible.

The maturation of the ISS capabilities provides the broad research community with a wide range of instrumentation and facilities in the microgravity of LEO. Presentations to the committee indicate that occupancy space within the ISS will be near 95 percent by the end of 2017, with Express Racks full by mid-2018. Similarly, occupancy of the external ISS racks and facilities is also nearly full. As the research capability on the ISS reaches maximum throughput in 2018, it will be important to focus on complete utilization of those facilities rather than developing additional government-funded facilities. The committee also recognizes the recent advancements in privately developed research facilities that can be used for research on the ISS. Budget data presented to the committee indicate that if fewer new facilities were to be developed, significant funds would become available for the conduct of experiments in existing facilities on the ISS.

[8] National Research Council, 2014, *Pathways to Exploration: Rationales and Approaches for a U.S. Program of Human Space Exploration*, The National Academies Press, Washington, D.C.

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Finding 2-8: While NASA has worked diligently to enhance ISS research capability since the 2011 decadal survey, the potential 2024 transition era of U.S. participation in the ISS may call for a change of strategy with respect to ISS facilities development, including use of privately developed research facilities.

Recommendation 5-7: NASA should direct an increasingly higher priority toward the conduct of science within existing International Space Station (ISS) hardware and research capabilities. Utilization of existing, including privately developed, ISS facilities should be maximized in recognition of the current funding limits, the ISS transition timeline, and the need for high-priority microgravity research.

NASA has appropriately initiated an internal activity to develop and assess various options for the ISS beyond 2024. At the time of this report, no conclusions have been drawn. NASA did provide the committee with a presentation on the status of this activity. To identify and assess potential ISS strategy options, NASA has developed a list of key considerations. These considerations include a long-term U.S. presence in LEO that is characterized by leadership of international partnerships, continuing support for commercial space activities in LEO, and an ecosystem that fosters ongoing deep space exploration objectives beyond LEO while continuing research and development for fundamental science and technology. The overarching consideration is affordability and sustainability.

Recommendation 5-10: It is essential that NASA as quickly as possible develop an International Space Station post-2024 strategy. This development factors strongly in the overall exploration strategy, space life and physical sciences research priorities, and resource allocation in terms of crew time, cargo delivery, and funding. This post-2024 strategy should address clear cost allocation among the various research activities and partners.

CHALLENGES AND OPPORTUNITIES OF A MULTISPONSOR SCIENCE PROCUREMENT APPROACH

The interests of the international participants and the ISS National Laboratory are all coordinated into a vast orbital enterprise that has many investors, constituents, and overseers. In particular, the emergence of CASIS as the manager of the ISS National Laboratory has brought major, non-exploration visibility and demands to the ISS. The committee considers the positive growth of research interest in the ISS to be a major accomplishment of both NASA and CASIS. Yet as NASA seeks to maximize space life and physical science on the ISS toward meeting its exploration needs, it must balance the challenges of managing that need with the many non-exploration demands on the ISS. This places at risk the progress of space life and physical sciences in support of exploration.

Finding 2-5: NASA coordination with ISS National Laboratory continues to develop in a laudable manner. However, as NASA meets the challenges of ISS National Laboratory implementation, its own exploration priorities face increased pressure to garner ISS research resources, such as crew time, cargo delivery, and ISS experiment prioritization. This pressure leaves NASA exploration research priorities at risk of not being met.

NASA openly seeks to make space accessible to research interests in other agencies; for example, its Science Mission Directorate has long been active in this regard. For the Human Exploration and Operations Mission Directorate (HEOMD) and SLPSRA, engaging other agencies in research is a relatively new venture. It has opened the realm of microgravity research, especially on the ISS, to other agencies. While this engagement is welcome and positive, it remains unlikely that other agencies will participate directly in the science of the exploration imperatives within NASA. Rather, other agencies will utilize NASA facilities to support their own research agendas.

Finding 2-6: NASA has initiated relationships and improved efforts to work with other U.S. agencies and nongovernmental institutions to explore the value of microgravity in science. The committee was unable to identify any other federal agency with a programmatic reason to absorb the full costs of developing and conducting experiments on the ISS, even in the face of rising interest in microgravity research.

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Finding 2-7: As NASA's ISS strategy evolves for the timeframe beyond the current 2024 commitments, there is a need for clear and objective analyses of all costs necessary to support research. Cost elements (e.g., launch, personnel, operations, facility maintenance) for ISS research projects and CASIS activities are important factors in the assessment of research activities and for determining appropriate partner funding levels.

Recommendation 5-4: Relationships with the National Institutes of Health, the National Institute of Standards and Technology, the National Science Foundation, the Department of Defense, the Department of Energy, and other agencies should be strengthened to better address the decadal survey and midterm review identified research priorities, especially exploration priorities. NASA should consider negotiation with the Center for the Advancement of Science in Space regarding International Space Station research allocations to better address NASA's exploration priorities.

ACTIONS TO BE TAKEN TO OPTIMIZE SCIENCE VALUE TO EXPLORATION, AND GUIDANCE FOR IMPLEMENTATION

As NASA has developed its deep space exploration plans, there has been tremendous refocusing on the science needed to best enable that exploration. The committee heard many presentations on the progress that has been made in space life and physical sciences since the 2011 decadal survey. The committee also heard from both NASA and the external science community that a large amount of science remains to be done to best support the move to deep space. In particular, the committee recognized the rather simple facts that all deep space mission plans absolutely require long periods of time in microgravity and will be outside the radiation protection of Earth's magnetosphere. There are no plans in any design reference missions to include large-scale induced gravity within the transit vehicle. If microgravity remains the operational environment for exploration transit, then humans and their attendant biological and physical support systems must function adequately and independently in microgravity for durations up to years. Yet the number of long-term space biology experiments on orbit has been minimal, and many physical systems currently employed on the ISS are known to require extensive maintenance. Therefore, understanding the fundamentals of long-term microgravity exposure for life and physical sciences is a top priority. Similarly, radiation, specifically operational exposure to actual deep space radiation, remains a major identified risk to humans and a large unknown risk to biological and physical systems in deep space. With the totality of human exploration experience beyond LEO restricted to the Apollo era, and the limited number of long-duration experiments conducted to date on the ISS, the need for microgravity and radiation science research is as strong now as ever.

Finding 3-1: Deep space exploration will take place for long periods of time in a microgravity and high-radiation environment; no large-scale induced gravity on the vehicle or station is foreseen with the current design reference missions, and the only shielding expected for radiation damage is that which is available on the space vehicle. Therefore, a fundamental understanding of both microgravity and radiation effects on biological and physical systems is essential for operational success and appropriate protection of astronauts.

Recommendation 3-1: As NASA continues to develop deep space mission scenarios involving long durations in microgravity, understanding the direct and interactive effects of radiation, microgravity, and small habitats on human biology and on the performance of biological and physical systems in space over long durations will need to have high priority in NASA science plans. NASA should also improve the coordination among the science research and engineering teams to better address the integrated effects in the design of the exploration elements and systems.

Finding 4-2: The current funding levels are insufficient to fully address the significant unknowns and risks of human exploration beyond LEO. Fundamental understanding of human health and behavior risks in microgravity, combined with fundamental microgravity physics and materials, in an integrated manner,

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is essential to extending the human neighborhood beyond LEO. Significant risks remain, particularly in understanding the radiation environment and its effects, environmental control and life support, human behavior, and protecting long-term crew health with integrated countermeasures. These risks are best addressed in the respective disciplines and in an increasingly integrated fashion.

Finding 4-3: NASA has pursued means to improve internal cross-organizational efforts across the research and technology development landscape, based on the linkages with the Technology Roadmaps.

Recommendation 5-11: NASA should aggressively lead in the 46 research priorities for deep space exploration identified in Table 4.1 of this midterm report to provide as much "pull" as possible for exploration enhancement using space life and physical sciences. NASA should, for example, lead in the development of microgravity-adapted biological and physical systems, making maximum use of all available platforms, including the International Space Station, specifically for the science behind the design and implementation of microgravity-optimized operation.

To help guide maximization of science return for exploration, especially considering the remaining available ISS time, the committee identified and ranked from among the high-priority recommendations in the decadal survey those exploration-enabling research priorities it deemed most critical for NASA to conduct. These rankings are summarized in Table 4.1. These rankings are accompanied by discipline science narratives that provide the underpinnings for each priority. Also presented are the criteria and considerations used for selecting and ranking the priorities.

The decadal surveys presented a total of 65 specific high-priority recommendations over 7 discipline areas. Using the criteria of the decadal survey that apply to deep space exploration, the committee determined that 46 of those recommendations were considered by the decadal survey committee to be the highest-priority recommendations related to exploration. The committee then ranked these 46 recommendations to produce Table 4.1, showing the 24 highest-, 12 higher-, and 10 high-priority recommendations targeted at exploration needs. The highest-priority recommendations recognize the need for long-term microgravity experiments, especially experiments that address interfaces between biological and physical systems that make use of the remaining time on the ISS. These recommendations also serve as a guide for initial deep space exploration planning, such as science that may be conducted on the Deep Space Gateway or similar exploration vehicles should they be implemented, where deep space radiation can be studied in its confluence with microgravity and all the issues assembled within a deep space transit vehicular system.

The committee wishes to highlight the tension between the "enabling" research and the research "enabled by" the unique microgravity environment. While recognizing the funding constraints, and therefore the need for NASA to prioritize microgravity research, the committee also recognizes the value of fundamental research such as the Alpha Magnetic Spectrometer and experiments utilizing the Cold Atom Lab. These fundamental science efforts can only be performed in a long-duration microgravity environment.

Recommendation 5-12: The committee recommends that a cautious approach be used when shifting the NASA research portfolio more toward those types of experiments necessary for deep space exploration, so as to maintain the benefits of important basic experiments, especially those uniquely enabled by International Space Station microgravity and already in progress, which may in the long term have the potential for major impacts in fundamental physical science.

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1

Introduction

The Committee on a Midterm Assessment of Implementation of the Decadal Survey on Life and Physical Sciences Research at NASA was tasked by the National Aeronautics and Space Administration (NASA) to review NASA's progress since the 2011 decadal survey, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era,*[1] to assess the high-priority research identified in the decadal survey in light of future human Mars exploration. The committee was further tasked to make recommendations on science priorities, specifically those priorities that best enable deep space exploration. The statement of task is reprinted in its entirety in Appendix A.

As is typical of midterm assessments, this review examines the progress NASA has made on the findings and recommendations of the 2011 decadal survey. This examination focuses on programmatic progress in Chapter 2 and on scientific progress in Chapter 3.

The statement of task goes beyond a simple assessment of progress and frames a specific set of tasks that are focused on that part of the decadal survey that explicitly deals with enabling space exploration. Those tasks ask for a deeper evaluation and additional ranking of those decadal survey recommendations that are a high priority for enabling exploration. Indeed, the statement of task required further prioritization of those high-priority, enabling recommendations, with an emphasis on deep space exploration and moving beyond low Earth orbit (LEO). Further, the statement of task referred to the year 2024 as a timeline marker against which the science of exploration into deep space was to be metered.

The committee, per its charge and as is the standard practice for midterm assessments, respected the findings and recommendations of the 2011 decadal survey. Rather than re-examine the decadal survey, the committee focused on the progress since the decadal survey and used the decadal survey priorities and findings as the basis for the midterm assessment. In addition, the committee also identified and ranked, "from among the highest-priority recommendations" in the decadal survey, targeted research priorities critical to the expansion of human exploration into deep space. These rankings anticipate future science needed for exploration and were based on the committee's assessments of the multiple data inputs addressing current research in space life and physical sciences. The evaluation and ranking of those recommendations that enable deep space exploration, together with the science that underpins those recommendations, are presented in Chapter 4.

[1] National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C., https://doi.org/10.17226/13048.

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Lastly, the committee was tasked with providing programmatic recommendations for implementation over the remainder of the decadal survey timeframe. Those recommendations appear in Chapter 5. It is expected that this effort will inform the next space life and physical sciences decadal survey, expected in the early 2020s.

As the 2011 decadal survey stated, "This review, or decadal survey, takes place in the context of the many remarkable achievements of the National Aeronautics and Space Administration (NASA) in exploring and studying space during the past 40 years. However, NASA's continued preeminence in this endeavor cannot be assumed, because other nations have both a keen interest, and the intent to make their own mark, in space."[2] This statement remains true at the midterm review. NASA's mission to extend human presence beyond LEO is work deserving of a nation that expects to remain the global leader in space exploration.

1.1 EXPLORATION IMPERATIVE SINCE THE DECADAL SURVEY

Much has changed since the 2011 decadal survey. NASA established the Space Life and Physical Sciences Research and Applications Division within the Human Exploration and Operations Mission Directorate. Assembly of the International Space Station (ISS) was completed. The space shuttle system was retired. Private enterprise has initiated cargo delivery service to the ISS. Development efforts are in the works for private enterprise transporting crew to the ISS. Suborbital vehicles have begun operation, providing additional research platforms. The U.S. ISS National Laboratory came to fruition, and its research content is determined by the Center for Advancement of Science in Space. While the decadal survey anticipated some of these developments, the breadth and depth of changes in the spaceflight landscape has been dramatic.

NASA's strategy for exploration has also evolved since the decadal survey (Figure 1.1). NASA is now focused on Mars as the horizon destination, and since the spring of 2017 has initiated planning for the Deep Space Gateway in cislunar space. This overarching exploration strategy establishes the context for the research priorities and the programmatic implementation. One key element that remains to be understood is the plan and strategy for the ISS beyond 2024 and the current international partnership funding agreements. This unknown affects how the research necessary for human space exploration will be accomplished.

1.2 STUDY CONTEXT AND CHALLENGES

With an evolving U.S. exploration strategy, the advent of private enterprise service providers for cargo and crew to the ISS, and availability of new research platforms, it was important that a midterm review of the 2011 decadal survey be carried out. It is important to note that the exploration strategy will continue to evolve; however, the fundamental scientific understanding for human survival in deep space, and the research needed to enable human space exploration, will continue beyond 2024 and throughout the execution of these exploration plans.

Space life and physical sciences research has been performed within NASA by various organizations. There has been significant change in the funding levels for this research due to other priorities within NASA. In the early 2000s, the funding was dramatically reduced to assure ISS assembly and to initiate the NASA human exploration activities beyond LEO. Since the early 2000s, NASA, given funding constraints, has gradually grown the funding for microgravity research. It is not back to the funding levels of the early 2000s; however, there is improvement.

NASA has also worked to focus research funding on the needs for human exploration, while maintaining some level of basic science research. The ISS provides a unique laboratory environment for basic science research.

This midterm review strictly uses the priorities identified in the 2011 decadal survey as its starting point. The committee also could not reasonably review all of the details of the research and the progress since the 2011 decadal survey in the time available, nor was such an examination called for in its charge. The committee made a concerted effort to review the key progress and issues with NASA, other government agencies, the Center for the Advancement of Science in Space (CASIS), the ISS Program, and the research community through an open National Academies of Sciences, Engineering, and Medicine event, a Community Input Symposium in April 2017.

[2] Ibid, p. 11.

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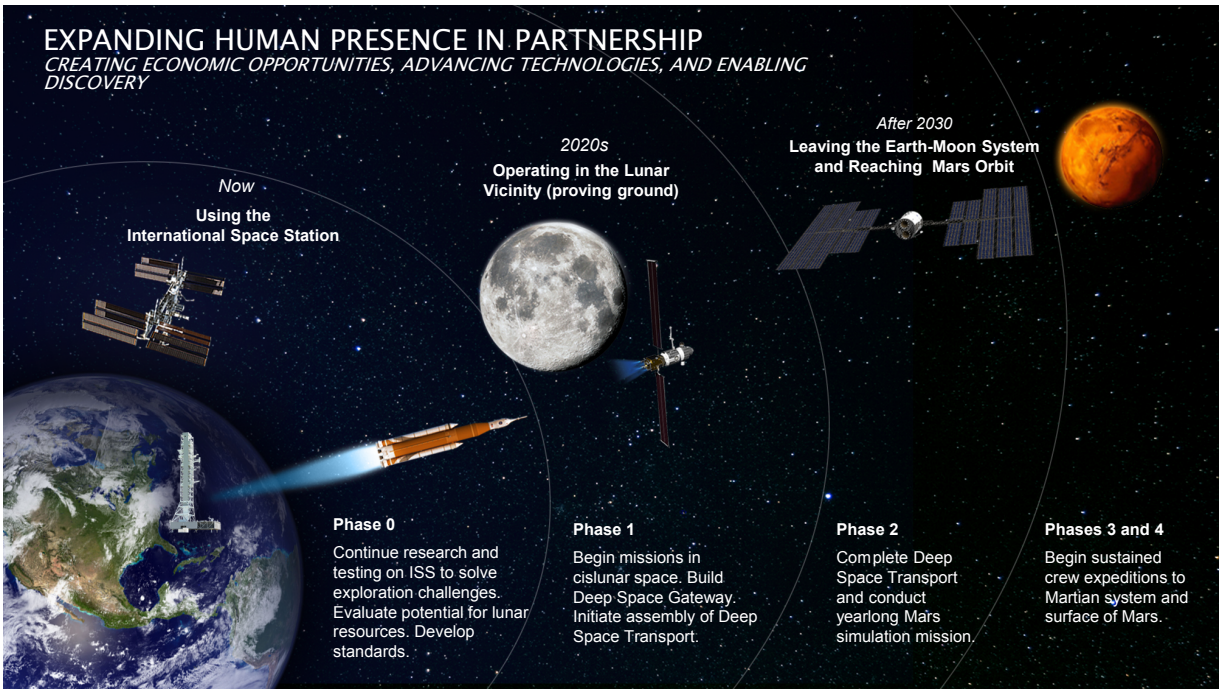

FIGURE 1.1 NASA exploration strategy. SOURCE: NASA, Ken Bowersox, NASA, "Human Exploration and Operations Committee Status," presentation on July 27, 2017, https://www.nasa.gov/sites/default/files/atoms/files/6-human_exploraton_operations_committee_report_tagged.pdf, p. 19.

The statement of task, shown in Appendix A, asks the committee to assess the progress since the 2011 decadal survey, to assess the high-priority research identified in the decadal survey in light of future human Mars exploration, and asks for recommendations on programmatic implementation for the remainder of the decadal survey timeframe, leading up to the next decadal survey.

In addition, four committee meetings were held to obtain briefings from NASA, CASIS, the National Institutes of Health, the National Institute of Standards and Technology, and the National Science Foundation. In all cases, the presenters were very responsive to the data requests and provided invaluable information and insight. NASA also provided significant support for the committee's analysis of the resources and funding levels for the research activities. Their support was exemplary.

1.3 REPORT ORGANIZATION

This report addresses the statement of task in an approach that draws multiple task elements together to provide a cogent understanding of the committee's assessment and recommendations. The committee recognized that the topics addressed in the report are significantly interrelated. Therefore, the report does not linearly follow the statement of task. Rather the committee has worked diligently to address the concepts of the statement of task and present its assessment, findings, and recommendations in a logical narrative that recognizes the interrelated nature of the subject matter.

The assessment begins in Chapter 2, where the overall programmatic and budgetary commitments of NASA to the guiding concepts presented in the decadal survey are examined. Chapter 2 addresses directly the approach and progress of NASA toward realizing the strategies, goals, and priorities of the decadal survey. Chapter 2 also discusses the plans for the ISS and LEO infrastructure, the opportunities and challenges of a multisponsor environ-

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ment, and resource constraints. Chapter 2 therefore presents a comprehensive view of the programmatic approach taken by NASA.

Chapter 3 considers the science developments in the research community that have occurred since the 2011 decadal survey in response to the programmatic commitments made by NASA. Chapter 3 serves as the scientific basis for assessing research progress in the context of that programmatic commitment. Chapter 3 also establishes the foundation from which the future needed science is prioritized.

The future direction of space life and physical sciences is considered in Chapters 4 and 5. Chapter 4 addresses the statement of task request to identify and rank from among the "highest-priority recommendations" a set of targeted research priorities that are critical for NASA to conduct in order to enable human exploration of deep space. Recommendations in Chapter 5 are structured in "themes" to account for the interrelated nature of the findings throughout the report. Findings and recommendations represent the consensus opinion of the committee members.

It should be noted that the number of pages devoted to individual discipline sections is not an indication of the priority of those disciplines. Rather, the length of discipline sections depends largely on the degree to which progress could be easily summarized across several related topics within that discipline, the growth in opportunities provided by new tools (such as omics technologies), and the extent of the enabling research needs that are yet to be addressed. For example, many of the high-priority animal and human biology research topics are being targeted aggressively in ISS experiments, as the critical relevance of this research to biomedical countermeasures is already widely understood at NASA. Conversely, physical sciences research on the ISS has been more limited and has seen less progress in addressing decadal survey recommendations. There is therefore correspondingly more discussion of unmet physical science priorities and their relevance to exploration. For this reason, while the scope and scale of this midterm study did not make it feasible to discuss the interface between microgravity physical phenomena and exploration systems in significant depth, the committee thought that it was important to provide in Chapter 4 details of a few relevant examples.

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The NASA Programmatic Approach and Strategy Addressing the 2011 Space Life and Physical Sciences Decadal Survey

The National Aeronautics and Space Administration's (NASA's) approach to addressing the 2011 decadal survey[1] has been multifaceted and, as described in later sections, has yielded significant research progress, given tight funding and resource constraints. Beginning with the establishment in 2011 of the NASA Division of Space Life and Physical Sciences Research and Applications (SLPSRA) within the Human Exploration and Operations Mission Directorate (HEOMD), NASA has engendered a rich and varied ecosystem that engages in various aspects of microgravity science. That ecosystem now includes the Center for the Advancement of Science in Space (CASIS) and the Space Technology Mission Directorate (STMD), as well as interfaces with other government agencies and the commercial spaceflight sectors. This broad range of players, each with its goals and aspirations, constitutes a team that requires complex coordination among offices within NASA and currently is focused on the International Space Station (ISS) Program.

Since the release of the 2011 decadal survey, NASA's overall exploration strategy has undergone significant evolution. Following the cancellation of the Constellation Program, NASA embarked on developing the Space Launch System (SLS) and Orion as the fundamental capabilities needed for exploration. Orion and SLS were originally targeted for initial test flights leading to the Asteroid Redirect Mission as part of NASA's focus on the "Journey to Mars." The 2014 National Research Council report *Pathways to Exploration: Rationales and Approaches for a U.S. Program of Human Space Exploration*[2] proposed various rationales and approaches for exploration for NASA to consider. Following a change in the administration in early 2017, the planning for the Asteroid Redirect Mission has given way to NASA's Deep Space Gateway strategy, which NASA announced to the NASA Advisory Council in public session on March 30, 2017. This strategy would have humans orbiting Mars in the mid-2030s. Missions to build a Deep Space Gateway station in cislunar space in the mid- and late 2020s, while NASA's commitments to the ISS potentially transition in a post-2024 timeframe, would be the next major step. These changes to NASA's exploration path ultimately influence the overall phasing and approach to the needed scientific research and development, albeit with many of the same fundamental questions needing answers. In addition, future planning for space life and physical sciences research is dependent on the strategy and

[1] National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C.

[2] National Research Council, 2014, *Pathways to Exploration: Rationales and Approaches for a U.S. Program of Human Space Exploration*, The National Academies Press, Washington, D.C.

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planning for the ISS beyond the currently funded 2024 timeframe. NASA appears to be working toward improved alignment between its exploration strategy and research priorities.

2.1 PROGRAM DEVELOPMENT SINCE THE 2011 DECADAL SURVEY

Following the 2011 decadal survey release, there have been significant changes affecting the fields of spaceflight and microgravity research. These changes include the methods of transporting research experiments to space and the ISS and the platforms available for research. These changes created both opportunities and challenges to the conduct of science in the field and especially to NASA programs in support of that science. The 2011 decadal survey specifically recognized a programmatic conclusion that, even in this era of change, the future of space exploration very much depends on life and physical sciences research being "central to NASA's exploration mission" and "embraced throughout the agency."[3] Moreover, the decadal survey stated that space life and physical sciences needed strong leadership and positioning within the overall NASA organizational structure to best enable NASA to engage science in prioritizations.

In a major programmatic response, NASA formed the SLPSRA Division in 2011, with NASA's Office of the Chief Scientist playing a leading role in its organization. This was a very important step in addressing the primary research needs and priorities described in the 2011 decadal survey. The formation of SLPSRA also directly addressed a primary programmatic recommendation of the decadal survey. SLPSRA includes the Human Research Program (HRP), the Space Biology Program, and the Physical Sciences Program. SLPSRA funding derives from ISS program elements. Therefore, within SLPSRA, the space life and physical sciences portfolio that it oversees is directly associated with the human exploration program within NASA. This association of the space life and physical sciences programs and the Human Research Program within SLPSRA nicely positions SLPSRA to conduct the necessary integration of research programs. SLPSRA has pursued such integration since its inception through strong leadership and highly dedicated science management efforts within the division, as well as strong coordination with the science leadership of the ISS.

The placement of SLPSRA within HEOMD, an operations directorate within NASA, presents the enormous opportunity for SLPSRA to exert influence on integration and mission planning to best deploy science in existing missions and within the developing new exploration mission options. The association of SLPSRA funding within the ISS budget line of HEOMD assures that SLPSRA has a voice "at the table"[4] in mission planning with the ISS, the major platform for space life and physical sciences research. This opportunity, however, co-exists with two important challenges. The first challenge is the advocacy of science within an operations directorate and the potential culture issues that can distinguish science and operations. This challenge is being addressed by SLPSRA via its efforts to integrate the science research results into the overall exploration planning within HEOMD. The second challenge is that the SLPSRA budget flows from the ISS Program budget, and the ISS is a platform with a potentially limited lifetime. SLPSRA long-term microgravity research is therefore dependent on the post-2024 ISS strategy.

In 2014, the National Academies of Sciences, Engineering, and Medicine established the Committee on Biological and Physical Sciences in Space (CBPSS) at the request of SLPSRA. The CBPSS is overseen jointly by the National Academies' Space Studies Board and the Aeronautics and Space Engineering Board. The overarching purpose of the committee is to support scientific progress in space research in the biological, medical, and physical sciences and assist the federal government, primarily NASA, in integrating and planning programs in these fields. As part of this work, the committee provides an independent forum for identifying and discussing issues in space life and physical sciences among stakeholders. It is also charged with monitoring progress in the implementation of the decadal survey recommendations.Thus, the ongoing and regular interactions between NASA and CBPSS, coupled with SLPSRA's extensive community engagement activities, signal a strong commitment to the science of the decadal survey.

[3] National Research Council, 2011, *Recapturing a Future for Space Exploration*, p. 3.
[4] Ibid.

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Finding 2-1: NASA addressed the overall strategy of the report *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era* in an appropriate and responsive programmatic manner through the formation of SLPSRA. NASA has enhanced community science input related to the decadal survey by developing a strong interaction with the Committee on Biological and Physical Sciences in Space.

2.1.1 The Diversifying Landscape of Entities Involved in Microgravity Research

The research portfolio called out in the decadal survey now is impacted by several organizations and entities beyond SLPSRA. One major entity is the U.S. ISS National Laboratory, designated as such by the U.S. Congress in the 2005 NASA Authorization Act (P.L. 109-155). CASIS was chosen in 2011 by NASA to be the manager of the laboratory. The presence of the ISS National Laboratory and its integration into the operations and management of the ISS has had a significant effect on ISS science inputs, outputs, and operations. A more complete description of the laboratory is presented in Section 2.3, along with additional discussion about the challenges and opportunities of this expanding and diversifying landscape of entities involved in science in low Earth orbit (LEO). CASIS, international partners, other NASA organizations (such as the Space Technology Mission Directorate and the Science Mission Directorate) and other government agencies, such as the National Institutes of Health, the National Institute of Standards and Technology, and the National Science Foundation, all have some interest in microgravity research.

The 6 years since the 2011 decadal survey has seen the development of commercial orbital and suborbital launch providers and an increasing economic development interest in microgravity research. These development interests, together with the interests of other agencies and international entities, all present NASA with a challenging set of interrelated interests that all have the potential to contribute to, and draw upon, microgravity research resources.

A resultant major change has been the capability of private enterprise to transport experiments to the ISS and for the private sector to present other spaceflight-related research capabilities. This growth has resulted in NASA developing new relationships and joint efforts with companies such as SpaceX and Orbital/ATK under cargo resupply contracts replacing capabilities previously provided by the space shuttle and international partners. These transportation services have largely maintained the volume of experiments delivered to the ISS.

In addition to these orbital commercial services, there has been significant growth in the suborbital research experiment efforts supplied by private enterprise companies. Appendix F includes the data set supplied by the Commercial Spaceflight Federation listing microgravity experiments since 2011 that have flown on balloons, parabolic, or suborbital missions. This is not considered a complete listing; however, it is representative of the growth in microgravity capability since the 2011 decadal survey. For example, there have been approximately 54 microgravity experiments flown on suborbital vehicles since the decadal survey. These new suborbital research platform opportunities were not available in 2011. These platforms provide an increase in the ability to conduct experiments measured in "minutes" of exposure to microgravity, a capability that joins with the "seconds" available in the parabolic aircraft flights and terrestrial drop towers. Therefore, the range of times in suborbital microgravity now available for science research provides a valuable continuum of research opportunities for the days, weeks, months, and years available on the ISS.

2.1.2 Research Inputs, Outputs, and Available Assessment Tools

This diversification of the entities contributing to the nation's microgravity portfolio creates an interesting challenge for measuring the progress of microgravity science. The committee recognizes that NASA, primarily through SLPSRA but also through the ISS Program Office, now coordinates a diverse set of organizations that each have a connected, but individually mandated, stake in the outcomes of microgravity and space life and physical science. A view of the various inputs, outputs, and stakeholders is represented in Figure 2.1. The positive manifestation of this stakeholder diversification includes wide-ranging support across agencies and the commercial marketplace. The challenges that arise include measuring and crediting progress, given the difficulties in managing diverse inputs and outputs. For the committee, the major challenges arise from the various ways in which the contributing entities fund, track, and evaluate research and research outcomes. For example, there are several mechanisms, even

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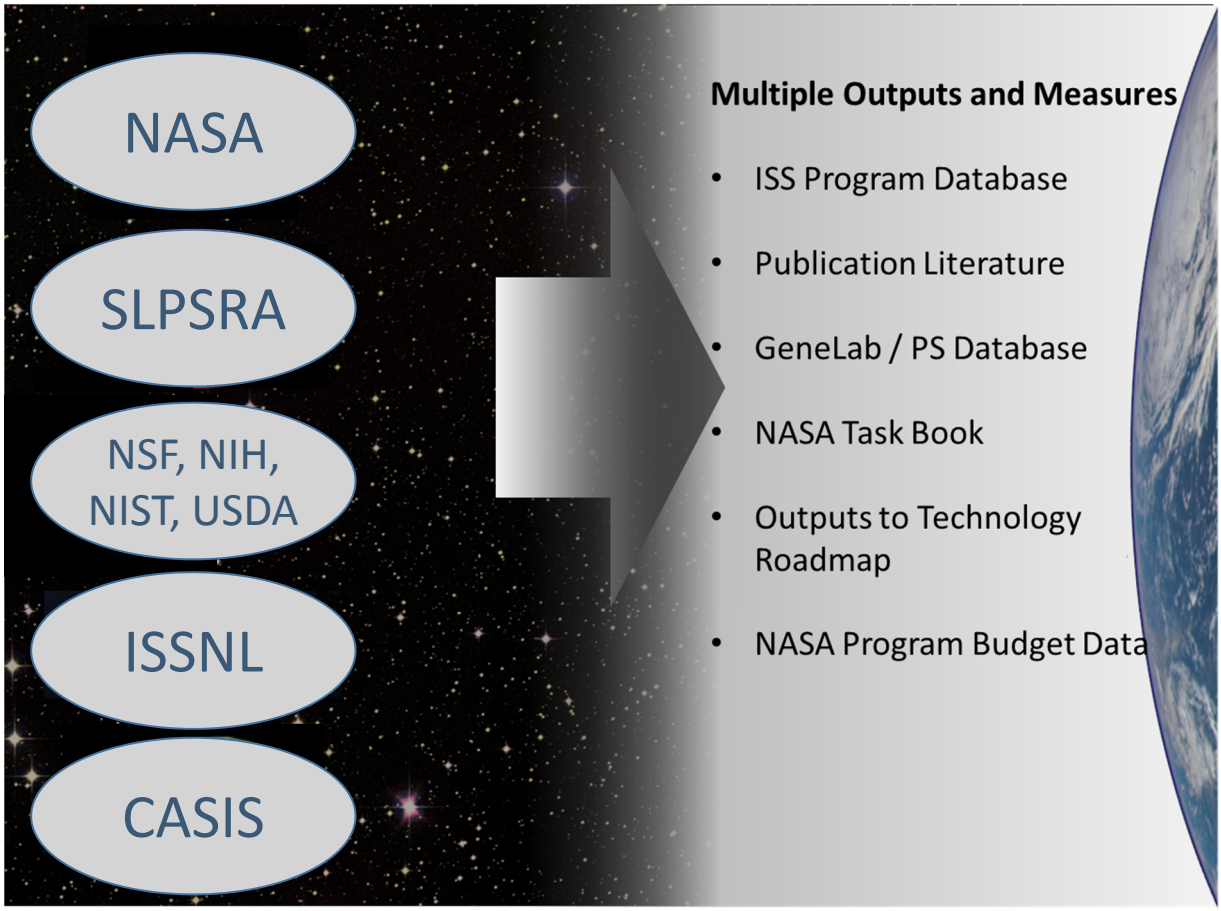

FIGURE 2.1 The evolving complexity and diversity of microgravity and space life and physical sciences inputs and outputs. As interest in microgravity research increases, NASA is increasingly sharing the science landscape with other government agencies (represented by but not limited to the National Science Foundation, the National Institute of Standards and Technology, and the National Institutes of Health). Much of that sharing occurs through the Center for the Advancement of Science in Space and the International Space Station U.S. National Laboratory. In addition to the complexity of entities interested and funding microgravity research (shown on the left), the outputs from that research have grown more complex and now include many measures beyond science publications (shown on the right).

within NASA, for tracking each space life and physical sciences related project. Within each of and among those tracking mechanisms there is yet to develop a codified taxonomy for mapping those projects and their outcomes against the recommendations of the decadal survey.

The committee therefore recognizes and appreciates the difficulties faced by NASA in tracking and measuring progress in this diverse and expanding research landscape. The committee was faced with collecting several data sets, each one useful and limited in its own way in generating a measure of progress, and combining those data sets into a general impression of progress in space life and physical science research. Each of these measures of progress is considered within the context of the entire space life and physical science research portfolio, with a recognition of each measure's limitations.

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2.2 NASA BUDGET AND FUNDING ENVIRONMENT SINCE THE 2011 SPACE LIFE AND PHYSICAL SCIENCES DECADAL SURVEY

One measure of commitment to space life and physical sciences is the overall NASA programmatic budget associated with microgravity science. For a top level assessment, the committee extracted the highest-priority recommendations from Table 13.1 of the decadal survey[5] and then updated the table with current funding information in order to develop an impression of budgets and tasks in the relevant aspects of space life and physical sciences within NASA. This budget information was also considered in the context of the availability of science missions to the ISS. At the next level, the *Space Life and Physical Sciences Research and Applications Division Task Book*[6] (referred to hereafter as the *Task Book*) was consulted for information on the numbers of principal investigators (PIs) and tasks, to evaluate trends over the past decade and within that larger budgetary and operations context. As a related measure, the numbers of proposal solicitations and selections within HEOMD were collated and compared annually. Then actual NASA budgets for external science grants were calculated. Finally, the committee attempted to collate the broader inputs and outputs throughout the agency that contributed to specific recommendations in the 2011 decadal survey. The committee recognizes that these measures are but one means of assessing progress in space life and physical sciences and, therefore, in Chapter 3 delves into the science outputs from the programs as another measure.

2.2.1 Top-Level View of NASA Budgets and Programs in Microgravity Science

Following a large decrease in funding in the 2004-2005 timeframe, NASA has gradually increased funding in space life and physical sciences. In fiscal year (FY) 2016, the funding had risen to about $215 million from approximately $170 million in FY2010. This increase demonstrates NASA's efforts to restore space life and physical science research following ISS assembly completion. However, as shown in Table 2.1, the funding environment has not been restored to its earlier levels of ~$780 million in 1996 and ~$410 million in 2001, all in FY2016 dollars.[7] Microgravity research remains underfunded relative to the funding levels of two decades ago. The years since the 2011 decadal survey have also seen an evolution in the number and organization of formal tasks and PIs involved in these areas of science. In FY1996, the budget for NASA's Office of Life and Microgravity Science and Applications (OLMSA) covered the science areas now in the SLPSRA and included by the decadal survey, in addition to the development of science hardware for the ISS. The large reduction in tasks and PIs followed the establishment of the Vision for Space Exploration in 2005,[8] with many of those tasks winding up in 2005-2006. Therefore, the end-state of that programmatic reduction occurred near 2008-2009, with only 363 reported tasks. This was essentially the status of the program when the 2011 decadal survey was developed. The timeframe from 2011 to the present is reflective of ISS assembly completion, and the partial revival of the science environment since the decadal survey and the formation of the SLPSRA within NASA's HEOMD.

Since the 2011 space life and physical sciences decadal survey and the establishment of SLPSRA, the numbers of formal research tasks and PIs have been on the rise, but remain below the robust portfolio funding and community science numbers seen in the early 2000s.

It is important to note that the budget drop from 1996 to 2001, at a time when the number of science tasks was actually increasing, was in part due to shifting funding lines within NASA. In that timeframe, the budget for ISS facility development was transferred from OLMSA to the ISS program in the Office of Space Flight while the grants to the external community remained in OLMSA.

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[5] National Research Council, 2011, *Recapturing a Future for Space Exploration*. Table 13.1 is reprinted in Appendix E.

[6] NASA, 2016, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*, NASA Research and Education Support Service, August, https://taskbook.nasaprs.com/publication/welcome.cfm.

[7] The fiscal year 2016 constant-year dollars were calculated using information at Bureau of Labor Statistics, "CPI Inflation Calculator," https://www.bls.gov/data/inflation_calculator.htm, accessed October 5, 2017.

[8] NASA, 2004, *The Vision for Space Exploration*, https://www.nasa.gov/pdf/55583main_vision_space_exploration2.pdf.

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TABLE 2.1 Budget and Task Summary from 1996 to 2016

Fiscal Year	Number of Tasks	Budget	Program Contents
1996	872	~500 million (FY2016$: $780 million)	Technology and applications for space research and human support in space, environmental health (microbiology, toxicology, barophysiology, and radiobiology), advanced life support, space human factors, advanced space suits, space biology research, plant biology, combustion science, materials science, fluids, fundamental physics, and supporting orbital operations and research
2001	1,014	~$300 million (FY2016$: $410 million)	Advanced human support, biomedical countermeasures, gravitational biology and ecology, microgravity research, materials science, environmental health, tissue engineering, telescience, human factors, radiation research
2010	364	~$150 million (FY2016$: $1,170 million)	Research supporting human exploration and International Space Station (ISS) space life and physical sciences research, including the Human Research Program (HRP) and the small portion of research within the Exploration Technology Demonstration Program that is related to life and physical sciences research
2016	485	~$215 million (FY2016$: $215 million)	Research includes the Human Research Program, Space Biology and Physical Sciences, including ISS research, ground based research, and research conducted on other flight platforms, funded by the Division of Space Life and Physical Sciences Research and Applications (SLPSRA)

NOTE: Budget numbers comments and tasks for 1996 to 2010 were obtained from Chapter 13 of the 2011 decadal survey (National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era,* The National Academies Press, Washington, D.C.). The 2016 task number was obtained from the NASA SLPSRA Task Book (NASA, 2016, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0,* NASA Research and Education Support Service, https://taskbook.nasaprs.com/publication/welcome.cfm, August 2017). Also note that task numbers reasonably mirrors numbers of principal investigators. The 2016 budget number was obtained from SLPSRA and is the combined 2017 HRP ($144.8 million) and Biological and Physical Sciences Program ($69.9 million) budget. FY, fiscal year.

2.2.2 ISS Timeline and Microgravity Science

The ISS is, by far, the most significant destination for microgravity research at this time. Therefore, the timeline and profile of ISS experimentation impacts progress in the science of the decadal survey.

The ISS has been operational as a science platform since the installation of the Destiny U.S. Laboratory in 2001, with the ISS being essentially assembly-complete since 2010. Microgravity space life and physical science within and associated with the ISS was highly active during the drive to station completion while the Space Shuttle Program was winding down. The Space Shuttle Program ended in 2011 when the shuttle *Atlantis*, on STS-135, used the Multi-Purpose Logistics Module to deliver supplies and experiments to the ISS. Hence, the bulk of microgravity science from the decadal survey has been taking place in the post-space shuttle era.

Since the retirement of the space shuttle, science to the ISS has been dependent on international and commercial missions to the ISS. The NASA Commercial Orbital Transportation Services (COTS) program has replaced the space shuttle for providing U.S. domestic access to the ISS. The COTS program consists, to this point, of commercial resupply missions from SpaceX and Orbital ATK. The SpaceX Dragon capsule offers science upmass (cargo to the ISS) and downmass (cargo returned to Earth), while the Orbital ATK Cygnus capsule offers only upmass to the ISS, with limited science as a free-flyer after unberthing.

SpaceX began the implementation flights of the COTS program with the CRS-1 mission. CRS-1 launched on October 8, 2012, and its Dragon capsule was docked to the ISS until October 28, 2012, and when it returned for splashdown in the Pacific Ocean. This flight established a SpaceX operational pattern of launch, berthing, and then

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remaining on station for approximately 30 days before return. Science experiments have regularly been part of the upmass and downmass. Orbital Sciences (now Orbital ATK) began COTS flights in 2013 with its Cygnus vehicle. The Cygnus operational pattern has been launch, berthing, and staying attached to the ISS for approximately 30 days, followed by detaching and reentry disposal. Science experiments have regularly been part of the upmass within Cygnus.

From 2010 through half of 2017, two to five missions per year (see Figure 2.2) have carried science experiments to and from the ISS. In both Cygnus and Dragon, there has been some upmass for science along with various upmass configurations, including controlled temperature storage. For Dragon only, there has been downmass available, again, also in controlled temperature storage.

2.2.3 Task Book Assessment of PIs and Tasks 2004-2016

The committee also reviewed the NASA *Task Book*[9] data for details on the number of PIs and number of tasks tracked by SLPSRA, as a measure of programmatic commitment to science. These data are freely available on the web[10] and the *Task Book* is easily searched, providing one means of assessing the state and trajectory of science managed by SLPSRA. In keeping with the budget numbers presented in Table 2.1, the number of PIs and tasks is recovering from a nadir in 2008-2009. However, the science project numbers and workforce remain well below the robust era of the early 2000s. The committee assessment shows a decline in the science workforce external to NASA, as measured by number of PIs, of almost 50 percent compared to 2004, with a steady but slow increase since 2009-2011, as shown in Figure 2.3.

The drop from 2004 in PIs and tasks was not spread evenly throughout SLPSRA. HRP suffered a ~30 percent loss of PIs, whereas Space Biology lost ~75 percent and Physical Sciences lost ~85 percent of their PI workforces. By 2016, HRP had recovered to 82 percent, Space Biology to 47 percent, and Physical Sciences to 16 percent of the pre-nadir levels. Looked at in terms of growth since the timeframe of the decadal survey, the 2016 numbers show clearly that there has been a small but continuing growth of the SLPSRA external PI community. HRP PIs have increased from 210 in 2010 to 235 in 2016. Space Biology PI numbers have increased from 43 to 65, and Physical Sciences PIs have increased from 48 to 71 over the same timeframe. This is shown in Figure 2.4.

2.2.4 Solicitations and Selections 2010-2016

The number of proposal solicitations[11] and the numbers of proposals selected for grant funding is a related top-level measure of NASA commitment to decadal survey science and recommendations. While the numbers of investigators and tasks (Figures 2.3 and 2.4) also measure top-level engagement, the number of solicitations represents NASA requests for that engagement (see Figure 2.5). It is clear that HEOMD has nearly doubled the number of solicitations in 2016, compared to 2010. The number of proposals selected for funding has similarly doubled.

2.2.5 External Grants Total Funding for Microgravity Science 2007-2016

Another measure of commitment to space life and physical sciences is the number and funding levels of grants to engage the external science community in solving space life and physical science related questions. The grant dollars invested by SLPSRA to the external science community to accomplish NASA research has shown an increase above its low of 2007-2009, shown in Figure 2.6. The total SLPSRA funding to acquire external science has risen from a low of ~$21 million in 2007 to ~$50 million in 2016. The HRP grants have risen from $17 million to $25 million, and the combined Space Biology and Physical Sciences grants have risen from $4 million to $20 million, with most of that increase happening in the period from 2011 to 2016. These figures show an increasingly robust use by SLPSRA of the expertise present in the science community formally outside of NASA.

[9] NASA, 2016, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*, NASA Research and Education Support Service, https://taskbook.nasaprs.com/publication/welcome.cfm, August 2017.

[10] Ibid.

[11] Formally known as NASA Research Announcements.

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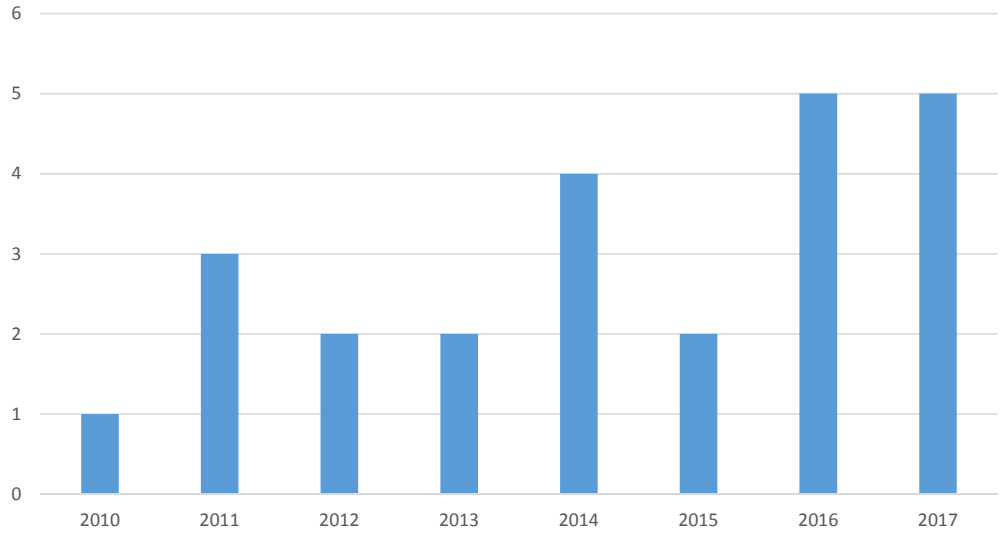

FIGURE 2.2 Science-capable U.S. missions to the International Space Station (ISS). This graph presents all vehicles docking with the ISS that were launched from the United States and that carried science to the ISS. These data include all Orbital ATK and SpaceX CRS missions, as well as the final space shuttle missions STS-130 through STS-135. NOTE: 2017 is complete through the launch of CRS-13, which is scheduled for December 2017. SOURCE: Data extracted from NASA materials, https://www.nasa.gov/feature/visiting-vehicle-launches-arrivals-and-departures.

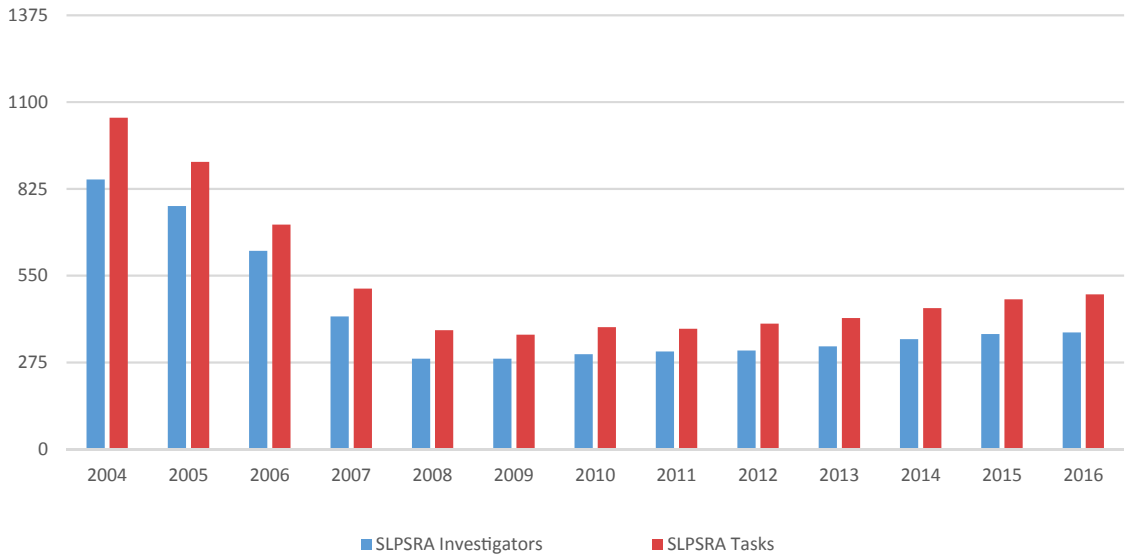

FIGURE 2.3 Number of principal investigators and tasks by year—all programs, Human Research Program (HRP) and Biological and Physical Sciences Program (BPS). SOURCE: NASA, 2016, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*, NASA Research and Education Support Service, https://taskbook.nasaprs.com/publication/welcome.cfm, August.

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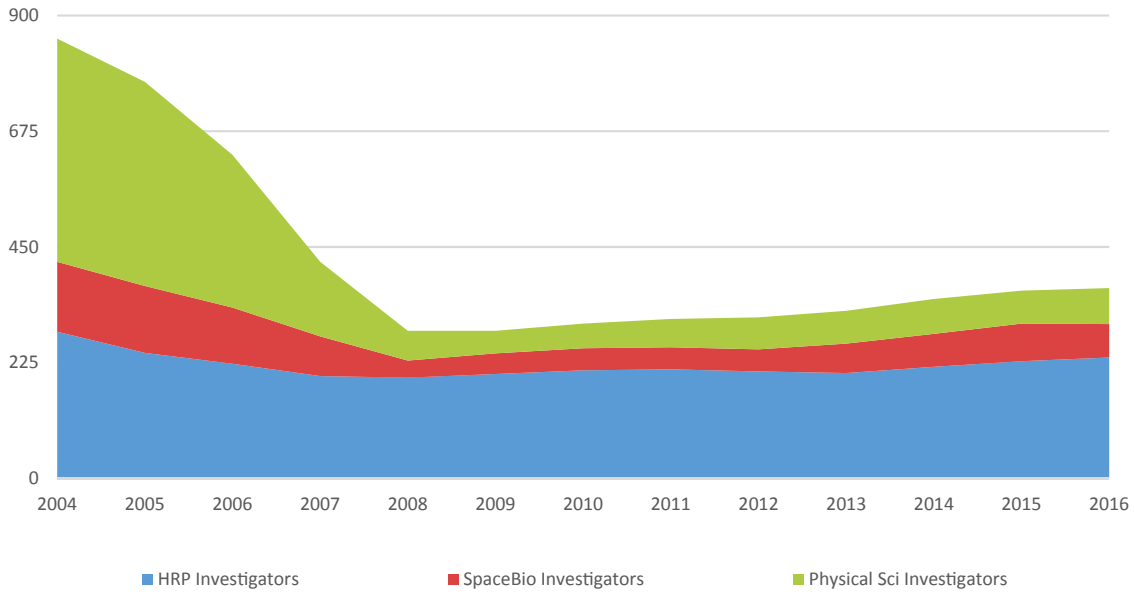

FIGURE 2.4 Number of principal investigators by program area and year—Human Research, Space Biology, and Physical Sciences. SOURCE: NASA, 2016, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*, NASA Research and Education Support Service, https://taskbook.nasaprs.com/publication/welcome.cfm, August.

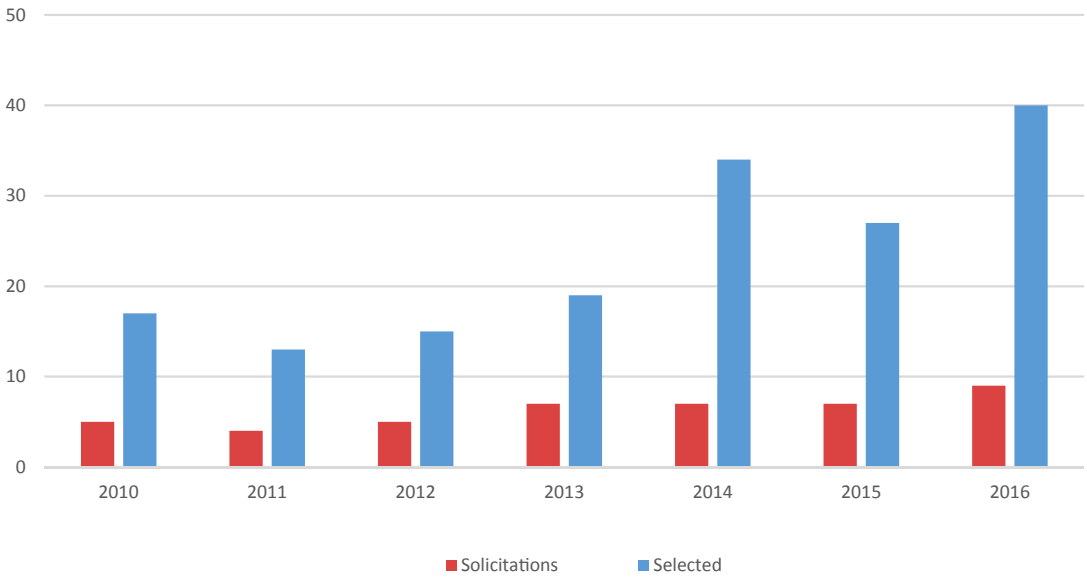

FIGURE 2.5 Announcements solicitating proposals and proposals selected for funding across the Human Exploration and Operations Mission Directorate. SOURCE: NASA Division of Space Life and Physical Sciences Research and Applications.

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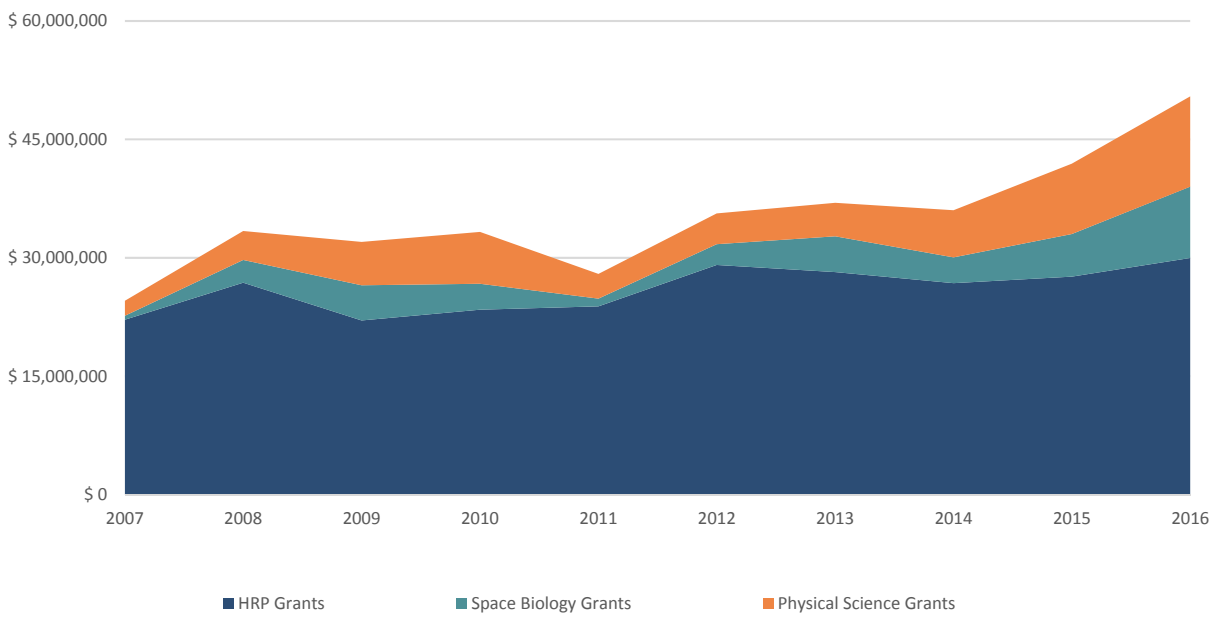

FIGURE 2.6 Grant dollars from Division of Space Life and Physical Sciences Research and Applications (SLPSRA) to engage the external science community for microgravity science in support of NASA. NOTE: HRP, Human Research Program. SOURCE: Data supplied by SLPSRA.

An important part of the microgravity research landscape in the external community has been the National Space Biomedical Research Institute (NSBRI), which was founded in 1997 on a 15-year contract, later extended for an additional 5 years until 2016. The institute was funded by a congressional budget line as a cooperative agreement within the overall NASA budget. NSBRI was originally designated as a research institute fully autonomous from NASA, focusing primarily on countermeasure research to prevent/mitigate physical and behavioral health problems during space missions. However, over the years, collaboration between the NSBRI and HRP evolved to form a greater partnership role. The NSBRI was a consortium of 12 research institutes, consisting of 7 research teams related to different domains of human health in space. While NASA Research Announcements (NRAs) generally included solicitations for both HRP and NSBRI-funded proposals, different topics for each organization were included in each NRA. Individual peer review panels evaluated both HRP and NSBRI proposals according to the same scientific quality criteria; however, within the pool of proposals that met threshold acceptable scientific scores, NSBRI selected proposals to fund based on their own scientific priorities. Because of the countermeasure focus, these priorities did not necessarily align with the highly recommended priorities of the 2011 decadal survey. Yet the NSBRI clearly has addressed some of these priorities. According to the 2016 *Task Book*, 73 of the 468 proposals listed were funded through the NSBRI. The decline in number funded by 2016 was due the fact that all NSBRI proposals had a 2016 end date. The Baylor College of Medicine–led Translational Research Institute (TRI) is a consortium funded as the successor to the NSBRI through a NASA cooperative agreement. The stated focus is on space health through facilitating technologies for astronaut health during deep space exploration missions. However, the impact of the establishment of the TRI and its role in microgravity research is not yet known.

The committee recognizes that dollars spent through NASA grants to the external science community (Figures 2.5, 2.6, and 2.7) are but one measure of the budgetary commitments necessary for the conduct of research within the portfolio of the 2011 decadal survey (Table 2.1). Additional recognized expenses include (but are not limited to) personnel costs for civil servant engineers, scientists, and managers required to perform the hardware design or development, assembly, and test, as well as the mission operations, grant management, project manage-

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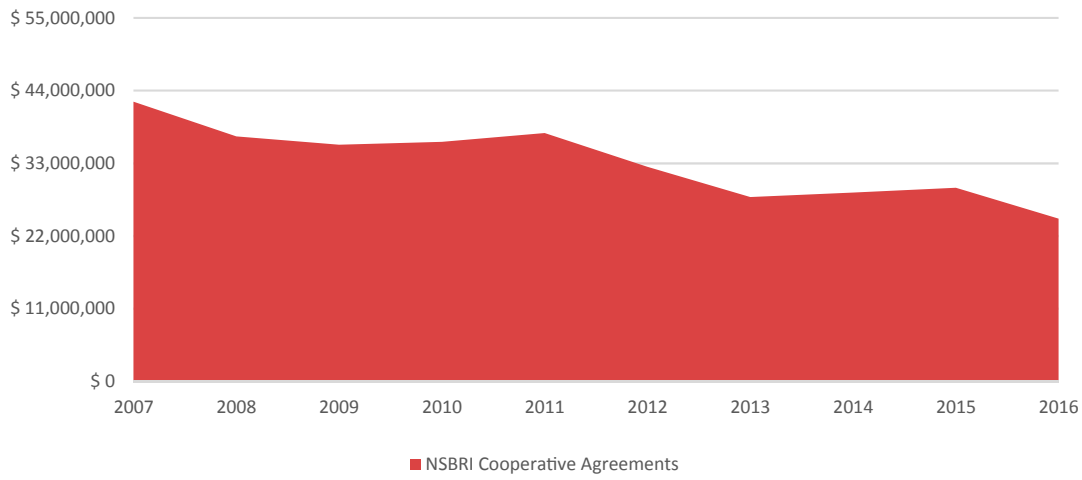

FIGURE 2.7 National Space Biomedical Research Institute (NSBRI) grant dollars plotted over time SOURCE: Data supplied by Division of Space Life and Physical Sciences Research and Applications.

ment, and associated tasks that are required for spaceflight projects. In addition to these necessary budget items within SLPSRA, the committee is cognizant that other elements within NASA contribute to the science of this portfolio. Examples of these contributions would be the large costs for projects such as Cold Atom Lab, mission operations and integration for projects, and databases such as Physical Sciences Informatics and GeneLab, which reside within budget lines outside of SLPSRA and potentially outside the data captured in Table 2.1.

Therefore, the committee broadly considered the science procurement by NASA from the external community, in the era of increased access to the ISS, as a significant measure of commitment to solving issues of space life and physical science. The increase in SLPSRA engagements of external science, as measured by both grant dollars and task/PI numbers, is seen as a very positive trend, and yet the ending of the NSBRI and the start of the TRI makes it difficult to calibrate the total science flow from those sources in this overall funding landscape. Moreover, the committee was made aware of concerns over the lack of regular cadence for NRAs from SLPSRA. At several points in the Community Input Symposium, participants clearly indicated that research programs suffer from an unpredictability and irregularity of proposal requests—a situation that imperils research programs and makes especially difficult the hiring and retention of students and postdoctoral associates within research grants. For example, Space Biology did not offer an NRA from 2014 through much of 2017, leaving many established programs unable to continue space biology projects. Similarly, participants in the Community Input Symposium indicated that SLPSRA has broken up the funding streams on grants into small increments that cover only parts of years, rather than fully funding proposals at one time or at least funding them on a yearly basis. This practice has left many important research programs lurching forward between small funding portions, rather than smoothly progressing through projects.

SLPSRA has gradually increased funding and external research engagements in space biology and physical sciences since the large decrease in funding in the 2007-2008 timeframe. In FY2016, the funding had risen to approximately $215 million from ~$170 million in 2010. This increase demonstrates NASA's efforts to re-emphasize space life sciences, and to a lesser extent physical science research, following ISS assembly completion. However, the funding environment, particularly in physical sciences, has not recovered from its earlier levels of ~$795 million in 1996 and ~$420 million in 2001, all in FY2016 dollars. In addition, there is no regular cadence of calls for research proposals; such a cadence is necessary to maintain the pool of science and researchers ready to support NASA needs over the long term.

Finding 2-2: NASA is supporting the science of the report *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era* by modestly increasing the budget and tasks for space life

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and physical sciences within overall funding constraints and the operational diversification of microgravity science across research platforms.

2.2.6 NASA Funding Aligned to Specific Decadal Survey Recommendations

The committee attempted to assess how the full range of priorities identified in the 2011 decadal survey was addressed by NASA. The committee also sought to evaluate inputs from CASIS, because CASIS has recognized the value in tracking elements of its science portfolio that address decadal survey recommendations. Both NASA and CASIS provided information through presentations to the committee to aid in the project mapping against specific decadal survey recommendations. The committee understands that NASA also uses the Small Business Innovative Research (SBIR) and Small Business Technology Transfer (STTR) programs to fund microgravity research. The committee has seen that microgravity research is included in the SBIR and STTR topic areas. However, NASA does not track the SBIR or STTR projects against the decadal survey priorities, and therefore neither NSBRI, SBIR, nor STTR research results entered into any attempts to map inputs to specific decadal survey recommendations.

The information from NASA regarding the mapping and tracking of science research to specific decadal survey recommendations took several forms and indicated a recognition of the value of tracking investments against recommendations. Research proposals to SLPSRA now require proposing PIs to identify decadal survey recommendations addressed by the proposed studies. Mapping of projects to decadal survey priorities is often a component of science reports within SLPSRA. Hence, it is clear to the committee that NASA has developed a culture of attending to decadal survey recommendations. That said, comprehensive tracking mechanisms that regularly map ongoing science research onto decadal survey recommendations does not yet exist.

The committee received reports on ISS Program Office efforts to create a database of ISS projects,[12] so as to increase research reporting to stakeholders within the government, the research community, the space community at large, and the general public. The addition of 2011 decadal survey research priorities to this database is particularly commendable, as it helps illustrate the magnitude of alignment among NASA's programmatic operations that address decadal survey recommendations. As of June 2017, the database is currently tracking 326 research projects, 249 (76.4 percent) of which have a PI sponsored within the SLPSRA program. Many of them are linked to specific 2011 decadal survey recommendations. Clearly, microgravity space life and physical sciences are major scientific reasons for the ISS as a research platform. The committee commends the ISS Program Office's effort to make this information available in an open web-based format and to summarize research findings in plain language for educational purposes. These metrics are impressive; however, they are limited and certainly not complete with respect to tracking the total commitment of NASA to microgravity space life and physical sciences priorities. This limitation in tracking to the decadal survey was recognized and openly discussed by both NASA and the committee.

The committee was further briefed on approximately 2,000 ground-based studies reported through the SLPSRA *Task Book* database: nearly 200 studies reported by CASIS; a large, yet undetermined, amount of funding for intramural directed intramural research projects; and hundreds, perhaps thousands, of ground- and space-based studies sponsored by NASA's international partners. Through individual queries to representatives of SBIR/STTR, the committee also learned of potentially hundreds of relevant SBIR/STTR research projects. Unfortunately, no mechanism currently exists, particularly in the *Task Book*, to summarize this vast body of research in a manner that allows mapping to specific priorities.

Thus, the current state of research reporting is impressive and rapidly improving, but remains fragmented and incomplete as far as a tool to assess the distribution of NASA commitments to the space life and physical sciences portfolio. Continued efforts to expand and improve research reporting—specifically, how the projects relate to the decadal survey—is essential for truly advanced program planning and maximizing research utility in the remaining years of the ISS and beyond. In this midterm review, therefore, the committee has primarily utilized program level data to assess NASA's attention to the decadal survey priorities, and relied on community input and the committee's own scientific expertise to assess commitments to specific priority areas.

[12] NASA, "International Space Station: Experiment List," https://www.nasa.gov/mission_pages/station/research/experiments/experiments_by_name.html, accessed October 5, 2017.

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Finding 2-3: NASA has a diverse approach to the reporting of investigation funding, published results, and experiments that is not fully coordinated among programs and offices, making overall tracking to decadal survey priorities difficult.

The ISS Program Office's efforts to increase research reporting to stakeholders within the government, the research community, and the community at large establishes an excellent model. Unfortunately, no mechanism exists to summarize this vast body of research across the agency. Thus, the current state of research reporting is fragmented and incomplete, but improving. Continued efforts to expand and improve research reporting are essential for program planning.

Finding 2-3 deals specifically with the reporting of science research and the codification of research results for the broader community. Other parts of this report deal with the need for more of the results from space life and physical sciences to map onto the NASA Technology Roadmaps[13] to enable exploration.

2.3 EXISTING AND EMERGING CHALLENGES TO AND OPPORTUNITIES FOR IMPLEMENTATION OF THE 2011 DECADAL SURVEY

NASA and its partners have had to address several additional key research resource constraints and opportunities beyond budget. Specific examples include the emergence of the ISS National Laboratory, crew time and cargo delivery as limiting resources, and potential interests of other government agencies. While these constraints are being addressed, the impending imperative to leave LEO for deep space carries with it an added and increasingly recognized pressure to retire NASA involvement in the ISS. The year 2024 is currently estimated as the potential beginning of the transition period for NASA involvement in the ISS.[14] With the clock always ticking, recognition of a potential end date for NASA ISS participation looms as a major consideration in microgravity science, particularly in accomplishing the space life and physical science needed to best enable exploration of deep space.

The NASA Transition Authorization Act of 2017 (P.L. 115-10) defined two primary objectives for the ISS program going forward: (1) to achieve the long-term goal and objectives under Section 202 of the NASA Authorization Act of 2010 and (2) to pursue a research program that advances knowledge and provides other benefits to the nation. NASA was tasked, in coordination with CASIS, ISS partners, the scientific user community, and the commercial space sector to "develop a plan to transition in a step-wise approach from the current regime that relies heavily on NASA sponsorship to a regime where NASA could be one of many customers of a LEO non-governmental human space flight enterprise." The act further states that NASA must provide an ISS transition report to Congress by December 1, 2017, with updates every 2 years until 2023.

Allocations of flight resources of upmass, downmass, and crew time are specified in intergovernmental agreements and U.S. legislation. A top-level representation of this allocation relationship is shown as a pie chart in Figure 2.8. In Figure 2.8, there are two pies, one for Russian research and one for the research on the international portion of the ISS that NASA and the other partners share. The U.S. and international research pie is divided into NASA's allocation and the allocation for the international partners. NASA's allocation is then split, due to U.S. legislation, into 50 percent for NASA research and 50 percent for the ISS National Laboratory (further details in the following section). Only a small portion of the NASA research slice is then made available to be allocated for the priorities outlined in the 2011 decadal survey.

This allocation of resources can lead to allocation difficulties, because the various entities having different, and sometimes competing, priorities negotiate for resources. As in the case for Space Biology, shown in Figure 2.9, while both CASIS and NASA may start off with similar fundamental science needs, the NASA needs for flight medicine to enable humans to go to Mars can pull research in a very different direction than the CASIS needs to produce medicinal results relevant back on Earth. Sometimes research can synergistically serve both needs; however, this creative tension between legitimate end goals does not always result in commonality of science needs or resource utilization.

[13] NASA, "NASA 2015 Technology Roadmaps," https://www.nasa.gov/offices/oct/home/roadmaps/index.html, accessed October 5, 2017.
[14] Presentation by NASA to the committee.

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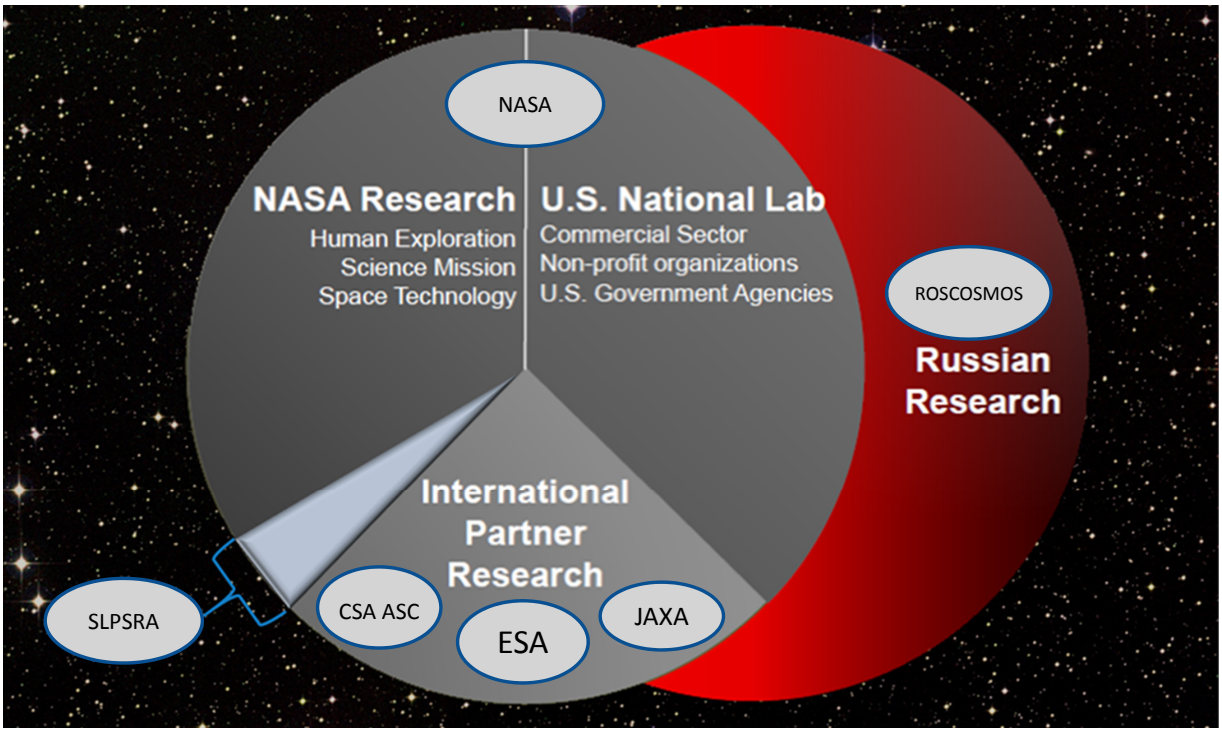

FIGURE 2.8 Allocation of resources within the International Space Station (ISS) and ISS programs. The Division of Space Life and Physical Sciences Research and Applications (SLPSRA) is the lowest priority within the slice of the pie that represents NASA research. Russian research takes place in the Russian facilities on the ISS. Within the U.S. and international segments, international partner research takes priority, and the remaining portion of the U.S. research capability is split between the U.S. ISS National Laboratory and NASA. Within the NASA segment, the Human Research Program takes precedent. Therefore, only a small portion of the capabilities of the ISS are available to SLPSRA and to SLPSRA research that enables exploration. The "size" of the SLPSRA slice of the ISS research is variable during any increment, hence, the slice in this graphic is meant only as a notional representation. SOURCE: Primary image courtesy of NASA.

These differences in research priorities pressure the science operations constraints. The ISS is constrained in several resources. Upmass, downmass, and crew time are all constrained to some extent, especially when flight delays occur that affect the resupply and return plan. In addition, operations scenarios, such as the major priority of keeping the ISS flying safely, can also disrupt research performance. There are also many costs to using the ISS, which include transportation costs, costs of schedule delays, costs and complexity of payload or facility development, and costs of implementation.

NASA provided estimates on downmass capability to the committee. In the space shuttle era, the downmass capability at four flights per year was approximately 36,000 kg. As NASA progresses into the use of the private enterprise space transportation providers (e.g., SpaceX, Boeing, Sierra Nevada), downmass capability is estimated to be less than one-third of the space shuttle era capability. This is an obvious limitation on the ability of researchers to return samples for terrestrial evaluation, recognizing the mass of the samples and the needed containers.

As a result of these constraints, considerations of areas of science that do not need ISS but still support exploration need careful attention and fostering. While an entire chapter of the 2011 decadal survey (Chapter 11) is dedicated to the capabilities of the ISS, it should be noted that the decadal survey recognizes that ground-based experiments are necessary in some areas, and some microgravity and spaceflight-related studies are well suited

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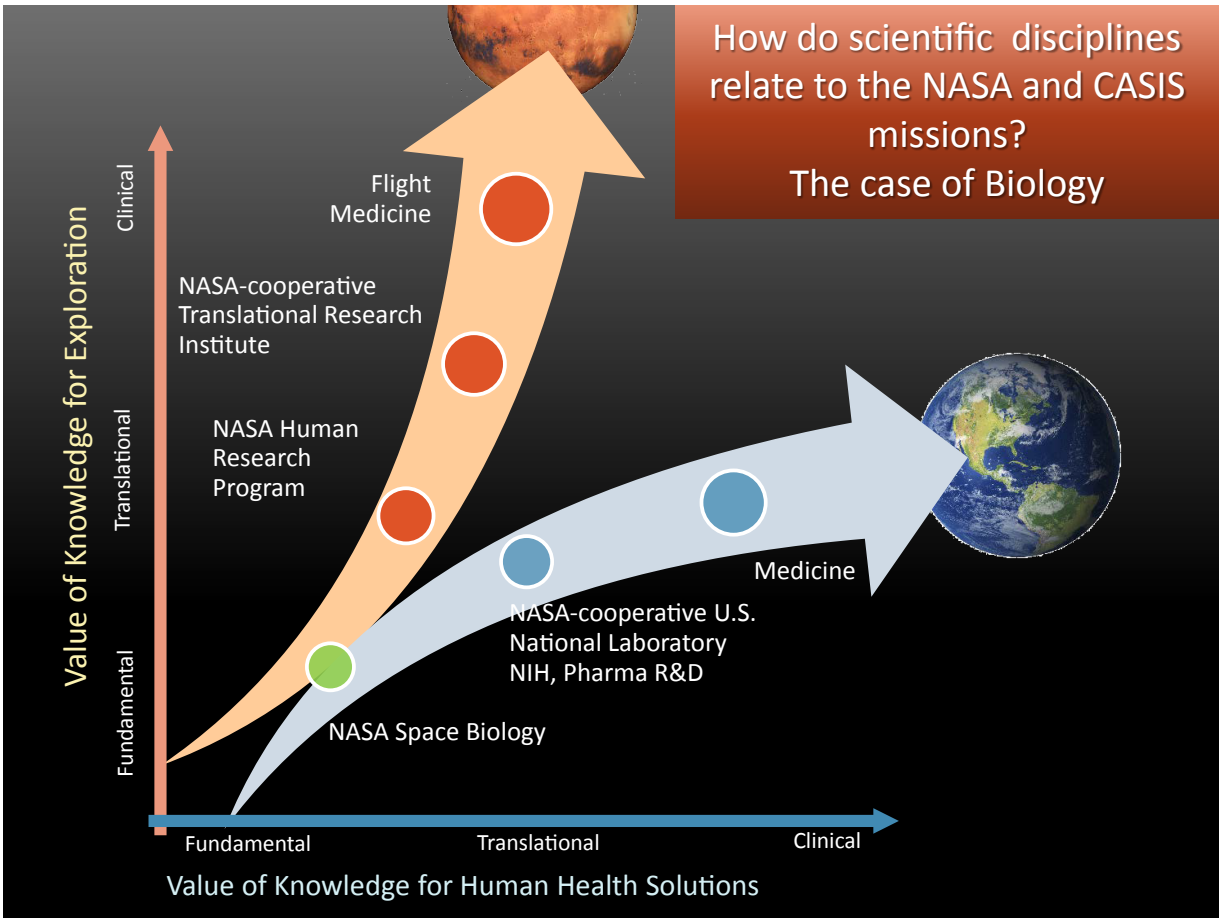

FIGURE 2.9 Commonalities and divergences in science research needs. This graphic illustrates that at the fundamental end of the spectrum NASA and the Center for the Advancement of Science in Space (CASIS) share many science needs and goals. However, as medical knowledge is translated to the clinical use end, the goals of NASA and CASIS are decidedly distinct. SOURCE: Julie A. Robinson, NASA, "I.S.S. Research: Past, Present and Future," presentation to the committee on February 8, 2017, slide 14.

for platforms other than the ISS. A robust consideration of ground studies and all available platforms would help ensure that the entire portfolio moves forward, targeting to ISS those experiments that specifically require its unique capabilities. However, the committee recognizes that space life and physical sciences research will need to be longer term—extending the duration of experiments to multiple years in microgravity—to best support deep space exploration. It is clear that the largest unknowns in exploration have to do with extended durations in microgravity, and these unknowns currently span many of the high-priority recommendations of the decadal survey. The HRP states that the high priorities will not be complete prior to 2024 and that work remains to address and optimize the responses to the risks after 2024.[15] Efforts in areas such as gene regulation, cognitive and behavioral conditions, and dietary strategies and food systems, discussed later in this report, clearly show the need for research beyond the 2024 timeframe.

[15] Steve Davison, presentation to the committee on February 7, 2017, slide 21.

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Finding 2-4: LEO science research needed for exploration will be required beyond 2024. Extended durations in microgravity, measured in years, will continue to be required to best meet deep space exploration research needs.

2.3.1 The ISS U.S. National Laboratory

The ISS National Laboratory has had a large impact on the science prioritized in the decadal survey. The impacts are felt at several levels—from the operations level where ISS crew time and resource priorities are laid out and executed, through to the public policy levels where science achievements are identified and recognized. The committee recognizes the need to work within a clear understanding of the genesis, role, and responsibility of the ISS National Laboratory and CASIS.

The ISS National Laboratory is a conceptual framework created by the U.S. Congress under the NASA Authorization Act of 2005 (P.L. 109-155), directing NASA to (1) designate the U.S. Orbital Segment (USOS) of the ISS as a "national laboratory" and (2) develop a plan to "increase the utilization of the ISS by other Federal entities and the private sector . . ." Components of the ISS constructed and operated by NASA and international partners, including the European Space Agency (ESA), the Canadian Space Agency (CSA), and the Japan Aerospace Exploration Agency (JAXA), are called the U.S. Orbital Segment, which currently consists of 11 pressurized components and various external elements.

The NASA Authorization Act of 2010 (P.L. 111-267) directed NASA to enter into a cooperative agreement with a not-for-profit entity—exempt from taxation under section 501(c)(3) of title 26 of the U.S. Code—to manage the ISS National Laboratory. To fulfill its mission as directed by the law, "ISS National Laboratory managed experiments shall be guaranteed access to, and utilization of, not less than 50 percent of the United States research capacity allocation, including power, cold stowage, and requisite crew time onboard the ISS through at least September 30, 2024. Access to the ISS research capacity includes provision for the adequate upmass and downmass capabilities to utilize the ISS research capacity, as available. The Administrator may allocate additional capacity to the ISS National Laboratory should such capacity be in excess of NASA research requirements."[16] Reciprocally, the NASA Authorization Act of 2010 enables NASA to negotiate with CASIS if it needs greater than 50 percent for exploration research.

In 2011, NASA selected CASIS[17] to be the manager of the ISS National Laboratory. As the nation's newest and only national laboratory orbiting Earth, the ISS National Laboratory strengthens relationships among NASA, other federal entities, private and public sector institutions, and new-to-space communities in the pursuit of national priorities for the advancement of science, technology, engineering, and education to benefit Earth. The ISS National Laboratory, as described under Section 202 of the NASA Authorization Act of 2010, serves as a gateway to space, defining new opportunities for commercial research and development, discovery-based science, and the exploration and economic development of LEO. The primary mission of CASIS is to maximize use of this unparalleled platform for innovation to benefit all humankind. As directed by the NASA Authorization Act of 2010, CASIS also "considers recommendations of the National Academies decadal survey on biological and physical sciences in space[18] in establishing research priorities and in developing proposed enhancements of research capacity and opportunities for the ISS National Laboratory."

The committee has come to understand the ISS National Laboratory and CASIS to be effective and critical partners in the utilization of the ISS. Further, the committee recognizes that CASIS may consider the decadal survey and may contribute to the science of the survey; however, CASIS is under no direct obligation to meet decadal survey priorities. The priorities of the decadal survey, especially those that are directed toward enabling exploration, largely remain the purview of NASA in general and SLPSRA in particular.

[16] 42 U.S. Code § 18354.

[17] See the U.S. National Laboratory/Center for the Advancement of Science in Space, Inc. website at http://www.iss-casis.org, accessed October 5, 2017.

[18] National Research Council, 2011, *Recapturing a Future for Space Exploration*.

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2.3.2 Crew on ISS and Crew Time Available for Science

Crew time available to science is often cited as the single biggest limitation to the conduct of science, including decadal survey science, on the ISS. This limitation is a result of two separate but related situations; the number of crew on station and the policies related to crew time distribution. Crew time is heavily oversubscribed and is the primary limiting resource for many types of research. For example, human research and rodent research demand is high and is crew-time intensive. Another challenge is the ISS National Laboratory/CASIS demand has grown to fully use the 50 percent allocation granted in the NASA Authorization of 2010 for crew time beginning in late 2015, which has required replanning of NASA-funded research. The distribution of crew time on the ISS is determined through a rather complicated set of prioritizations. According to presentations from NASA to the committee, the prioritization of crew time puts SLPSRA and a large portion of the 2011 decadal survey science at the bottom, making it difficult for SLPSRA science to be conducted in a timely fashion.

With the end of the space shuttle program, the United States and all of the other ISS partners became dependent on the reduced capability of the Russian Soyuz to transport crews to and from the station. The Soyuz is a three-person transport system. Therefore, at any one time the expedition crew size is a maximum of five or six persons when two Soyuz are docked to the ISS. At any one time, generally three persons are on the ISS within the U.S. segment. Although the NASA Commercial Crew Program has been initiated in the United States to provide more crew transportation capability, the ISS is spending its prime years literally understaffed. As a result, the ISS has yet to reach its envisioned science potential.

According to data presented to the committee, the total crew time available for science is approximately 35 hours per week (Figure 2.10). Agreements reached in 2017 that allow NASA contractors to buy additional seats on Soyuz, increasing the number of crew on the U.S. segment to four, greatly increase the crew time available for science. These additional Soyuz seats are expected to be available through 2019, potentially easing the transition from the current Soyuz to the Commercial Crew Program. The Commercial Crew Program was initiated by NASA to stimulate the development of privately operated crew vehicles that can service launches to LEO and the ISS. At present, Commercial Crew launches to the ISS have not yet occurred, and test flights are not planned until 2018. The advent of the Commercial Crew Program is expected to increase crew size to a full ISS capacity, so that nearly an additional person-week of crew time effort for science can be performed, potentially doubling the crew time availability for science to ~60 hours per week.

All too often, space life and physical science waits in a queue for crew time (see next section). Some experiments that are highly important to the decadal survey portfolio need crew time beyond their program allocation and, therefore, never make it to the queue or may be dropped. The importance of crew time as a limitation cannot be overstated—the amount of crew time available for space life and physical science is the single biggest factor in accomplishing the science needed before sending humans into deep space. While NASA and CASIS have been working to identify potential synergies and are working to more effectively coordinate research, CASIS priorities, by definition and law, need not align with NASA priorities nor the 2011 decadal survey recommendations. Nevertheless, the committee recognizes the laudable efforts of CASIS to integrate consideration of the decadal survey into their processes, as well as their work with NASA and the microgravity research community to advance issues of common interest.

Finding 2-5: NASA coordination with ISS National Laboratory continues to develop in a laudable manner. However, as NASA meets the challenges of ISS National Laboratory implementation, its own exploration priorities face increased pressure to garner ISS research resources, such as crew time, cargo delivery, and ISS experiment prioritization. This pressure leaves NASA exploration research priorities at risk of not being met.

2.3.3 Interagency Multisponsorship and Commercial Interactions

The first half of this decade has seen some major shifts in NASA and commercial plans regarding the human occupancy of LEO, and those shifts present both opportunities and challenges to the paradigm that had guided space life and physical science research through 2010. The emergence of commercial interest in space had been

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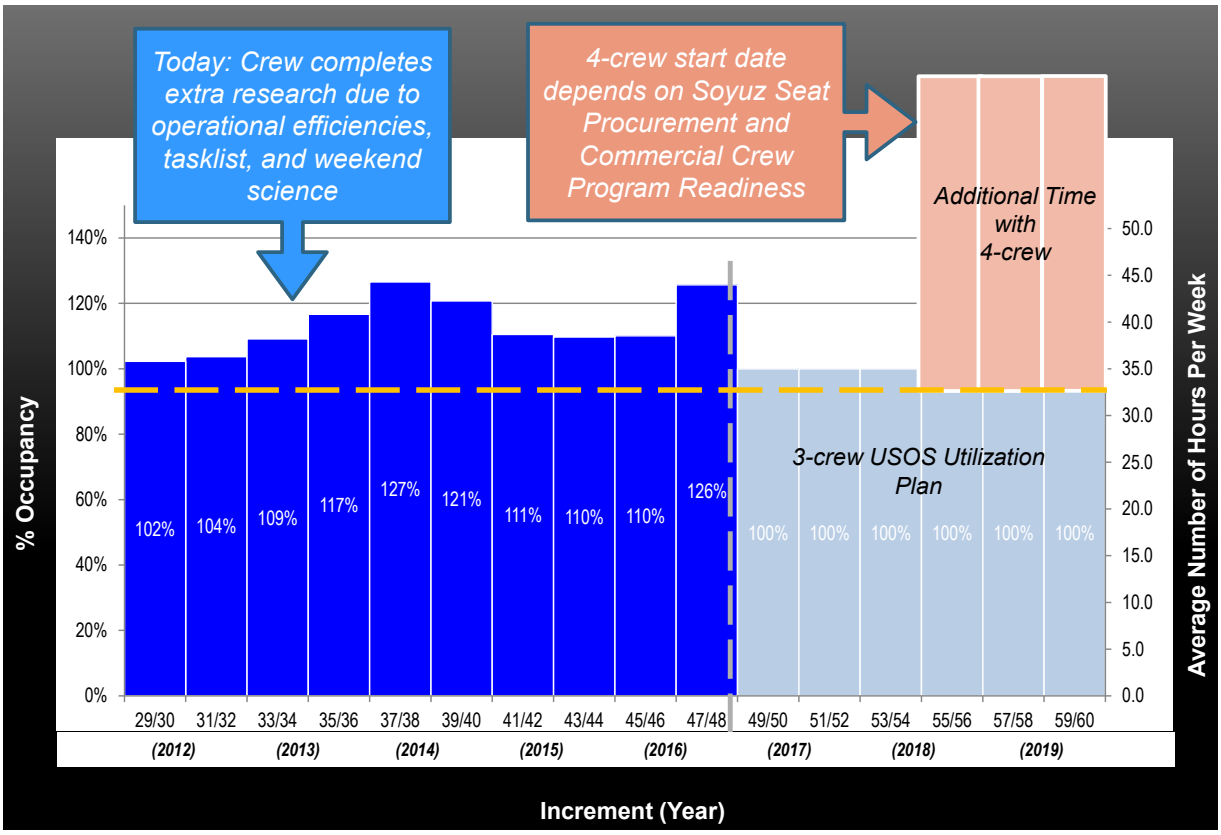

FIGURE 2.10 Crew time available for science on the International Space Station (ISS). The percent crew occupancy is indicated on the left y axis, and the total hours available for science is indicated on the right y axis. SOURCE: Julie A. Robinson, NASA, "I.S.S. Research: Past, Present and Future," presentation to the committee on February 8, 2017, slide 19.

a significant development. Commercial spaceflight now influences research, space tourism, and space manufacturing, all of which can make use of LEO. While space tourism has been considered one potential return on investment for some commercial interests, the potential for unique research and manufacturing opportunities has fueled a wider interest among a broad scope of U.S. industries, ranging from biomedical processes all the way through the physical sciences involved in manufacturing. Much of the role of CASIS has been perceived to be to generate, enable, and grow private interest in the use of the ISS. CASIS has been successful in generating and enabling such interest, yet this interest actually competes for resources with NASA exploration science, because the goals of the research within the ISS National Laboratory need not be aligned with NASA exploration needs or the recommendations of the 2011 decadal survey.

Over the same half decade, NASA is developing an approach aimed at building closer interactions with other government agencies such as the National Institutes of Health (NIH) and the National Science Foundation (NSF). In part, this set of interactions is viewed as a logical outgrowth of regular access to the ISS National Laboratory. However, an additional goal of developing these interactions stems from a desire by NASA that other government agencies might develop a strong interest in pursuing their own microgravity programs, programs that would be enabled by the ISS National Laboratory as well as the potential growth in commercial LEO platform providers. The committee understands NASA's desire that some of these other government agencies or private laboratories might be interested in taking over part of the research portfolio of the decadal survey, freeing up NASA resources for deep space exploration.

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Presentations made to the committee suggest that NASA is actively engaged in searching for research synergies with other government and nongovernment agencies. There is evidence of clear interaction in the form of signed memoranda of understanding (MOUs) and even some shared funding opportunities. These newer MOUs continue a practice of interagency cooperation that has existed for years with much of NASA science, and interactions in human health that are particularly relevant to NASA exploration. Examples of such agreements include the following:

- *CASIS*. In 2015, SLPSRA signed an MOU with CASIS in order to develop a formal framework of collaboration in the use of the ISS, and to co-develop use campaigns. From 2012 to 2016, CASIS has awarded 196 projects with companies, schools, universities, and other government agencies for use of the ISS National Laboratory manifest.
- *Alfred P. Sloan Foundation*. In 2016, NASA entered into a nonreimbursable Space Act Agreement (SAA) with the Sloan Foundation to provide a structure for interactions that support research on the microbial ecosystems of the ISS. This SAA provided a framework that produced a postdoctoral associate program specifically to advance knowledge of the Microbiology of the Built Environment (MoBE)—particularly, because MoBE can be uniquely advanced by studies in the environment of the ISS and microgravity.
- *NSF*. In 2010, NASA entered into a memorandum of agreement with NSF on space, Earth, and biological sciences cooperative activities to strategically plan and conduct joint scientific research projects. One outcome related to this agreement was the ISS/PK4 interagency research agreement that cooperatively supports the Dusty Plasma program, which utilizes the ESA Plasma-Krystal 4 facility on the ISS. The Dusty Plasma program resulted in a joint solicitation between HEOMD and NSF.
- *National Institute of Standards and Technology (NIST)*. In 2015, NIST signed an MOU with NASA to provide technical interchange to enhance the impact of NASA's MaterialsLab microgravity materials science programs. This is an agreement to exchange expertise and data.
- *NIH*. NASA and NIH have a strong history of science interactions. This history includes the BioMed-ISS solicitation in 2011 that built upon a 2007 MOU signed by NIH and NASA. In 2017, NASA and NIH signed a letter of agreement creating a management intent within both agencies to look for opportunities for interaction, for cross support of science, to encourage space-related health research, and to engage in other joint research activities. The letter of agreement is also an outgrowth of successful NIH research opportunities that have been directed to space-related science.

There exists, therefore, a clear context and opportunity for joint science programs among NASA, CASIS (consistent with 2010 NASA Authorization Act) and other government and nongovernment agencies and institutions. To date, these joint programs have been designed for the mutual and shared interests of the parties, and there has been no shift of microgravity sciences responsibilities from NASA to those other agencies and institutions. At present, those other agencies and institutions look to NASA to provide unique access to the microgravity environment. NSF in particular was very direct in its characterization of the ISS as a facility provided to the community by NASA. In each case of actual joint science, the science made sense from the standpoint of already existing needs of that other agency, rather than that agency taking on science traditionally within the NASA domain. Scientific collaboration that leverages special expertise and addresses existing needs is apparently valued most in these interactions. The collaboration is built around the unique facilities provided by NASA.

One of the major challenges to these interactions is that other agencies, as well the commercial sector, look to NASA to provide launch and operations costs for microgravity research. As NASA attempts to free itself from those launch and operations costs in order to explore deep space, they will have to be absorbed by those entities wishing to conduct research in microgravity. The question of who pays for future launch and operations costs remains the single largest unknown with regard to expanding interest in microgravity research to other government agencies.

Finding 2-6: NASA has initiated relationships and improved efforts to work with other U.S. agencies and nongovernmental institutions to explore the value of microgravity in science. The committee was unable to identify any other federal agency with a programmatic reason to absorb the full costs of developing and conducting experiments on the ISS, even in the face of rising interest in microgravity research.

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2.3.4 Lead, Collaborate, Watch, and Park Opportunities

The resources available for NASA research are inadequate to support the development of the broad array of research recommended by the decadal survey. One approach for improving the allocation of research resources would be to use a modified version of an approach applied by the Army Research Laboratory (ARL). ARL has classified each of the technologies in its 2015-2035 Science and Technology Campaign Plans[19] as falling into one of three categories: lead, collaborate, or watch. In the 2016 National Academies report *NASA Space Technology Roadmaps and Priorities Revisited*,[20] a recommendation was made to use this system with modified definitions and one additional category: park. To help improve the traceability between this decadal survey research portfolio and the technology portfolio, it is suggested that NASA consider use of these definitions for space life and physical sciences, modified slightly to reflect research rather than technology priorities. These four categories can help NASA determine the level of cooperative development with others and thus reduce their research expenditures.

- *Lead*. NASA's needs and timing for a given expected research result are so unique that advancing the research will require NASA investment without substantial shared investments by others. Maintaining in-house expertise and infrastructure for this research while engaging scientists outside the agency is critical to NASA's unique needs.
- *Collaborate*. NASA establishes an interdependent partnership with other organizations (government, industry, academia, or international partners) to pursue research using shared investments. This collaboration can take several forms. A common example is NASA and another government agency coordinating research and communicating the results to each other. Another form is a public–private partnership in which NASA provides part of the funding with cost sharing by the industry partner. NASA can also provide its research partners with access to unique infrastructure, research advances, and in-house expertise that significantly influence the direction of the collaboration. Collaborating allows NASA's in-house technical experts to develop research that they may not have otherwise been afforded the opportunity to do so.
- *Watch*. NASA maintains high vigilance monitoring emerging research and corresponding efforts within industry, academia, and international markets. Research in this category will most likely be research that is not unique to NASA missions. It is important that NASA stay actively engaged in the national and international scientific dialog to remain poised to react to developments that meet NASA needs. One means of staying actively engaged in the national and international scientific dialog is the attendance at and the participation in scientific conferences by NASA researchers.
- *Park*. Pursuing research advancement requires better definition of mission or operational requirements before proceeding. NASA would minimize effort for research in this category until better definition is achieved.

2.4 PLANS FOR THE INTERNATIONAL SPACE STATION; ISS POST 2024

NASA's stated goal of moving exploration beyond LEO to deep space is laudable and exciting, while also placing enormous strain on the microgravity science needed to best support human spaceflight. With the task of examining microgravity science in relation to this shift in emphasis toward exploring deep space, the committee carefully considered both the future plans for ISS and the timelines for the NASA transition to deep space exploration. These overarching policy and program plans were further considered in the context of advances within the science portfolio of the decadal survey in Chapter 4.

NASA has developed excellent collaborations with the ISS partners. For example, joint HRP bed-rest studies with ESA and the German Space Agency (DLR); SLPSRA space biology "Bion" collaborations with Russia, ESA,

[19] U.S. Army Research Laboratory, 2014, *S&T Campaign Plans 2015-2035*, Adelphi, Md., September, http://www.arl.army.mil/www/default.

[20] National Academies of Sciences, Engineering, and Medicine, 2016, *NASA Space Technology Roadmaps and Priorities Revisited*, The National Academies Press, Washington, D.C., doi: 10.17226/23582.

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CNES,[21] and JAXA; combustion physics efforts such as the Saffire project (ESA, Japan, Russia and American scientists); and the Flare project (led by JAXA in Japan, the United States, and ESA).

The future of the ISS is an important consideration for microgravity research. NASA is currently evaluating the potential strategies for the ISS beyond 2024. Currently, all the ISS partners have committed funding for ISS activities through 2024. The 2017 NASA Transition Authorization Act requires NASA to provide a report to Congress by December 1, 2017, on the ISS strategy post-2024. Based on the science outlined as high priority in this report, the committee does not believe that the necessary research to extend human presence beyond LEO will be complete prior to the 2024 timeframe. Thus, the committee believes that ISS planning beyond 2024 is critical to the overall research necessary for deep space exploration.

NASA has appropriately initiated an internal activity to develop and assess various options for ISS beyond 2024. At the time of this report, no conclusions have been drawn. Conclusions and recommendations are expected in the report to Congress due December 1, 2017. NASA did, however, provide the committee a presentation on the status of this activity. To identify and assess potential ISS strategy options, NASA has developed a list of key considerations. These considerations include a long-term U.S. presence in LEO that is characterized by leadership of international partnerships, and continuing support for commercial space activities in LEO that occur in an ecosystem that (1) fosters ongoing deep space exploration objectives requiring LEO while (2) continuing research and development for fundamental science and technology. The overarching consideration is affordability and sustainability.

NASA is evaluating options to address the considerations above, options that include extension of the current operating model, use of various public–private partnerships, and incorporation of private modules for private enterprise purposes. Transitioning to each of these options is also being assessed keeping in mind the potential of a permanent and/or short duration crew-tended platform. NASA's options include the reuse of existing, on-orbit ISS hardware. Long-term microgravity research and utilization requirements for NASA, together with the impacts to the National Laboratory, are apparently factored into the NASA analysis. There has been, and continues to be, significant assessment of the various public–private partnerships. For example, in May 2017, the National Academies hosted a "Symposium on America's Future in Space." A key element of this symposium was discussion of public–private partnerships, how they have been applied in areas other than space exploration, history of application in the space exploration industry, and possible outcomes. This discussion provided an excellent opportunity to broaden the knowledge base on public–private partnerships for consideration in the area of space exploration. Lessons-learned and ideas from these and similar discussions will be valuable as NASA develops the ISS post-2024 strategy.

ISS hardware can support additional needed research beyond 2024 based on NASA's extensive analysis. NASA has determined that significant margin and life remains in the ISS hardware through the 2020s and into the 2030s.

In conclusion, the committee applauds NASA's efforts in developing the ISS strategy beyond 2024. However, it is critical that the strategy be developed in a timely manner to identify the necessary science for the highest priority research needs identified in Chapter 4. In addition, it is not clear to the committee that the necessary business case analyses or use scenarios exist to appropriately assess the commercial development or private enterprise options for LEO, as well as the potential eventual scope of partnerships with other government agencies. In all cases, the full costs associated with access to the ISS, operations of experiments, maintenance of the ISS facilities, and experiment data/sample return must be considered along with the costs of the specific research experiments.

Finding 2-7: As NASA's ISS strategy evolves for the timeframe beyond the current 2024 commitments, there is a need for clear and objective analyses of all costs necessary to support research. Cost elements (e.g., launch, personnel, operations, facility maintenance) for ISS research projects and CASIS activities are important factors in the assessment of research activities and for determining appropriate partner funding levels.

[21] The Centre National d'Études Spatiales is the French government space agency.

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2.4.1 Budget Allocation in Light of Limited Time Remaining on the ISS

NASA has worked diligently to develop the ISS research capabilities since the 2011 decadal survey. Figure 2.11 shows the current and planned increase in Major Internal Research Facilities for the ISS and the EXPRESS Rack Expansion from 2016 through 2018. This is significant growth in capability given the tight fiscal constraints and the challenges of developing on-orbit flight hardware. Presentations to the committee indicate that occupancy space within the ISS will near 95 percent by the end of 2017, with EXPRESS Racks full by mid-2018 (Figure 2.12). Similarly, occupancy of the external ISS racks and facilities is also nearly full. As the research capability on the ISS reaches maximum capacity through 2018, it will be important to focus on complete utilization of those facilities, rather than developing new facilities. Budget data presented to the committee indicate that if fewer new facilities were to be developed by NASA, significant funds would become available for the conduct of experiments in existing facilities on the ISS. As the research capability is increased through 2018, it is important to realize that the available research time from 2018 to 2024 is limited, given the crew time limitations. NASA is proactively working to maximize these resources; however, the costs required for additional research capabilities must be traded off against the funding of high-priority research. Cost for additional research capability typically exceeds the cost of individual research investigations by at least an order of magnitude due to the essential engineering, development, and crew safety precautions. It will therefore be important for NASA to focus remaining funding and crew time through the 2024 timeframe on the high-priority research investigations rather than developing new research facilities and capabilities.

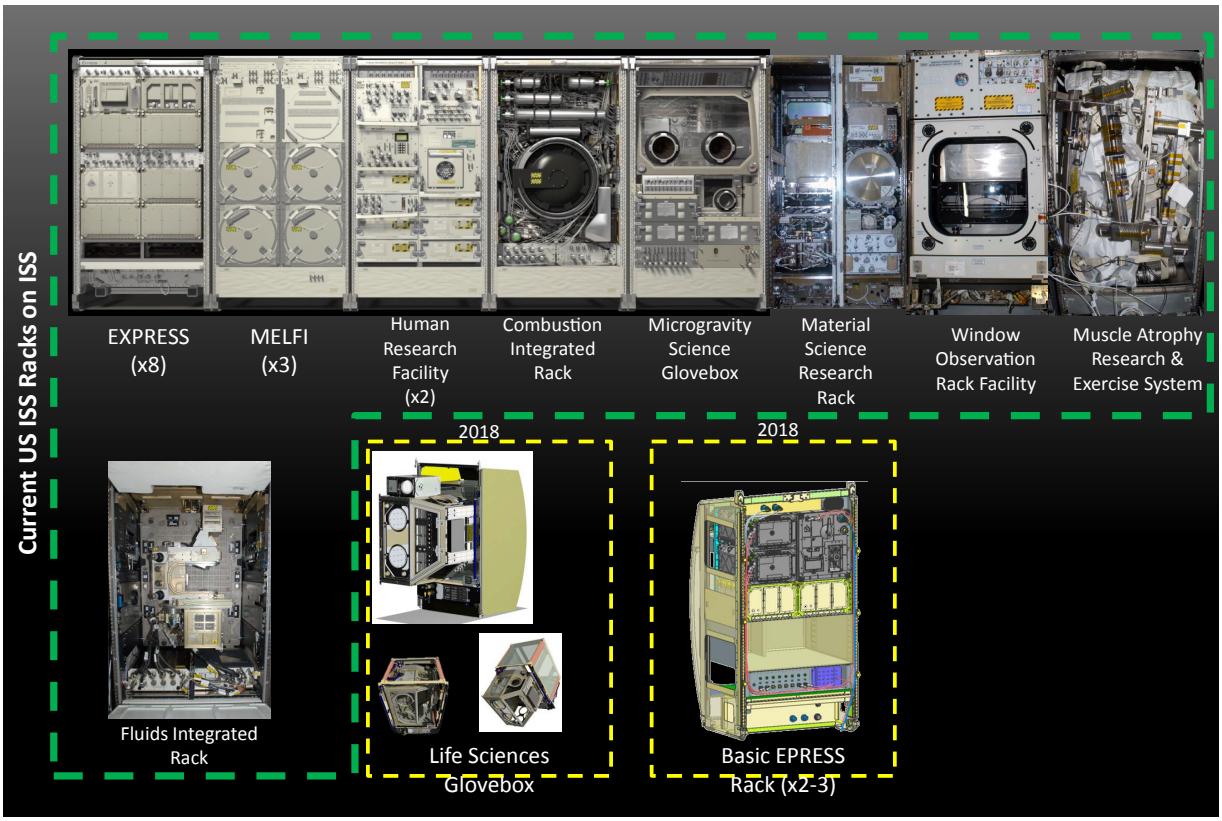

FIGURE 2.11 Current major internal research facilities on International Space Station (ISS) and planned improvements through 2018. SOURCE: Julie A. Robinson, NASA, "I.S.S. Research: Past, Present and Future," presentation to the committee on February 8, 2017, slide 28.

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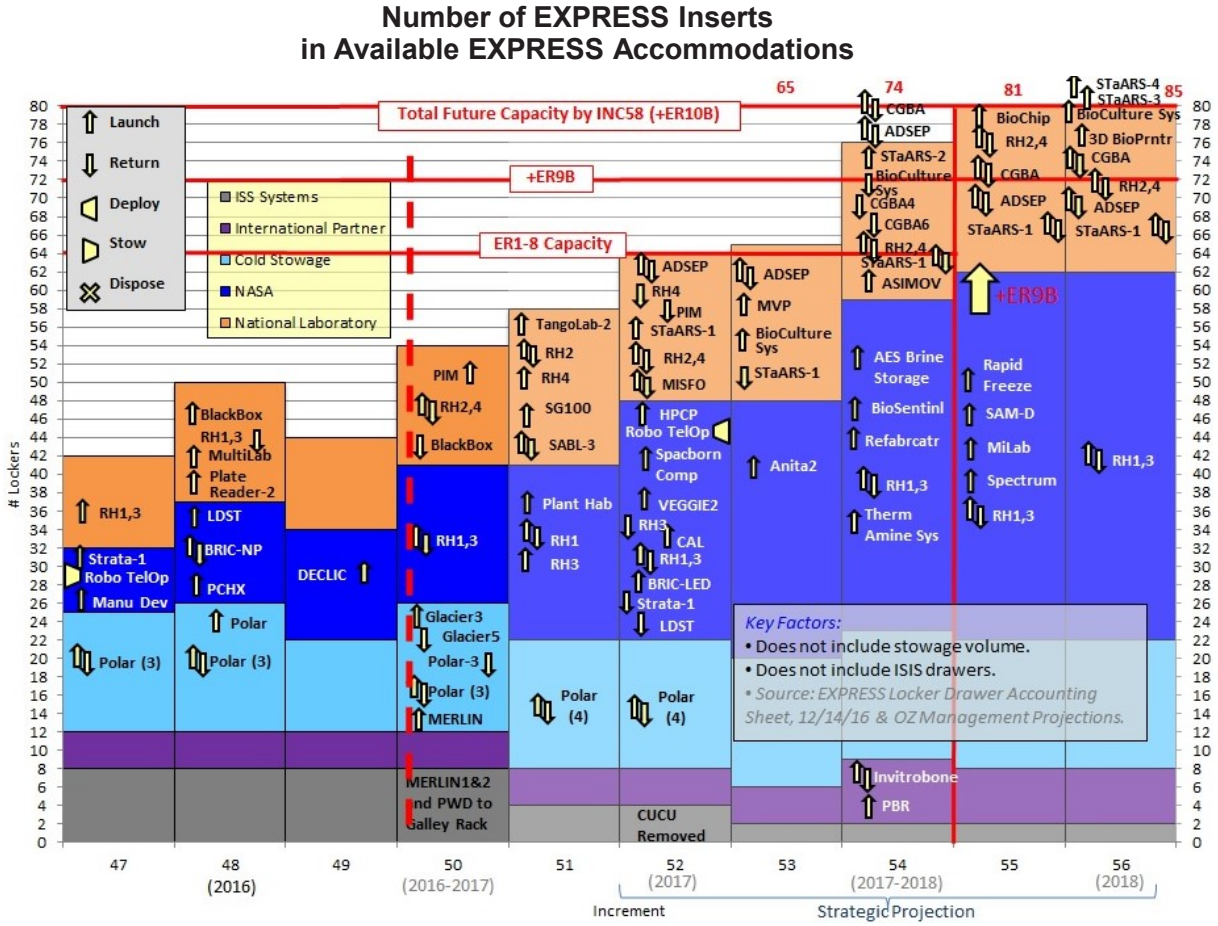

FIGURE 2.12 Number of Expedite the Processing of Experiments for Space Station (EXPRESS) inserts in available EXPRESS accommodations. EXPRESS Racks are defined as multipurpose payload rack systems that store and support research aboard the International Space Station (ISS). These racks provide structural interfaces, power, data, cooling, water, and other items needed to operate any science experiments, irrespective of discipline. Individual racks can accommodate multiple specialized payloads as inserts. There are clear trends that indicate a filled ISS capacity in the 2018 timeframe. SOURCE: Julie A. Robinson, NASA, "I.S.S. Research: Past, Present and Future," presentation to the committee on February 8, 2017, slide 29.

Finding 2-8: While NASA has worked diligently to enhance ISS research capability since the 2011 decadal survey, the potential 2024 transition era of U.S. participation in the ISS may call for a change of strategy with respect to ISS facilities development, including use of privately developed research facilities.

Development of new capabilities is expensive compared with use of existing capabilities, and a greater portion of limited funds could be directed toward science in order to maximize return of data from the ISS in its remaining years.

2.4.2 Open Databases and Reference Experiments

One mechanism envisioned for enhancing the science value of all NASA and NASA interactive science is the establishment of well-designed and well-populated databases of research in microgravity. The establishment

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and use of open databases with machine-readable data is supported by a 2013 Executive Order[22] that made this practice the default for government data. This executive order, among other things, mandated open-access publications and spurred initiatives for the archiving of big data. For NASA HEOMD, this order led to the establishment of the Physical Sciences Informatics (PSI) data repository and GeneLab. PSI arose along with the NIST MOU, with the PSI data repository designed to be the major mechanism for making physical sciences data from SLPSRA and the ISS broadly available to the science community. GeneLab and its biological data repository are designed to be the major mechanism for making large omics-related data sets similarly available to the science community.[23] Both PSI and GeneLab are viewed as extremely positive outcomes in support of open science within NASA. PSI and GeneLab point toward and accomplish the more rapid dissemination of raw and processed data from spaceflight experiments and are quite successful in that regard. Both of these initiatives have driven science toward more data-rich experiments that have the real potential to provide useful data for years after the data is gathered.

The PSI and GeneLab initiatives are viewed as both enlightened and already successful. The committee clearly recognizes that there has been a recognizable shift toward high-data-density experiments that provide data more quickly than in the past to the databases, and those kinds of databases are being used by the community. The SLSPRA request for proposals to use GeneLab data, the Innovation Awards Research Solicitation, was well received by the external science community, with 6 of 34 proposals receiving funding after scientific merit and technical feasibility reviews.

One question that arises is that of database persistence—Are there provisions, financial and administrative, for maintenance and continuous updating of GeneLab and PSI for the indefinite future? Furthermore, is there a plan to add to the data a guide for users showing who already has analyzed the data, and in what fashion, to avoid duplication of effort? The value of the database to users might be greatly increased by including such follow-ups with the data. The committee was presented with cogent future plans for GeneLab that included continued interaction with the user community. Database persistence might be an issue to consider in such interactions.

Although the concept is not clearly developed and articulated by SLPSRA, reference experiments can be used in support of open databases. In one version of the concept, a reference experiment was described as being an experiment of high data density that does more than address a single narrow hypothesis and is proposed by an interactive and extensive team of scientists to collect data that would be suitable for later analysis by multiple groups of investigators. In such an experiment, the resulting data would have the benefit of multiple PI inputs into design and implementation, with the intent of creating large data sets with wider-ranging applicability and extensive post-flight usage than a single-hypothesis experiment. This sort of implementation concept for reference experiments is seen by the committee as having value, although an important question is who will perform the experimental work after the reference experiment is agreed upon. A full process has yet to be developed. For example, the creation of RNA-sequencing libraries, with appropriate technical and biological repeats, would be valuable, but a process needs to be in place to determine the appropriate experts to perform the sequencing and to ensure the latest and best-validated methods and equipment are used. It would be important also to establish who should perform the initial bioinformatic quality controls (for consistency of technical and biological replicates, deduplication, quality check against genomic data, and so on), although the process by which to do this is yet to be developed. Other versions of the reference experiment concept include the use of science definition teams and contracted sample processing. As NASA still is developing its approach to reference experiments, the committee is not in a position to make firm recommendations, other than to encourage the further development of the concept in coordination with the relevant external communities that possess the needed technical and analytical expertise.

[22] White House, Office of the Press Secretary, 2013, "Executive Order—Making Open and Machine Readable the New Default for Government Information," May 9, https://obamawhitehouse.archives.gov/the-press-office/2013/05/09/executive-order-making-open-and-machine-readable-new-default-government-.

[23] The term omics broadly refers those biology technologies ending in -omics, including genomics, proteomics, or metabolomics.

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3

Science Progress Toward the Goals and Priorities of the 2011 Space Life and Physical Sciences Decadal Survey

In this chapter, the committee has assessed the progress made in the respective science disciplines against the priorities identified in the 2011 space life and physical sciences decadal survey.[1] This assessment occurred within the context of the overall advancement of space life and physical sciences and against the backdrop of advancement in related fields outside of the space sciences. This assessment also drew upon a wide-ranging discussion of the high-priority recommendations from the decadal survey and incorporated presentations from NASA and CASIS (Center for the Advancement of Science in Space) and from the literature so that the conclusions would encompass programmatic and science considerations. For each of the discipline sections below, the committee utilized data from the literature and the many presentations to the committee in order to assess progress made on the highest-priority research recommendations summarized in Chapter 13 of the decadal survey. These assessments also informed the development of Findings 2-1 through 2-4 in Chapter 2. Areas for improvement in the respective science disciplines, together with the impact of those areas on the selection of the highest-priority recommendations, are discussed in Chapters 4 and 5. Note that the length of the discipline sections in this chaper and in Chapter 4 generally reflects the degree to which progress could be easily summarized across several related topics within that discipline. Growth in opportunities provided by new tools (such as omics technologies), and the extent of the enabling research needs that are yet to be addressed, also influenced the length of the respective discipline sections. Thus areas of research where a larger proportion of the decadal survey recommendations have been addressed may require less discussion than areas where less progress has been made.

3.1 PLANT AND MICROBIOLOGY

The 2011 decadal survey called out three high-priority areas in Plant and Microbial Science (P). As identified within the decadal survey the specific priority items are as follows:

- P1—Establish a microbial observatory program on the International Space Station (ISS) to conduct long-term, multigenerational studies of microbial population dynamics.

[1] National Research Council (NRC), 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C.

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- P2—Establish a robust spaceflight program of research analyzing plant and microbial growth and physiological responses to the multiple stimuli encountered in spaceflight environments.
- P3—Develop a research program aimed at demonstrating the roles of microbial-plant systems in long-term life support systems.

There has been strong response by NASA to these priorities. In response to these recommendations, NASA has been the sponsoring space agency, starting in 2012, for 34 grants in plant science and 26 in microbiology, as categorized in the NASA *Task Book*.[2] Progress has been good also in terms of publications (although these, as expected, lag the proposals funded by several years), with about a quarter of the articles on flight experiments, half on ground studies, and a quarter being review articles.

Novel discoveries both "enabling" and "enabled by" have been made in all three areas. An example related to P1 is the discovery of changes in virulence and growth of potentially pathogenic bacteria in the ISS environment,[3] including activation of regulatory genes not seen in ground-based controls. Examples for P2 include recognition that stress responses and cell wall biosynthetic responses dominate gene expression changes in plants in the space environment as compared to ground-based controls,[4] and that plant genotype, even one-gene differences, can have a strong differential effect on plant gene expression in space. For long-term life support related to P3, the "Veggie" Veg-01 experiment successfully showed the growth of edible plants on the ISS in a specifically designed set of plant growth modules and that food processing was developed and the attendant regulatory hurdles were cleared in order to allow those plants to be eaten by the crew on orbit. In particular, the life sciences programs related to P1, P2, and P3 have been responsive to changing technologies for gene and gene expression omics data, following the rapid development of these new approaches and methods in the broader science community.

The new U.S. ISS National Laboratory, managed by CASIS, has contributed directly to the plant science area, such as funding of the NanoRacks-Algal Growth and Remediation project, meant to overcome barriers to use of algae for remediation of carbon dioxide to oxygen, and the Characterizing Root Attractions in Arabidopsis projects, to study how growing roots sense and respond to gravity.

NASA has implemented its plans for GeneLab and a focus on missions using omics technologies, with plant and microbial biology contributing significantly to GeneLab. As noted in Chapter 2, the GeneLab database[5] is an integrated repository and bioinformatics system for omics data from space-based research and its ground-based controls as well as and space-relevant experiments. These implementations derive strongly from the 2011 decadal survey plant and microbial recommendations and extend to other discipline areas. Progress to date includes public website access to the repository with searchable and downloadable tables of results from many space missions and sets for ground controls, including some legacy data, and progress toward setting standards for data and metadata. The GeneLab data, Release 1.0.18 deployed on June 30, 2017, presents a total of 133 distinct data sets/studies and more than 6.5 Tb of compressed primary data, together with the associated metadata. Flight experiments have been performed to specifically add data to the GeneLab database, and several principal investigator (PI) experiments have been augmented to provide supplemental data to GeneLab. A new set of grants were conceived and awarded in 2017 to more deeply explore GeneLab data, thus making exceptional use of the omics approaches called out in the decadal survey.

In addition to the research funded directly and fully by NASA, plant and microbial investigations related to P1, P2, and P3 have been supported by CASIS. In areas of biotechnology with respect to plants and microbes, CASIS

[2] National Aeronautics and Space Administration (NASA), 2015, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*, NASA Research and Education Support Service, https://taskbook.nasaprs.com/publication/welcome.cfm.

[3] The grant is to Cheryl Nickerson, "Titled RNA Deep Sequencing and Metabolomic Profiling of Microgravity-Induced Regulation of the Host-Pathogen Interaction: An Integrated Systems Approach," NNX13AM01G, in NASA, 2015, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*.

[4] These grants are to Anna-Lisa Paul, NNX09AL96G, "The Biological Impact of Spaceflight and Extraterrestrial Environments on Molecular Signaling and Gene Expression in Plants," and to Robert Ferl NNX15AB12G, "Early Stage Plant Adaptation to Spaceflight: Molecular Responses of Arabidopsis to the Transition from Terrestrial Environment to Space," in in NASA, 2015, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*.

[5] See the NASA GeneLab website at https://genelab.nasa.gov, accessed October 5, 2017.

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has supported significant studies—more than a dozen mission experiments that relate to the recommendations of the 2011 decadal survey. Through its Good Health initiative, CASIS has partnered with NASA on the major goals of establishing a Microbial Observatory within the ISS, helping to fully address recommendation P1. By exploiting existing discovery research relevant to the National Institutes of Health (NIH) Human Microbiome Project, CASIS is working to expand the breadth and depth of microbiome research in space, in part for the benefit of Earth but also in support of tying space research to the National Microbiome Initiative.[6]

3.2 BEHAVIOR AND MENTAL HEALTH

The 2011 decadal survey identified four priority areas in Behavior and Mental Health (B) research. The priorities targeted (1) mission relevant performance, (2) integrated translational research on individual and group functioning conducted through simulations of long-duration missions, (3) individual differences in resilience to stressors to inform an individualized medicine approach, and (4) enhancement of team performance, including effectiveness of multinational crews. The specific priority areas are as follows:

- B1—Develop sensitive, meaningful, and valid measures of mission-relevant performance for both astronauts and mission control personnel.
- B2—Conduct integrated translational research in which long-duration missions are simulated specifically for the purpose of studying the interrelationships among individual functioning, cognitive performance, sleep, and group dynamics.
- B3—Determine the genetic, physiological, and psychological underpinnings of individual differences in resilience to stressors during extended space missions, with development of an individualized medicine approach to sustaining astronauts during such missions.
- B4—Conduct research to enhance cohesiveness, team performance, and effectiveness of multinational crews, especially under conditions of extreme isolation and autonomy.

The Human Research Program's (HRP) Behavioral Health and Performance (BHP) element has been strongly responsive to the recommendations of the 2011 decadal survey and has made considerable progress in reaching these goals. HRP made a strong research and development investment on projects within the BHP domain (i.e., the Behavioral Medicine (BMed), Team, and Sleep sections) through annual announcements, and in 2014, a second BHP-specific NASA Research Announcements (NRAs), primarily in the BMed and Team areas. Some solicitations included projects that were funded through the National Space Biomedical Research Institute (NSBRI). The specific research priorities listed on these NRAs focused on the recommendations of the 2011 decadal survey. The 2016 *Task Book*[7] lists a total of 76 BHP/Human Factors Behavioral Performance (HFBP) funded projects; 49 of these projects address 2011 decadal survey priorities: B1, 16 projects; B2, 14 projects; B3, 6 projects; B4, 13 projects. Findings have been published in scientific journals, technical reports, and conference proceedings.

A major advance in reaching the goals of the decadal survey is the BHP priority to develop or make use of isolated, confined, and extreme environments, or enclosed simulation facilities, as analogs for different aspects of a Mars mission (B2). The Hawai'i Space Exploration Analog and Simulation (HI-SEAS) habitat located on a Mars-like surface on the Mauna Loa volcano on the Big Island of Hawaii is an example. HI-SEAS includes the experience of both living in confinement and performing in an isolated and potentially dangerous environment by doing simulated space walks outside of the habitat. Beginning with an initial 2013 HI-SEAS 4-month mission, subsequent missions (currently Mission V) have ranged up to 1 year of confinement. Each mission has consisted of a diverse gender, ethnic, and cultural team; several groups of investigators studied each team over the course of their confinement, following the research protocol of their own NASA funded analog project. The combination of results from different groups of investigators, each pursuing their own projects in this setting, has enabled the

[6] Executive Office of the President, 2016, "FACT SHEET: Announcing the National Microbiome Initiative," May 13, https://obamawhitehouse.archives.gov/the-press-office/2016/05/12/fact-sheet-announcing-national-microbiome-initiative.

[7] NASA, 2015, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*.

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overall assessment of numerous individual and group variables. Further, use of the Human Exploration Research Analog (HERA) facility at NASA Johnson Space Center (JSC) and access to National Science Foundation (NSF) Antarctic stations also have provided settings for multiresearch team investigations simulating aspects of a long duration mission. The various simulation studies in isolated environments (e.g., Hi-SEAS, or HERA) address both B2 and B4 priorities. Also, there are both laboratory-based and analog setting (HERA, Antarctica) projects being conducted to identify various cardiovascular, blood-based, and saliva-based biomarkers of stress and resilience. Projects in Antarctica also are focusing on brain changes in structure and function in isolation (B3).

The Cognition test developed by Basner[8] and colleagues was completed and is being evaluated in flight as well as analog studies as a measure of cognitive performance (B1). HRP support for team projects assessing the functioning of multiteam systems is a significant advance; ongoing projects are being carried out in both laboratory and field settings (B4). Other ongoing projects in the team area are developing computer models (composition algorithms) to predict different factors of team performance and inputting data collected on current and past studies. In addition, an important team development is the inclusion of mission controllers in a number of projects.

While not specifically addressing a decadal survey priority, other funded projects focus on specific risk factors designated in the 2006 report *A Risk Reduction Strategy for Human Exploration of Space: A Review of NASA's Bioastronautics Roadmap*.[9] Significant progress has been made, primarily by projects funded through the NSBRI, to develop countermeasures to prevent or mitigate emotional distress. Computer-interactive intervention programs for dealing with depression, anxiety, and conflict resolution are in their final stages of refinement. The development of virtual reality technologies to simulate Earth experiences and avatars to increase motivation to exercise are additional examples of innovative countermeasures that are being developed and empirically tested to deal with the stresses of a Mars mission.

Progress in the sleep area is reflected in the installation of an extensively researched lighting system on the ISS to help entrain circadian rhythms. Another sleep area project is investigating the predictive validity of a set of relevant, valid, and reliable biomarkers for distinguishing those who are more resilient versus those who are more susceptible to the adverse neurobehavioral effects of the combination of high-performance demands and total sleep deprivation stressors—two conditions commonly experienced in spaceflight.

3.3 ANIMAL AND HUMAN BIOLOGY

The 2011 decadal survey identified many high-priority areas of research for Animal and Human Biology (AH). That list (below) represents an ambitious set of priorities under conditions of limited resources and opportunities on orbit.

- AH1—The efficacy of bisphosphonates should be tested in an adequate population of astronauts on the ISS during a 6-month mission.
- AH2—The preservation/reversibility of bone structure/strength should be evaluated when assessing countermeasures.
- AH3—Bone loss studies of genetically altered mice exposed to weightlessness are strongly recommended.
- AH4—New osteoporosis drugs under clinical development should be tested in animal models of weightlessness.
- AH5—Conduct studies to identify underlying mechanisms regulating net skeletal muscle protein balance and protein turnover during states of unloading and recovery.
- AH6—Conduct studies to develop and test new prototype exercise devices and to optimize physical activity paradigms/prescriptions targeting multisystem countermeasures.
- AH7—Determine the daily levels and pattern of recruitment of flexor and extensor muscles of the neck, trunk, arms, and legs at 1 *g* and after being in a novel gravitational environment for up to 6 months.

[8] M. Basner, A. Savitt, T.M. Moore, A.M. Port, S. McGuire, A.J. Ecker, J. Nasrini, et al., 2015, Development and validation of the *Cognition* test battery for spaceflight, *Aerospace Medicine and Human Performance* 86:942-52.

[9] NRC, 2006, *A Risk Reduction Strategy for Human Space Exploration of Space: A Review of NASA's Bioastronautics Roadmap*, The National Academies Press, Washington, D.C.

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- AH8—Determine the basic mechanisms, adaptations, and clinical significance of changes in regional vascular/interstitial pressures (Starling forces) during long-duration space missions.
- AH9—Investigate the effects of prolonged periods of microgravity and partial gravity (3/8 or 1/6 *g*) on the determinants of task-specific, enabling levels of work capacity.
- AH10—Determine the integrative mechanisms of orthostatic intolerance after restoration of gravitational gradients (both 1 *g* and 3/8 *g*).
- AH11—Collaborative studies among flight medicine and cardiovascular epidemiologists are recommended to determine the best screening strategies to avoid flying astronauts with subclinical coronary heart disease that could become manifest during a long-duration exploration-class mission (3 years).
- AH12—Determine the amount and site of the deposition of aerosols of different sizes in the lungs of humans and animals in microgravity.
- AH13—Multiple parameters of T cell activation in cells should be obtained from astronauts before and after re-entry to establish which parameters are altered during flight.
- AH14—Both to address the mechanism(s) of the changes in the immune system and to develop measures to limit the changes, data from multiple organ/system-based studies need to be integrated.
- AH15—Perform mouse studies of immunization and challenge on the ISS using immune samples acquired both prior to and immediately upon re-entry to establish the biological relevance of the changes observed in the immune system. Parameters examined need to be aligned with those in humans influenced by flight.
- AH16—Studies should be conducted on transmission across generations of structural and functional changes induced by exposure to space during development. Ground-based studies should be conducted to develop specialized habitats to support reproducing and developing rodents in space.

In response to these recommendations, NASA has funded, starting in 2012, 36 grants in Animal Biology (Vertebrate and Invertebrate) as listed in the NASA *Task Book*. These new proposals have been funded after severe cut backs in the budget for NASA's Fundamental Biology Program from 2004 to 2010. However, 25 new tasks are listed for basic research in Animal and Human Biology as compared to 123 for applied research in the HRP. In terms of publications, progress has improved over papers published from research grants during 2004 to 2010, although papers from new grants typically lag the grant start-up date by 1 to 4 years.

Since the decadal survey, there have been more opportunities and funding for fundamental Animal and Human Biology research. The number of investigators for fundamental Animal and Human Biology has increased since 2004, but still is much smaller than investigators for HRP where there are annual calls for more applied, risk-based human research on the ISS (Figure 2.4). Support from the HRP for rodent research has been available for high-priority research related to vision impairment and other high-risk areas related to prolonged spaceflight. International collaborations on free flyers such as the Russian Bion mission have provided spaceflight opportunities for the space rodent community (10 NASA-funded tasks). However, rodent research in space has been confined to mice to allow higher sample sizes for the limited number of rodent habitats on orbit. Rodent research on orbit is resource- and crew-time intensive, which significantly limits opportunities on the ISS. Certain opportunities to develop and support a new generation of young researchers are available such as NSBRI's 32 postdoctoral fellowships, but this support is mainly for more applied research related to the HRP. Only one postdoctoral task was listed for new investigators in fundamental Animal and Human Biology, and none were listed for graduate students.

The HRP Human Health and Countermeasures (HHC) element has been responsive to the recommendations of the 2011 decadal survey and has made progress in reaching these goals. It should be noted, however, that high-priority areas identified in the decadal survey are not identical to the highest risk factors identified by the HRP for exploration of deep space.

HRP made a strong research and development investment on projects within HHC. Some solicitations included projects that were funded through the NSBRI. The specific research priorities listed on these NRAs focused on the recommendations of the 2011 decadal survey. The 2016 *Task Book*[10] lists a total of 22 HHC- and NSBRI-funded

[10] NASA, 2016, *The Space Life and Physical Sciences Research and Applications Division Task Book 7.0*.

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projects; 9 of these projects address 2011 decadal survey priorities (AH1, 1; AH2, 1; AH4, 1; AH6, 1; AH9, 2; AH11, 1; AH13, 1; and AH14, 1).

Data are available to document the efficacy of bisphosphonates combined with exercise on the Advanced Resistive Exercise Device (ARED) to maintain bone density as compared to previously available exercise hardware. This progress was funded by the HRP at NASA JSC.[11] Some data on bone reversibility are available from Sibonga[12] and co-workers and Shirazi-Fard[13] and associates. In simulated microgravity, bone density, bone architecture, and muscle mass correlate with mechanical loads.[14] Also during simulated microgravity, sclerostin antibody treatment increases bone mass by increasing bone formation in both normally loaded and unloaded environments.[15] New osteoporosis drugs under clinical development are being tested in animal models on the ISS by CASIS.

Other than documenting some benefits of ARED over interim Resistive Exercise Device (iRED) for the musculoskeletal system, the large individual variability in skeletal muscle changes with long-duration stays on the ISS is again an example of the challenges with muscle health with spaceflight.

Some limited progress is available concerning the effects of prolonged periods of microgravity on the determinants of task-specific, enabling levels of work capacity, but no data concern the effects of partial gravity.

3.4 CROSS-CUTTING ISSUES FOR HUMANS IN THE SPACE ENVIRONMENT

Translating knowledge from basic laboratory discoveries to human spaceflight is a challenging, two-fold task: *Horizontal integration* requires multidisciplinary and transdisciplinary approaches to complex problems; *vertical translation* requires meaningful interactions among basic, preclinical, and clinical scientists to translate fundamental discoveries into improvements in the health and well-being of crew members in space and in re-adaptation to gravity. There are significant scientific gaps in many horizontal or cross-cutting areas that span multiple physiological systems that can affect human mental health, safety, and performance. Many of these interactions also involve interactions with behavioral factors, such as effects on cognitive function. Cross-cutting, thematic areas require the collaboration of teams of experts representing different areas of physiological, behavioral, nutritional and engineering expertise. The goal of these approaches is to mobilize fundamental and applied scientific knowledge and resources in a way that optimizes their utility for human spaceflight crews.

3.4.1 Cross-Cutting Issues Unrelated to Radiation Biology

The 2011 decadal survey listed eight high priorities in the cross-cutting (CC) area unrelated to radiation biology, primarily involving human research. (Radiation is discussed in the following section.) These decadal survey cross-cutting priorities are as follows:

- CC1—To ensure the safety of future commercial orbital and exploration crews, quantify postlanding vertigo and orthostatic intolerance in a sufficiently large sample of returning ISS crews, as part of the immediate post-flight medical exam.
- CC2—Determine whether artificial gravity (AG) is needed as a multisystem countermeasure and whether continuous large-radius AG is needed or intermittent exercise within lower-body negative pressure or

[11] A. LeBlanc T. Matsumoto, J. Jones, J. Shapiro, T. Lang, L. Shackelford, S.M. Smith, et al., 2013, Bisphosphonates as a supplement to exercise to protect bone during long-duration spaceflight, *Osteoporos Int*. 24(7):2105-2114.

[12] J.D. Sibonga, H.J. Evans, H.G. Sung, E.R. Spector, T.F. Lang, V.S. Oganov, A.V. Bakulin, L.C. Shackelford, and A.D. LeBlanc, 2007, Recovery of spaceflight-induced bone loss: bone mineral density after long-duration missions as fitted with an exponential function, *Bone* 41(6):973-978.

[13] Y. Shirazi-Fard, C.E. Metzger, A.T. Kwaczala, S. Judex, S.A. Bloomfield, and H.A. Hogan, 2014, Moderate intensity resistive exercise improves metaphyseal cancellous bone recovery following an initial disuse period, but does not mitigate decrements during a subsequent disuse period in adult rats, *Bone* 66:296-305. doi:10.1016/j.bone.2014.06.005; erratum in 2014 *Bone* 71:265.

[14] R. Ellman, J. Spatz, A. Cloutier, R. Palme, B.A. Christiansen, and M.L. Bouxsein, 2013, Partial reductions in mechanical loading yield proportional changes in bone density, bone architecture, and muscle mass, *J. Bone Miner. Res*. 28(4):875-885.

[15] J.M. Spatz, R. Ellman, A.M. Cloutier, L. Louis, M. van Vliet, L.J. Suva, D. Dwyer, M. Stolina, H.Z. Ke, and M.L. Bouxsein, 2013, Sclerostin antibody inhibits skeletal deterioration due to reduced mechanical loading, *J. Bone Miner. Res*. 28(4):865-874.

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short-radius AG is sufficient. Human studies in ground laboratories are essential to establish dose-response relationships, and what gravity level, gradient, rotations per minute, duration, and frequency are adequate.

- CC3—Conduct studies on humans to determine whether there is an effect of gravity on micronucleation and/or intrapulmonary shunting or whether the unexpectedly low prevalence of decompression sickness on the space shuttle/ISS is due to underreporting. Conduct studies to determine operationally acceptable low suit pressure and hypobaric hypoxia limits.
- CC4—Determine optimal dietary strategies for crews and food preservation strategies that will maintain bioavailability for 12 or more months.
- CC5—Initiate a robust food science program focused on preserving nutrient stability for 3 or more years.
- CC6—Include food and energy intake as an outcome variable in dietary intervention trials in humans.
- CC10—Expand understanding of gender differences in adaptation to the spaceflight environment through flight- and ground-based research, particularly potential differences in bone, muscle, and cardiovascular function and long-term radiation risks.
- CC11—Investigate the biophysical principles of thermal balance to determine whether microgravity reduces the threshold for thermal intolerance.

There has been a strong response by NASA to some of these priorities, although it should be noted that essentially there has been no effort to date to pursue the recommendation to assess artificial gravity as a multisystem countermeasure (CC2).

Progress appears to have been made in quantifying orthostatic intolerance, and research continues to examine postlanding vertigo and its effects on performance. According to the *Task Book*, the 10 projects funded in this general area have focused on recovery in a broader sense: injury risk, vestibular function, sensorimotor adaptation and performance, and recovery of performance (CC1). There are no studies listed dealing with artificial gravity (CC2). Two funded projects were in the general area of assessing the effects of gravity on micronucleation and/or intrapulmonary shunting, one of which studied musculoskeletal-induced nucleation during decompression stress (CC3). One dietary strategy and food science study researched the extended preservation of pharmaceutical agents (CC4). Six projects focused on the food science program in which individual projects studied the role of vitamins, predictive models related to vitamin degradation, nutrient content, food acceptability, food storage, food stability, shelf life, and mass reduction (CC5).

3.4.2 Cross-Cutting Issues Related to Radiation Biology

NASA has funded 61 research proposals that are focused on radiation issues listed as being priorities from the 2011 decadal survey. Of these research grants, 7 have been NSCORs (NASA Specialized Center of Research)—large multi-institutional program projects that have permitted a team of investigators with different expertise to focus on questions of radiation significance from several different perspectives. The NSCORs have been extremely important because of the value in a team approach in understanding the complexities of radiation physics, biology, and chemistry in space and also because some new investigators have been brought into the field of space radiation studies through this mechanism. In addition, while these applications have focused almost exclusively on radiation, there are several others that include radiation combined with other mixed stresses such as microgravity, lack of physical activity, and others. The decadal survey had noted the need to study several stressors at one time to achieve a better understanding of the space radiation environment.

The 2011 decadal survey listed three main priorities related to radiation biology. These involved studies at all levels from cellular in vitro work to animal experiments to human studies. The following is a summary of those goals:

- CC7—Conduct longitudinal studies of astronauts for cataract incidence, quality, and pathology related to radiation exposures to understand both cataract risk and radiation-induced late tissue toxicities in humans.
- CC8—Expand the use of animal studies to assess space radiation risks to humans from cancer, cataracts, cardiovascular disease, neurologic dysfunction, degenerative diseases, and acute toxicities such as fever, nausea, bone marrow suppression, and others.

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- CC9—Continue ground-based cellular studies to develop endpoints and markers for acute and late radiation toxicities, using radiation facilities that are able to mimic space radiation exposures.

The decadal survey lists radiation as a key factor potentially limiting space travel. The quality of radiation exposures in space is predominantly galactic cosmic rays (GCR) composed of protons (89 percent), helium (approximately 9 percent), and highly charged ions called HZE (high Z (atomic number) and energy) particles. These HZE particles have an atomic mass greater than helium and a high kinetic energy and, consequently, have great penetration and damaging power. While they are not as abundant as other particles found in space, they are much more damaging and of greatest concern for the long-term safety of the crew.

The decadal survey placed a focus on studies of GCR because although Earth-based X-rays and gamma-rays have been well studied since their discovery in 1895, GCR exposures have only been found important during the space travel era and therefore the consequences of exposure are not as well understood. Nevertheless, the decadal study noted that comparisons of GCR with Earth-type exposures could be valuable in understanding GCR effects. NASA has developed a Space Radiation Laboratory at Brookhaven National Laboratory to mimic the space exposure conditions; this is accomplished primarily because current in-flight exposures on the ISS and other spacecraft are shielded at least in part from GCR exposure by Earth's Van Allen belts and also because doses on the ISS are much lower than those encountered during a Mars-type mission. The facility has the capability of doing rodent and cell exposures. The goals of the NASA program are to understand and where possible mitigate the risk of space radiation exposure.

Most estimates of doses during flight to and from Mars are in the range from 0.6 to 2 Sv. Cancer is considered a stochastic effect of radiation and therefore can occur at any dose with an increase in incidence with increasing dose. Based on this concept, any exposure to radiation can be associated with some risk of developing cancer. Most radiation-induced leukemias occur within 2 to 20 years following exposure, while solid cancers take somewhat longer to develop with a latent period of 15 years or more (except thyroid cancers from radioactive iodine incorporation which may occur as early as 8 years after exposure). This means that the cancer risk to astronauts would begin sometime after their return to Earth and is not likely to be an in-mission risk. NASA has considered this risk to be important, and the decadal survey recommended that risks associated with cancer development continue to be studied. In addition, the risk is significantly different for males and females in space, with females being more susceptible, predominantly because lung cancer risk is higher in females than males for the ages considered to be flight eligible. As a consequence, NASA has invested funding in understanding sex-based differences in cancer induction. Studies on cancer-related exposures have received less funding in the past several years and have been de-emphasized in light of possible degenerative risks that are even more poorly understood; nevertheless, the risk of cancer has not really been decreased by different approaches.

3.4.2.1 Cataracts and Deterministic Studies in Humans (CC7)

Deterministic effects of radiation exposure have a dose threshold and the incidence of effects is usually expressed as a threshold-sigmoid curve. For high-dose effects such as acute radiation syndromes (ARS), the thresholds for induction are usually considered to be higher than those that would be encountered in a Mars mission. While NSBRI conducted some studies on ARS in animal models during the early part of the decadal survey, these were phased out recently because the doses needed to induce ARS are lower than would be encountered in a worst-case scenario in a Mars-type mission.

In addition, human studies relevant to cataracts (CC7) have been given a lower priority in the past 2 to 3 years. While cataract development appears to be a deterministic effect of radiation with a dose threshold, cataracts are easily treated and are not likely to threaten a mission. Long-term follow-up of the astronauts would continue to be important, but the priority placed on it would be reduced.

3.4.2.2 Animal Models of Human Disease (CC8)

In the decadal survey, deterministic effects were focused on identified endpoints that are more poorly understood and possibly mission-threatening, such as disturbances in cognition or cognitive function due to radiation

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effects on the central nervous system (CNS). Studies have been done in NASA-funded programs using normal mice and mice that are models for cognitive disorders (such as mice that are knocked-out for the ApoE gene) and have demonstrated that even low doses of HZE radiation are associated with disruption of neurons, inflammation in the CNS, and stress responses in precursors of neural cells. Functional disorders in mice that are related to neurocognitive endpoints in humans have also been identified by many groups, but their studies with radiation effects are being exploited by NASA-funded investigators.

The relevance to spaceflight is unknown, and it is not clear that endpoints in mice necessarily relate well to functional endpoints in humans; nevertheless, the possible consequences could be mission-threatening and therefore require additional study. Recent NASA work has placed a priority on CNS effects of HZE exposure, examining both single-ion and mixed-field exposures. The radiation research programs of NASA have focused on some degenerative endpoints, but stronger emphasis needs to be placed on these degenerative risks of radiation exposure that could be mission threatening, such as CNS effects. This work cannot be done in humans, and therefore, an emphasis on mechanisms is needed so that extrapolation from animal models to humans is feasible.

3.4.2.3 In Vitro Studies to Mimic Space Radiation Effects (CC9)

Many deterministic effects have not been identified, but a priority was placed on examining endpoints relevant to low-dose HZE exposures, alone or in combination with space radiation conditions (mixed field, with microgravity, etc.), including inflammation, reactive oxygen species generation, and DNA damage, all of which may play a role.

3.4.3 Other Potential Risks

It should be noted that a few studies recommended by the decadal survey are focused on cardiovascular effects. While it is known that high-dose exposures can increase the risk of heart attack and stroke, some evidence from Japanese atomic bomb survivors (which would be predominantly low- linear energy transfer [LET] radiation) has suggested that this may be possible even for low-dose exposures. Nevertheless, this risk would probably be deterministic in mechanism and thus would not be considered a mission-threatening exposure as the CNS effects could possibly be.

Recent work on topics highlighted by the decadal survey has transitioned from mechanisms-oriented work to work that is focused on mitigation. This is in contrast to past approaches for radiation programs that focused on mechanisms first and various countermeasures as a result of mechanisms-based work. This work on countermeasures is somewhat uneven because there are few countermeasures developed to be effective following radiation exposure, and those selected for study appear to be questionable in their relevance to spaceflight. The National Institute of Allergy and Infectious Diseases (NIAID) and the National Cancer Institute (NCI) in NIH have strong radiation mitigator and countermeasures programs oriented toward identifying new agents that can be used both to prevent and to decrease radiation effects for acute toxicities as well as for carcinogenesis (particularly in NCI programs). While the focus is on low-LET Earth-bound types of exposures, NASA should be able to dove-tail with those studies to include some high-LET testing of significant compounds. The countermeasures component of the radiation program currently under way was not defined in the original decadal survey and seems to be disconnected from other mitigation work on-going in the United States.

3.5 FUNDAMENTAL PHYSICAL SCIENCES IN SPACE

The 2011 decadal study provided the following four broad recommendations that should characterize NASA's low-gravity fundamental physics (FP) research activities:

- FP1—Research on complex fluids and soft matter. Microgravity provides a unique opportunity to study structures and forces important to the properties of these materials without the interference caused by Earth-strength gravity.

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- FP2—Understanding of the fundamental forces and symmetries of nature. High-precision measurements in space can test relativistic gravity, fundamental high-energy physics, and related symmetries in ways that are not practical on Earth. Novel effects predicted by new theoretical approaches provide distinct signatures for precision experimental searches that are often best carried out in space.
- FP3—Research related to the physics and applications of quantum gases. The space environment enables many investigations, not feasible on Earth, of the remarkably unusual properties of quantum gases and degenerate Fermi gases.
- FP4—Investigations of matter near a critical phase transition. Experiments that have already been designed and brought to a level of flight readiness can elucidate how materials behave in the vicinity of thermodynamically determined critical points. These experiments, which require a microgravity environment, will provide insights into new effects observable when such systems are driven away from equilibrium conditions.

Specifically, the decadal study states that "a successful exploration program in physical science necessitates first of all a ground-based fundamental sciences program."[16] Other recommendations emphasized the importance of international collaborations and partnering with other agencies. These recommendations were followed by four prioritized programmatic areas for NASA research: Soft Condensed Matter Physics and Complex Fluids, Precision Measurements of Fundamental Forces, and Symmetries, Quantum Gases, and Critical Phenomena. The decadal survey report further concluded that in order to maintain a robust program in the fundamental physical sciences a "critical mass of at least 100-150 funded investigators" is necessary to provide coverage of the areas of physical science of importance to NASA.[17]

Development and progress in the fundamental physical sciences in space since 2011 has been quite substantial. The design, construction, and evaluation phases of the Cold Atom Laboratory (CAL) facility have been completed and CAL is expected to be transported to the ISS in late 2017 where it will become operational immediately. Achieving microgravity Bose-Einstein condensate (BEC) will be a high-priority project while other cold atom studies should soon make contributions to quantum information and computing. The ground-based research supporting ACES (Atomic Clock Ensemble in Space) is proceeding well. Colloids are now offering new metamaterials opening the way of controlling light propagation in completely new ways. The understanding of the glass transition is progressing through the many studies of the jamming transition in granular materials, foams, and colloids.

The following comments concern projects listed in the NASA *Task Book* as current fundamental physics research projects. In the decadal survey report, the projects belonging to FP1, FP2, and FP3 are all rated as being of high relevance in terms of overall importance, but are rated as being of low to moderate importance to enabling exploration. The projects in FP4 concerning critical fluids received a lower overall rating.

3.5.1 Soft Condensed Matter Physics and Complex Fluids (FP1)

The primary focus for this area of microgravity research concerns the behavior of colloids and proteins near phase transitions while in a microgravity environment. An important rationale for these studies arises from observations that in a normal gravitational environment, even very small colloidal particles may aggregate during the crystallization/gelation process. This aggregation of crystals or gels is affected by gravity in subtle ways, producing deleterious inhomogeneities in the process. The suppression of gravitationally induced inhomogeneities was found to facilitate the formation of ordered phases, an important property that is desired for photonic devices used in optical communication systems.

The projects looking at protein crystallization in microgravity are generally in their early stages. Their objective is to investigate the possible role of protein aggregates in the crystallization process. There are also a number of active projects on protein crystallization managed by CASIS that are aiming to improve drug purification, delivery, or storage, or provide a better understanding of protein structures with a role in disease.

[16] NRC, 2011, *Recapturing a Future for Space Exploration,* p. 262.
[17] NRC, 2011, *Recapturing a Future for Space Exploration*. p. 262.

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A new liquid crystal project will address two-dimensional hydrodynamics, including thermocapillarity in the absence of convection. A foam project is proceeding in collaboration with a group of European researchers. This project concerns the hydrodynamics of "wet" foams, which are impossible to stabilize on Earth because of gravity drainage, and the investigation of the jamming transition of bubbles. Viewed collectively, a robust research program has been established—one that will obtain important properties of soft condensed matter in a microgravity environment.

3.5.2 Precision Measurements of Fundamental Forces and Symmetries (FP2)

The choice of a boundary between FP2 and FP3 is ambiguous at best. The vast majority of the ISS-based activities in these two areas will utilize the unique capabilities of CAL, which is expected to be completed, tested, and placed on ISS during the latter part of 2017. This research, especially the pico-Kelvin BEC experiments, will complement and dramatically extend Earth-based BEC experiments—very much in consonance with the decadal survey priority of establishing robust ground- and space-based fundamental physical science programs. The CAL is the very major component of fundamental physics located on the ISS, and its complexity essentially requires either ground-based or autonomous operation.

Overall, the 2016 NASA *Task Book* indicates 15 research projects that fall under one of two program recommendations related to precision measurements of fundamental forces and symmetries. Of these, 8 are fully carried out on the ISS, 3 are fully ground based, and 4 have both flight and ground components.

A number of the research programs involve ISS-based and ground-based clocks and precision measurements and comparisons. This work has the potential to provide improved sensitivity tests of weak equivalence and sensitive tests of symmetry breaking. The dramatic improvement in ground-based clock accuracy that has occurred over the past decade will likely not be implemented on the ISS, but even so, there is good reason to pursue observation of possible symmetry breaking.

The idea of pursuing entangled networks of clocks that include space-based clocks (with European Space Agency participation) has the potential of producing significantly improved tests of the possible time variation of fundamental constants.

3.5.3 Quantum Gases (FP3)

The CAL provides unique opportunities to investigate BEC in microgravity. It is especially responsive to the decadal survey's emphasis on robust ground- and space-based experiments that are mutually supportive and enable unique new science. Specifically, only in a microgravity environment will it be possible to produce condensates at temperatures in the pico-Kelvin range. Achieving such an unprecedented low temperature is a unique opportunity, available only as a consequence of a microgravity environment; the microgravity environment ensures that future revolutionary BEC advances that take place on the ground can be further enhanced when carried out on the ISS. Future experiments involving condensates on the ISS can attain even lower temperatures than the enabling ground-based research. Similar benefits will likely also accrue to planned atom interferometry studies.

3.5.4 Critical Phenomena (FP4)

Critical fluids are interesting fluids with properties that change substantially when close to a critical point. The changes in these fluid properties can be observed most effectively by monitoring them as a function of temperature in the vicinity of a critical point. For example, microgravity experiments using such fluids have allowed for a better understanding of the wetting of the walls of containers, an important issue for fluids in space. At present, NASA has one active project in this area, carried out in collaboration with CNES (Centre National d'Études Spatiales). However, as noted in the introduction, the decadal survey evaluated this area as one with a lower overall relevance compared to FP1 to FP3.

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3.6 APPLIED PHYSICS

The decadal survey called out three high-priority areas in Applied Physical Sciences (AP): fluid physics, combustion, and materials research. Research in these areas is deemed central to many new exploration technologies, enabling new exploration capabilities and yielding new insights into a broad range of physical phenomena in space and on Earth. It is envisioned that research focused on applied physical sciences will result in fundamental and practical improvements for propulsion, power generation, life support, fire safety, and advanced materials extraction, synthesis, and processing. The following key areas were identified in the decadal survey as having the highest priority in the applied physical sciences:

- AP1—Reduced-gravity multiphase flows, cryogenics and heat transfer database and modeling, including phase separation and distribution (i.e., flow regimes), phase-change heat transfer, pressure drop, and multiphase system stability.
- AP2—Interfacial flows and phenomena (including induced and spontaneous multiphase flows with or without phase change) relevant to storage and handling systems for cryogens and other liquids, life support systems, power generation, thermal control systems, and other important multiphase systems.
- AP3—Dynamic granular material behavior and subsurface geotechnics to improve predictions and site-specific models of lunar and Martian soil behavior.
- AP4—Development of fundamentals-based strategies and methods for dust mitigation during advanced human and robotic exploration of planetary bodies.
- AP5—Experiments on the ISS to understand complex fluid physics in microgravity, including fluid behavior of granular materials, colloids and foams, biofluids, non-Newtonian and critical point fluids, etc.
- AP6—Fire safety research to improve methods for screening materials for flammability and fire suppression in space environments.
- AP7—Combustion processes research, including reduced-gravity experiments with longer durations, larger scales, new fuels, and practical aerospace materials relevant to future missions.
- AP8—Research on numerical simulation of combustion to develop and validate detailed single phase and multiphase combustion models for interpreting and facilitating combustion experiments and tests.
- AP9—Reduced-gravity research on materials synthesis and processing and control of microstructure and properties, to improve the properties of existing and new materials on the ground.
- AP10—Development of new and advanced materials that enable operations in harsh space environments and reduce the cost of human space exploration.
- AP11—Fundamental and applied research to develop technologies that facilitate extraction, synthesis, and processing of minerals, metals, and other materials available on extraterrestrial surfaces.

3.6.1 Physical Science Informatics (Open-Access Database)

Subsequent to the publication of the decadal survey, NASA initiated a new mandatory public archive requirement for physical science investigator teams who must now upload relevant research products and data to NASA's Physical Science Informatics (PSI) public data archive (http://psi.nasa.gov). This mechanism offers significant opportunities to expand impacts of the microgravity research on science and technology development on Earth. Moreover, the addition of ground-based theoretical and experimental studies funded through the PSI program provided extended benchmark data for model development. NRAs for fluids, complex fluids, combustion, and material science have been released with 16 investigations funded to mine the PSI database expanding and extending impacts of the space experiments.

3.6.2 Fluid Physics

The research aims, themes, and priorities within the Fluid Physics discipline of NASA's Physical Sciences Research Program have remained largely unchanged from the time of the 2011 decadal survey, which draws from

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earlier, nearly identical, reviews done by the National Research Council (now the National Academies of Science, Engineering, and Medicine) dating back to the early 2000s.[18,19] In fact, most fluids research themes (>90 percent) presently under pursuit by NASA targeting research needs for exploration have been clearly identified for many decades. Current NASA leadership has internalized the fluids research needs and priorities and has directed the currently limited resources of fluid physics funding, launch opportunities, flight platforms, and ground-based facilities to address the general exploration-critical topics of multiphase transport phenomena in variable acceleration fields—especially the nearly weightless environment common aboard spacecraft. As examples, the design challenges are manifold for managing the water cycle for life support, plant and animal habitats, and systems involving liquids fuels, propellants, and coolants, requiring a concerted effort to establish a practical theoretical foundation with which to design the next generation of highly reliable fluids systems for bolder yet safer and more affordable space exploration.

Making good use of limited resources, the current fluids program is on track for completing, preparing, and initiating new low-gravity (low-*g*) multiphase fluids research investigations targeting mission-enabling forced and unforced capillary flows (i.e., CCF,[20] CFE[21,22]), flows in packed beds (i.e., BPRE[23]), cryogenic fluids management (i.e., ZBOT 1, 2, and 3[24]), forced flow with boiling and condensation (i.e., FBCE[25]), forced flow phase separations (i.e., TPSE[26]), enhancements (STP-H5 EHD[27]), pool boiling heat transfer (MABE[28] and BXF[29]), and others. At least 6 flight investigations are currently funded contributing to about 25 fluid physics journal publications per year. In addition to adding to the fundamental understanding of low-*g* fluid phenomena (i.e., CVB[30]), such investigations have the potential to inspire practical advanced systems design. For example, NASA-supported capillary flow research performed on the ISS (i.e., CCF and CFE) and in terrestrial facilities has matured to the point of direct application to next generation spacecraft fluid systems design. This work is now being applied to omics in space using the three-dimensional (3D) printer on the ISS,[31] semi-passive CO_2 removal,[32] condensing

[18] NRC, 2000, *Microgravity Research in Support of Technologies for the Human Exploration and Development of Space and Planetary Bodies*, The National Academies Press, Washington, D.C.

[19] NRC, 2003, *Assessment and Directions in Microgravity and Physical Sciences Research at NASA*, The National Academies Press, Washington D.C.

[20] M. Dreyer, Principal Investigator, 2017, "Capillary Channel Flow (CCF)," December 6, https://www.nasa.gov/mission_pages/station/research/experiments/303.html.

[21] M. Weislogel, Principal Investigator, 2017, "Capillary Flow Experiment (CFE)," December 6, https://www.nasa.gov/mission_pages/station/research/experiments/978.html.

[22] M. Weislogel, Principal Investigator, 2017, "Capillary Flow Experiment (CFE-2)," December 6, https://www.nasa.gov/mission_pages/station/research/experiments/459.html.

[23] B. Motil, Principal Investigator, 2017, "Packed Bed Reactor Experiment (PBRE)," July 19, https://www.nasa.gov/mission_pages/station/research/experiments/1111.html.

[24] M. Kessemi, Principal Investigator, 2014, "Zero Boil-Off Tank experiment (ZBOT)," last updated September 18, https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/msg/zbot.

[25] I. Mudawar, Principal Investigator, 2017, "Forced Flow Boiling Experiment (FBCE)," last updated May 12, 2017, https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/fbce.

[26] N. Hoyt, M.F. Kang, A. Kharraz, J. Kadambi, and Y. Kamotani, Cyclonic two-phase flow separator experimentation and simulation for use in a microgravity environment, *Journal of Physics: Conference Series* 327(1), http://iopscience.iop.org/article/10.1088/1742-6596/327/1/012056/meta.

[27] J. Didion, Principal Investigator, 2017, "Electro-Hydro Dynamics," December 13, https://www.nasa.gov/mission_pages/station/research/experiments/1993.html.

[28] J. Kim, Principal Investigator, 2017, "Microheater Array Boiling Experiment (MABE)," December 20, https://www.nasa.gov/mission_pages/station/research/experiments/16.html.

[29] V. Dhir, Principal Investigator, 2012, "Boiling eXperiment Facility (BXF)," December 5, https://www.nasa.gov/mission_pages/station/research/experiments/BXF.html.

[30] P. Wayner, Principal Investigator, 2017, "Confined Vapor Bubble Experiment," December 20, https://www.nasa.gov/mission_pages/station/research/experiments/465.html.

[31] S. Wong, AI Biosciences, Principal Investigator, 2017, "A 3D Printer Enabled, High Performing, Microgravity Compatible, and Versatile Sample Preparation Platform," NASA PH I SBIR, https://sbir.nasa.gov/firm/node/59039.

[32] J. Graf, CapiSorb, Principal Investigator, 2018, "Capillary Structures for Exploration Life Support," March 14, https://www.nasa.gov/mission_pages/station/research/experiments/2364.html.

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heat exchangers, contingency urine collection,[33] brine drying,[34] plant/water delivery systems,[35] and many others. Similar application impacts are anticipated for planned research relating to packed bed reactors, phase change heat transfer systems, propellant thermostatics and dynamics, phase separations, and new exploration-enabling space fluid physics investigations. More investigations along these lines are anticipated, and improved analytical and numerical tools for advanced system design are becoming available for broader development and application.

To maximize research output, NASA is increasingly evaluating and prioritizing its fluid physics research projects, based on multiuser potential, the potential for subsequent follow-on research of greater breadth and/or depth, the value and applicability of the open-archive data to the community, and the value of further experiments with modified hardware to other communities (i.e., fluid physics experiments re-tooled and re-purposed to address further applications in life support, such as NASA Glenn Research Center's Packed-Bed Reactor Experiment, which is under consideration for new investigations along different lines by NASA Johnson Space Center). These efforts are intended to achieve the greatest science return and engineering impact for each dollar spent. Such efforts are in motion for fluid physics, with specific demonstrations of success within the decadal survey review timeframe, with hopes of increasing technology readiness level for low-*g* system design.

3.6.3 Combustion Research

There are two thrusts in combustion research in the decadal survey: research that enables human space exploration and research that is enabled by the unique microgravity environment (decadal survey priorities AP6 to AP8).

Spacecraft safety measures include fire prevention, detection, and suppression. The emphasis of fire prevention is on material (mostly solids) flammability and selection. The Burning and Suppression of Solids (BASS) experiment used an existing flow tunnel in the microgravity glovebox aboard the ISS. The tests were carried out from 2010 to 2014. Although the testing conditions were limited by an oxygen range equal or less than 21 percent at 1 atmospheric pressure, and by small samples, this series was successful in investigating the effect of low-speed forced flow on flame spread, burning, and extinction for burning solids. The test database is already available in the NASA PSI archive.

The Spacecraft Fire Safety Demonstration (Saffire) is a large-scale microgravity fire experimental series. The first series of tests was conducted in the unmanned Cygnus supply vehicle before it entered the atmosphere and was destroyed. Large-scale fire tests are a high-priority objective in the decadal survey. NASA accelerated the schedule anticipated by the decadal survey for such a large-scale test and also formed a team of U.S. and international investigators to plan and conduct the work.[36] The funding for Saffire is provided by NASA's Advanced Exploration Systems (AES) Division rather than the Division of Space Life and Physical Sciences Research and Applications (SLPSRA). Both BASS and Saffire projects provide valuable data for model calibration and validation.

The Solid Fuel Ignition and Extinction Experiment (Sofie) consists of five individual projects to study solid flammability under a variety of pressure and oxygen atmospheres to be carried out inside the ISS Combustion Integrated Rig (CIR). Sofie will extend the test condition to an enriched oxygen and different pressure environment, with improved velocity control and better diagnostics. Sofie is currently in a queue waiting for CIR to become available. Due to crew availability and other factors, the flight experiment may not be started until 2020.

The Flame Extinguishment Experiments (FLEX) were conducted to assess the effects of gravity on fuel burning rates, soot formation, and flame extinction using liquid droplets and to gain basic understanding of combustion science to enable energy efficient space exploration. The experiments can provide insights on fire suppression for next-generation crew exploration vehicles. The FLEX-2 experiments conducted droplet burning of pure fuels, fuel mixtures, droplet arrays, and convective droplet burning. The FLEX-2 investigation unexpectedly discovered

[33] R. Jenson, IRPI, LLC, Principal Investigator, 2015, "Passive, Collapsible Contingency Urinal for Human Space Flight," NASA Ph I SBIR, August 1, https://ntrs.nasa.gov/search.jsp?R=20160005400.

[34] M. Sargusingh, CapiBRIC, Prinicipal Investigator, 2018, "Capillary Structures for Exploration Life Support," March 14, https://www.nasa.gov/mission_pages/station/research/experiments/2364.html.

[35] R. Adamson, 2017, "NASA Partners With Techshot, Tupperware to Update ISS Plant Production System," *Industry News*, July 26, http://blog.executivebiz.com/2017/07/nasa-partners-with-techshot-tupperware-to-update-iss-plant-production-system.

[36] A European cargo ship was initially planned as the experiment platform.

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"cool flames" of droplet burning after radiation extinction of the hot flame. These cool flame experiments provided excellent microgravity data for the study of low-temperature chemistry of liquid fuels, which plays a critical role in advanced internal combustion engines. These results drew significant attention in the combustion research community and have sparked renewed research interest in cool flames. FLEX-ICE-GA (FLEX-Italian Combustion Experiment for Green Air) also tested surrogate fuel mixtures as defined by the Italian Space Agency (ASI) by using the FLEX-2 experimental configuration. FLEX-2J experiment is a joint effort between NASA, the Japanese Aerospace Exploration Agency, and universities to study fuel droplet motions during flame spreading along a one-dimensional droplet array. The international collaboration provides a greater platform and extends the science impact on a broader community for microgravity combustion research.

ACME (Advanced Combustion via Microgravity Experiments) focuses on combustion science and technology using the CIR hardware in the ISS. The primary goal is to improve efficiency and reduce pollutant emission in terrestrial combustion. The secondary goal is fire prevention. ACME has five independent experiments to investigate the burning rates and soot formation in laminar, gaseous, and non-premixed flames, with and without an electric field. The ground-based research has led to a large number of publications. On-orbit testing is planned to begin in 2017. PSI-funded ground experiments and modeling have provided a generic flame regime diagram for both cool flames and hot flames in premixed and diffusive combustion systems.

3.6.4 Materials Research

The NASA *Task Book* indicates that a limited number of PIs (around 27 PIs for the year 2017) are currently supported for ground-based, flight experiment, and/or simulation investigations in materials sciences. These investigations build upon and complement the wealth of data and observations of more than 70 materials sciences researchers who were supported by NASA in 2004. Indeed, as stated in the 2011 decadal study, "Collectively, through the shuttle era, research in the physical sciences generated an impressive number of peer-reviewed publications, landmark measurements, and discoveries, all of which could only be achieved through access to space." [37]

The decadal survey has further emphasized that materials science and engineering are central to NASA's exploration mission, both crewed and uncrewed. The decadal survey further recommends emphasis on materials synthesis and processing and the control of microstructure and properties, as well as emphasis on advanced materials to enable the NASA mission.[38]

The research aims of the current investigations primarily focus on the crucial understanding of the influence of processing in reduced gravity on the structure and properties of materials. However, because of the sparsity of research being funded in the materials science area, it is difficult to identify or describe specific research themes or directions for the research activities being funded. Notwithstanding, the research themes of the current investigations can be roughly categorized as follows:

- Modeling and simulation of crystal growth, dendritic microstructures, and liquid phase sintering;
- Effects of convection on microstructural transition, mushy zones and macrosegregation,
- Coarsening of solid-liquid mixtures and dendrites;
- Electromagnetic or electrostatic levitation of liquids;
- Directional solidification;
- Crystal growth of II-VI compound semiconductors; and
- Eutectics.

Continuation of the above-mentioned research activities is crucial for future space exploration and is highly recommended. At the same time, additional investments need to be made in mission-critical materials science and engineering arena. As clearly recognized in the decadal study, studies in "lightweight materials, self-healing materials, and other new materials tailored to NASA's demanding missions, are central to the success of future

[37] NRC, 2011, *Recapturing a Future for Space Exploration,* p. 20.
[38] Ibid., Table 9.1.

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human and robotic exploration."[39] As predicted in that report, research in reduced gravity has certainly led to new insights about how various processing methods are affected, as is the internal structure and, ultimately, the properties of such materials.

In summary, given the limited opportunities for conducting significant materials research aboard the ISS over the past 8 years, progress and technological competence in this critical area will remain sluggish and uncertain.

3.7 TRANSLATION TO SPACE EXPLORATION SYSTEMS

The 2011 decadal survey called out 16 high-priority areas in Translation to Space Exploration Systems (TSES). As identified in the decadal survey, the specific high-priority items are the following:

- TSES1—Conduct research to address issues for active two-phase flow relevant to thermal management.
- TSES2—To support zero-boiloff propellant storage and cryogenic fluid management technologies, conduct research on advanced insulation materials, active cooling, multiphase flows, and capillary effectiveness, as well as active and passive storage, fluid transfer, gauging, pressurization, pressure control, leak detection, and mixing destratification.
- TSES3—NASA should enhance surface mobility; relevant research includes suited astronaut computational modeling, biomechanics analysis for partial gravity, robot-human testing of advanced spacesuit joints and full body suits, and musculoskeletal modeling and suited range-of-motion studies, plus studies of human-robot interaction (including teleoperations) for the construction and operation of planetary surface habitats.
- TSES4—NASA should develop and demonstrate technologies to mitigate the effects of dust on extra-vehicular activity (EVA) systems and suits, life support systems, and surface construction systems. Supporting research includes impact mechanics of particulates, design of outer-layer dust garments, advanced material and design concepts for micrometeoroid mitigation, magnetic repulsive technologies, and the quantification of plasma electrodynamic interactions with EVA systems; dynamics of electrostatic field coupling with dust; and regolith mechanics and gravity-dependent soil models.
- TSES5—NASA should define requirements for thermal control, micrometeoroid and orbital debris impact and protection, and radiation protection for EVA systems, rovers, and habitats and develop a plan for radiation shelters.
- TSES6—NASA should conduct research for the development and demonstration of closed-loop life support systems and supporting technologies. Fundamental research includes heat and mass transfer in porous media under partial gravity and microgravity conditions and understanding the effect of variable gravity on multiphase flow systems.
- TSES7—NASA should develop and demonstrate technologies to support thermoregulation of habitats, rovers, and spacesuits on the lunar surface.
- TSES8—NASA should perform critical fire safety research to develop new standards to qualify materials for flight and to improve fire and particle detectors. Supporting research is necessary in materials qualification for ignition, flame spread, and generation of toxic and/or corrosive gases and in characterizing particle sizes from smoldering and flaming fires under reduced gravity.
- TSES9—NASA should develop a standard methodology for qualifying fire suppression systems in relevant atmospheres and gravity levels and would benefit from strategies for safe post-fire recovery. Specific research is needed to characterize the effectiveness of fire suppression agents and systems under reduced gravity and to assess the toxicity of various fire products.
- TSES10—Research should be conducted to allow regenerative fuel cell technologies to be demonstrated in reduced-gravity environments.
- TSES11—To support the development of new energy conversion technologies, research should be done on high-temperature energy conversion cycles, device coupling to essential working fluids, heat rejection systems, materials, etc. Research is also required on more efficient surface-base primary power and on

[39] Ibid., p. 265.

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the technologies to enable solar electric propulsion as an option to transfer large masses of propellant and cargo to distant locations.

- TSES12—To make fission surface power systems a viable option, research is needed on high-temperature, low-weight materials for power conversion and radiators and on other supporting technologies.
- TSES13—Development and demonstration of ascent and descent system technologies are needed, including ascent/descent propulsion technologies, inflatable aerodynamic decelerators, and supersonic retro propulsion systems. The required research includes propellant ignition, flame stability, and active thermal control; lightweight flexible materials; and rocket plume aerothermodynamics and vehicle dynamics and control.
- TSES14—Research is required to support the development and demonstration of space nuclear propulsion systems, including liquid-metal cooling under reduced gravity, thawing under reduced gravity, and system dynamics.
- TSES15—Research is needed to identify and adapt excavation, extraction, preparation, handling, and processing techniques for a lunar water/oxygen extraction system.
- TSES16—NASA should establish plans for surface operations, particularly In Situ Resource Utilization (ISRU) capability development and surface habitats. Research is needed to characterize resources available at lunar and martian surface destinations and to define surface habitability systems design requirements.

As identified in the original decadal survey, these recommendations were meant to "identify technologies needed to enable exploration missions to the Moon, Mars, and elsewhere as well as the foundational research in life and physical sciences."[40] In the decadal survey, science and technology development areas were recommended to support near-term objectives and operational systems (i.e., prior to 2020) and objectives and operational systems for the decade beyond 2020. While technologies and operational systems may be near term (prior to 2020) or longer term (2020 and beyond), supporting research is necessary for both near term and longer term during the timeframe of this decadal survey. The decadal survey not only defined research priorities in science, but also included discussions about establishing the technological knowledge required to ensure the orderly transition of new technology into operational space exploration systems.

Starting in parallel and since the publication of the decadal survey, NASA and the National Academies have expended significant effort on the NASA technology portfolio. The NASA Authorization Act of 2010 directed NASA to create a program to maintain its research and development base in space technology. In response, NASA created a set of 14 draft space technology roadmaps to guide the development of space technologies. These roadmaps were the subject of a comprehensive external review by the National Academies, which in 2012 issued the National Research Council report *NASA Space Technology Roadmaps and Priorities: Restoring NASA's Technological Edge and Paving the Way for a New Era in Space*.[41]

NASA then began a reexamination and updating of its 2010 draft technology roadmaps,[42] resulting in a new set of roadmaps in 2015.[43] A significant aspect of the updating was the effort to assess the relevance of the technologies by showing their linkage to a set of mission classes and design reference missions (DRMs) from the Human Exploration and Operations Mission Directorate and the Science Mission Directorate. The new set of roadmaps also included a roadmap that addresses aeronautical technologies.

In the spring of 2015, the updated roadmaps were released to the public for review and comment. Also in 2015, the National Academies were asked to assemble a committee to evaluate the new technologies in the updated set of 14 space technology roadmaps; this resulted in the 2016 report *NASA Space Technology Roadmaps and Priorities Revisited*.[44] Per that study's statement of task, the aeronautics roadmap was not included in the most recent

[40] Ibid., p. 299.

[41] NRC, 2012, *NASA Space Technology Roadmaps and Priorities: Restoring NASA's Technological Edge and Paving the Way for a New Era in Space,* The National Academies Press, Washington, D.C.

[42] NASA, "NASA 2015 Technology Roadmaps," https://www.nasa.gov/offices/oct/home/roadmaps/index.html.

[43] Ibid.

[44] National Academies of Sciences, Engineering, and Medicine, 2016, *NASA Space Technology Roadmaps and Priorities Revisited*, The National Academies Press, Washington, D.C., doi: 10.17226/23582.

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National Academies study because the 2010 space technologies roadmaps had no such aeronautics roadmap. Other key tasks of this most recent National Academies review were:

- Identifying technologies in NASA's 2015 roadmaps that were not evaluated by the 2012 NRC report,
- Prioritizing those technologies using the same process documented in the 2012 NRC report, and
- Recommending a methodology for future independent reviews of NASA's technology roadmaps.

That technology committee did not asses the progress made by NASA since the originally published 2012 report. Thus, in reviewing the progress made on the 16 TSES priority items identified in the decadal survey, this committee did not have a broad and prior National Academies review to depend on, but only the knowledge contained within the committee.

The decadal survey focused its science and technology needs into seven topic categories: space power and thermal management; space propulsion; extra-vehicular activity (EVA); life support; fire safety; space resource extraction, processing, and utilization; and planetary surface construction. These areas represent a subset of the 15 technology roadmaps (see Figure 3.1). To understand the intersection of the decadal survey and these technology roadmaps, the 16 resulting high-priority TSES items from the decadal survey are best mapped into the most

FIGURE 3.1 NASA Technology Roadmap Technical Areas. SOURCE: NASA, https://www.nasa.gov/sites/default/files/thumbnails/image/oct_roadmap_tabs_0.jpg, accessed October 5, 2017.

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TABLE 3.1 Mapping of Decadal Survey Translational Priorities (TSES) against Technology Priorities Identified in 2012 and 2016 Reviews of NASA's Technology Roadmaps.

	TSES	Main TA #	Other TA #	17 Highest	88 High
1	Two-phase flow for thermal management	14	2	14.1.2	14.1.2
2	Cryogenic fluid management	14	2	14.1.2	2.4.2, 14.1.2
3	Mobility rovers and robotic systems switch to surface mobility	4, 7	6	none	4.4.2, 4.4.8, 6.2.1, 6.2.2, 7.3.2
4	Dust mitigation systems	7	6	none	6.2.1, 7.1.2, 7.6.3
5	Thermal, micro meteoroid, radiation and orbital debris protection for EVA , rovers, and habitats	6	7, 12	X.1, X.2	6.1.4, 6.2.1, 6.5.3, 12.1.1
6	Closed-loop life support systems	6		X.3	6.1.1, 6.1.2, 6.1.3, 6.2.2
7	Thermoregulation technologies of habitats, rovers, and spacesuits	14		none	none
8	Fire safety: materials standards and particle detectors	6		none	6.4.2
9	Fire suppression and post-fire strategies	6		none	6.4.2, 6.4.4
10	Regenerative fuel cells	3	10	none	none
11	Energy conversion technologies	3	10, 12	3.1.3, 3.1.5	3.1.3, 3.1.5, 3.3.3, 3.3.5
12	Fission surface power	3		X.2, 3.1.5	3.1.5, 10.1.1
13	Ascent and descent systems	9	12	X.2, X.4	9.1.1, 9.1.2, 9.1.4, 9.2.7, 9.4.5, 12.1.1, 12.2.1
14	Space nuclear propulsion	2		2.2.3	2.2.3
15	Lunar water and oxygen extraction systems	7		none	7.1.2, 7.1.3, 7.1.4
16	Plans for surface operations including in situ resource utilization and surface habitats	7	12	X.2	4.3.6, 7.1.2, 7.1.3, 7.1.4, 7.4.2, 7.4.3, 7.6.2, 12.2.1

NOTES: Main TA # denotes which technology roadmap maps closest to the TSES; Other TA # denotes other technology roadmap(s) that are relevant to the TSES; 17 Highest denotes which highest-priority Level 3 technology maps to the TSES; 88 High denotes which high-priority level 2 or 3 technology maps to the TSES; None indicates that there were no highest- or high-priority technologies associated with the TSES. Several of the 17 highest technologies (denoted by X.1, X.2, etc.) are combinations of related Level 3 high-priority technologies (see Table 3.2 and the report for detailed descriptions of the X.x "technologies").

relevant technology roadmaps by denoting what high-priority item maps onto what roadmap using the TSES (decadal survey) and TA (technology roadmap) designations. The relationship of the decadal survey priorities to the 17 highest-priority technologies and 88 high-priority technologies published in the 2016 National Academies technology roadmaps review[45] is shown in Table 3.1 (the technologies are listed in numerical order, not priority order). Table 3.2 provides the names of the Level 3 technologies listed in Table 3.1.

A majority (9 of 16) decadal survey TSES high-priority recommendations do align with the 2015 review's 17 highest priorities. Several (7 of the 16) do not align with the 17 highest priorities because the decadal survey considered only those technologies that required supporting research in physical and biological sciences. Thus the National Academies 2012 and 2015 roadmap reviews provide a more complete picture of overall NASA technology development priorities.

As a part of supporting this committee's assessment, NASA was asked to provide data on all of the projects in its biological and physical sciences portfolio and indicate how they mapped to the decadal survey. Only three

[45] See Tables S.1 and 2.1 in National Academies of Sciences, Engineering, and Medicine, 2016, *NASA Space Technology Roadmaps and Priorities Revisited*.

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TABLE 3.2 Key for Level 3 Technologies Listed in Table 3.1

TA Level 3 Number	TA Level 3 Name
X.1	Radiation Mitigation for Human Spaceflight - includes:
	6.5.1 Radiation Risk Assessment Modeling
	6.5.2 Radiation Mitigation
	6.5.3 Radiation Protection Systems
	6.5.4 Radiation Prediction
	6.5.5 Radiation Monitoring Technology
X.2	Lightweight and Multifunctional Materials and Structures - includes:
	10.1.1 (Nano) Lightweight Materials and Structures
	2.1.1 Materials: Lightweight Structures
	12.2.1 Structures: Lightweight Concepts
	12.2.2 Structures: Design and Certification Methods
	12.2.5 Structures: Innovative, Multifunctional Concepts
X.3	Environmental Control and Life Support System (ECLSS) - includes:
	6.1.1 ECLSS: Air Revitalization
	6.1.2 ECLSS: Water Recovery and Management
	6.1.3 ECLSS: Waste Management
	6.1.4 ECLSS: Habitation
X.4	X.4 Guidance, Navigation, and Control (GN&C) - includes:
	4.6.2 Relative Guidance Algorithms (for Automation Rendezvous and Docking)
	5.4.3 Onboard Autonomous Navigation and Maneuvering (for Position, Navigation, and Timing)
	9.2.7 Terrain-Relative Sensing and Characterization (for Descent and Targeting)
	9.2.8 Autonomous Targeting (for Descent and Targeting)
2.2.3	(Nuclear) Thermal Propulsion
2.4.2	Propellant Storage and Transfer
3.1.3	Solar Power Generation (Photovoltaic and Thermal)
3.1.5	Fission Power Generation
3.3.3	Power Distribution and Transmission
3.3.5	Power Conversion and Regulation
4.3.6	Robotic Drilling and Sample Processing
4.4.2	Supervisory Control
4.4.8	Remote Interaction
6.1.1	Air Revitalization
6.1.2	Environmental Control and Life Support System Water Recovery and Management
6.1.3	Environmental Control and Life Support System Waste Management
6.1.4	Habitation
6.2.1	Extravehicular Activity Pressure Garment
6.2.2	Extravehicular Activity Portable Life Support System
6.4.2	Fire Detection and Suppression
6.4.4	Fire Remediation
6.5.3	Radiation Protection Systems
7.1.2	In Situ Resource Utilization Resource Acquisition
7.1.3	In Situ Resource Utilization Products/Production
7.1.4	In Situ Resource Utilization Manufacturing/Infrastructure Emplacement
7.3.2	Surface Mobility
7.4.2	Habitation Evolution
7.4.3	Smart Habitats

continued

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TABLE 3.2 Continued

TA Level 3 Number	TA Level 3 Name
7.6.2	Construction and Assembly
7.6.3	Dust Prevention and Mitigation
9.1.1	Rigid Thermal Protection Systems
9.1.2	Flexible Thermal Protection Systems
9.1.4	Deployment Hypersonic Decelerators
9.2.7	Terrain-Relative Sensing and Characterization
9.4.5	EDL Modeling and Simulation
10.1.1	(Nano) Lightweight Materials and Structures
12.1.1	Materials: Lightweight Structure
12.2.1	Structures: Lightweight Concepts
14.1.2	Active Thermal Control of Cryogenic Systems

projects, the Smolder Transition and Flaming (STaF) and Smolder Evaluation Test (SET), the Zero Boil-Off Tank (ZBOT) Experiment, and Direct Computational Simulations and Experiments for Internal Condensing Flows System-Instabilities/Dynamics in Microgravity and Terrestrial Environments, identified with TSES8, TSES2, and TSES11. This small number was somewhat surprising in that it might be expected that many areas that identify as enabling exploration would have linkages to these 16 TSES high-priority recommendations. However, the committee could anecdotally identify other projects that could be tied to these 16 priorities, so there is likely an issue of not formally identifying research project connections to the translational research priorities. This lack of formal traceability may hamper NASA in its effort to ensure that new knowledge of biological and physical behaviors is translated into the development of technology systems, including systems prioritized in the technology roadmaps, affected by those behaviors.

One of the challenges that the original decadal survey identified that appears to continue today is adequate two-way communication between the scientists doing the basic research and the technologists using those research results. Two-way communication between the two groups is critical in order to integrate their efforts. This communication process requires that engineers understand and apply the research results of scientists and that the scientists work within the context of specific missions. Successful transition from research results to implemented technology requires that program managers, engineering leaders, and research leaders create an environment where scientists interact with engineers on the specifics of system requirements. The 2011 decadal survey, in Chapter 12, "Linking Science to Mission Capabilities Through Multidisciplinary Translational Programs,"[46] contains an in-depth discussion of the issues associated with creating organizations that can successfully translate research into technologies and technologies into exploration systems. That chapter discusses ideas such as multidisciplinary teams of scientists and engineers working these challenges together, improved databases and data sharing, improved commercial sector interaction, as well as collaboration with other agencies and international partners across both science and engineering together to summarize a few key suggestions. This advice remains as important today as when the decadal survey was released.

3.8 SCIENCE STATUS IN LIGHT OF EXPLORATION MISSION DEVELOPMENT

As promised in the introduction to this chapter, the committee has assessed in discussions above the progress made in the respective science disciplines against the priorities identified in the 2011 decadal survey. This assessment occurred within the context of the overall advancement of space life and physical sciences and against the backdrop of advancement in related fields outside of the space sciences. The committee's assessment of science

[46] NRC, 2011, *Recapturing a Future for Space Exploration*, p. 369.

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in the research community resulted in a picture of strong science advancement and achievement since the 2011 decadal survey. This assessment strongly suggests that the programmatic and budgetary commitment NASA and SLPSRA have made to address the decadal survey recommendations has produced results that translate into discoveries in support of exploration.

When considering these achievements in the context of planned deep space exploration missions, as requested in the charge to the committee and discussed in Chapter 2, the committee was struck by the fact that all concepts for deep space transit missions include prolonged periods of microgravity in a high radiation environment. No concepts for large-scale shielding or induced gravity exist in the defined mission plans. Rather than avoiding microgravity and radiation, NASA seeks to understand and mitigate the effects of those two spaceflight conditions in both the biological and physical systems involved in the extended missions in deep space.

Finding 3-1: Deep space exploration will take place for long periods of time in a microgravity and high-radiation environment; no large-scale induced gravity on the vehicle or station is foreseen with the current design reference missions, and the only shielding expected for radiation damage is that which is available on the space vehicle. Therefore, a fundamental understanding of both microgravity and radiation effects on biological and physical systems is essential for operational success and appropriate protection of astronauts.

Recommendation 3-1: As NASA continues to develop deep space mission scenarios involving long durations in microgravity, understanding the direct and interactive effects of radiation, microgravity and small habitats on human biology, and on the performance of biological and physical systems in space over long durations will need to have high priority in NASA science plans. NASA should also improve the coordination among the science research and engineering teams to better address the integrated effects in the design of the exploration elements and systems.

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4

Prioritizations and Rankings to Optimize and Enable the Expansion of Deep Space Human Exploration

NASA's goal of moving beyond low Earth orbit (LEO), when considered against the current timeline and plans for the International Space Station (ISS), creates a sense of urgency for science accomplishment in support of deep space exploration. As generally acknowledged by NASA and within the broader microgravity community, and as described in examples given in Chapter 3, there is not sufficient research time on the ISS to fully explore all aspects of the 2011 space life and physical sciences decadal survey[1] priorities within the station's planned lifespan. To meet the task of identifying the highest-priority recommendations as targeted research priorities, the committee considered the decadal survey recommendations within the context of ISS implementation plans, while utilizing the ranking criteria from the decadal survey. In addition to those criteria, the committee also considered the progress in microgravity research and programmatic considerations, together with available microgravity platforms, to produce a specific set of targeted research priorities to meet the committee's statement of task. Once ranked, those targeted priorities were then considered within the context of the broader and general advances in relevant scientific disciplines. The committee recognized that budget limitations play a key role in the need for identifying top priorities for exploration.

4.1 TARGETING DECADAL SURVEY EXPLORATION RESEARCH PRIORITIES

4.1.1 The Decadal Survey Approach to Organizing Priorities

The 2011 decadal survey was carried out by the Committee for the Decadal Survey on Biological and Physical Sciences in Space (survey committee), with prioritized disciplinary recommendations determined by seven supporting disciplinary panels, with one member of the survey committee also serving as a member of each panel.

Each panel identified and prioritized program elements that were especially important to their area of coverage and reported this prioritized list to the survey committee. The survey committee took the highest priorities of all seven panels and characterized each individual panel recommendation according to whether it enabled space exploration and/or was enabled by space exploration. Given that there were several NASA exploration policy options that were undecided, the survey committee further ranked each of these high-priority program elements

[1] National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C.

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as "high," "medium," or "low," according to its degree of relevance to each of eight possible prioritization criteria. Those results were tabulated in tables within Chapter 13 of the 2011 decadal survey report. Among the eight criteria, several were particularly relevant to exploration needs.

The statement of task for this midterm assessment study specifically requested further prioritization *only* for those decadal survey recommendations that were determined to be the highest priority for exploration, and therefore, it was important that the current committee consider the current landscape of exploration options being discussed within NASA.

4.1.2 Overarching Space Exploration Strategy

The overall space exploration strategy continues to evolve with the recently announced Deep Space Gateway. Figures 4.1 and 4.2 lay out the current top-level NASA Exploration Strategy and Reference Phase 1 plan. NASA's space exploration strategy is intended to be flexible and account for technical lessons learned from expanding into deep space and findings from microgravity research investigations. As NASA progresses along its planned strategy, it might find it helpful to implement a periodic review process to address known and potential exploration strategy changes and to develop options to react to such changes. Results of the ISS post-2024 strategy and the exploration progress over time could be incorporated into these periodic reviews.

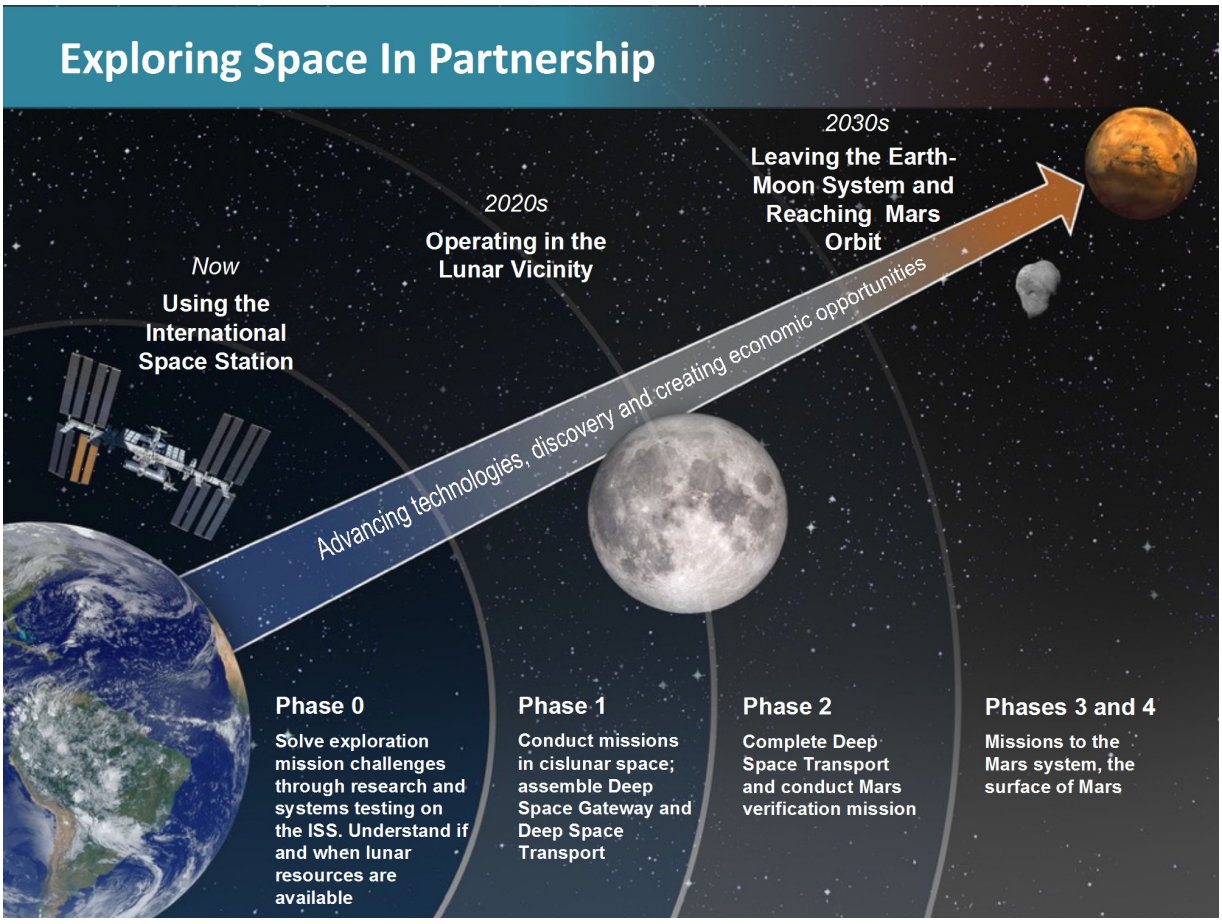

FIGURE 4.1 Current top-level NASA Exploration Strategy. SOURCE: Craig Kundrot, presented to committee on April 18, 2017, slide 5.

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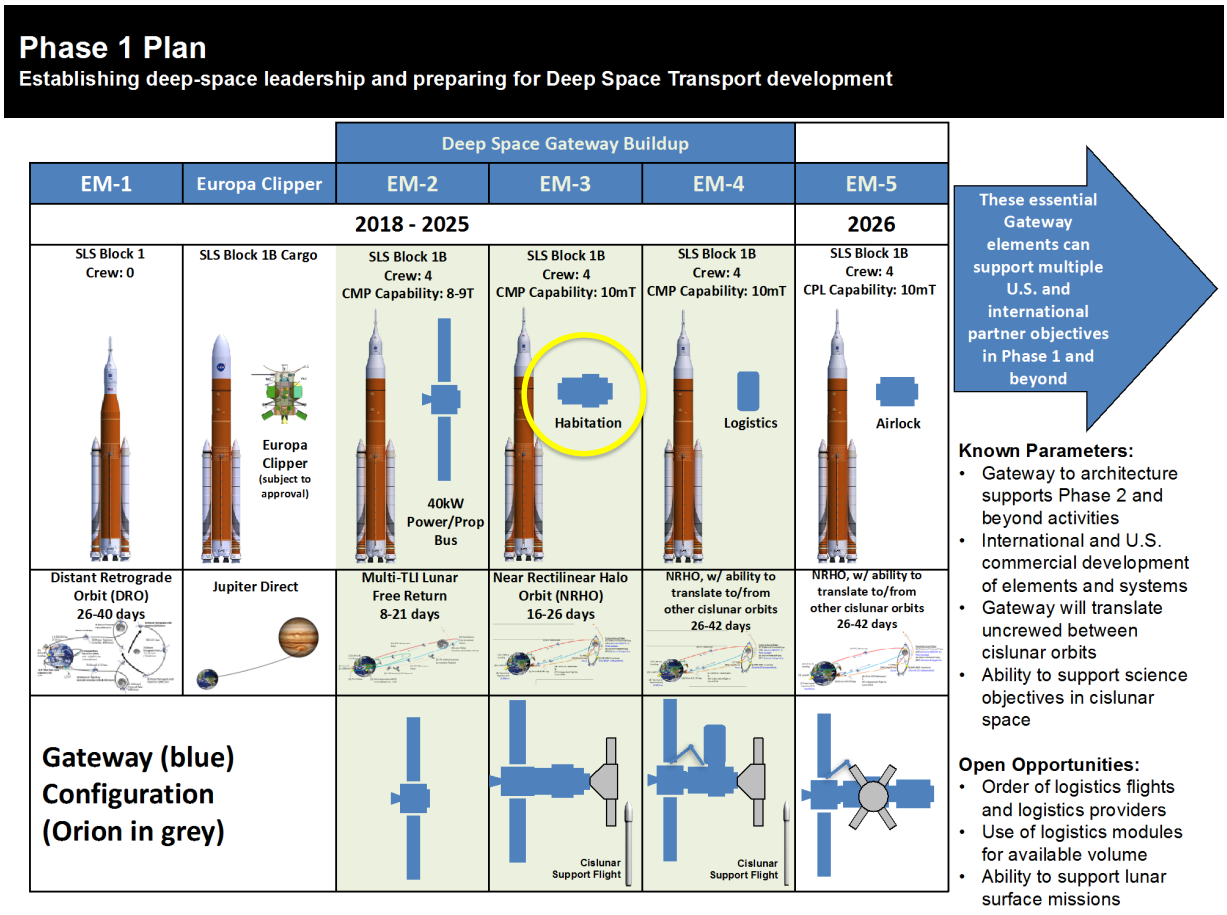

FIGURE 4.2 Reference plan for Phase 1 of the NASA Exploration Strategy. SOURCE: Craig Kundrot, presented to committee on April 18, 2017, slide 6.

The 2011 decadal survey developed "Bounding Policy Options" to assess the research priorities. Also, the NASA exploration strategy has evolved over time from the "Vision for Space Exploration," including the Constellation Program, to the "Journey to Mars," with the Asteroid Redirect Mission and the Space Launch System and Orion, to the current "Deep Space Gateway" strategy. It is clear from this history that NASA's exploration strategy can be a changing and moving target. NASA works diligently to align the resources and priorities based on its strategy at the time. It is also clear to the committee that such changes can, and will, impact the criteria and priorities for research. For each strategy, the detailed requirements found in a Design Reference Mission are a useful way to guide research priorities and to assess potential design options based on what is learned from the research.

Finding 4-1: Several exploration pathways have been proposed by NASA since the 2011 decadal survey, involving the Moon, asteroid missions, Mars, and cislunar space. NASA has also used a separate set of Design Reference Missions to develop its Technology Roadmaps. These frequent exploration strategy changes result in unclear traceability between research investigations and exploration needs.

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4.2 CRITERIA FOR MIDTERM RANKING OF THE HIGH-PRIORITY RECOMMENDATIONS FOR EXPLORATION

As noted above, Chapter 13 of *Recapturing a Future for Space Exploration* laid out a specific set of criteria that were used to categorize research along the continuum of basic, applied, or translational science within that document. Those decadal survey criteria are reprinted in Appendix E. In order to address the charge for this report, the committee made extensive use of these criteria to assess how relevant the high-priority recommendations were to exploration—specifically, by giving weight to those criteria that most prominently support the exploration agenda. As such, the committee considered Criteria 1, 2, 3, and 5 (listed below) to be the most directly relevant to exploration needs. These criteria were used in the decadal survey to rank those recommendations that were a high priority for enabling exploration. The midterm committee respected the rankings in the decadal survey and determined that these criteria still accurately capture the bulk of the principles that most enable the exploration agenda. These criteria best distinguish the science that is needed for exploration from the science that is enabled by microgravity.

The following prioritization criteria from the 2011 decadal survey guided the top-level identification of targeted research priorities for exploration for this midterm assessment:

- *Prioritization Criterion 1:* The extent to which the results of the research will reduce uncertainty about both the benefits and the risks of space exploration (*positive impact on exploration efforts, improved access to data or to samples, risk reduction*).
- *Prioritization Criterion 2:* The extent to which the results of the research will reduce the costs of space exploration (*potential to enhance mission options or to reduce mission costs*).
- *Prioritization Criterion 3:* The extent to which the results of the research may lead to entirely new options for exploration missions (*positive impact on exploration efforts, improved access to data or to samples*).
- *Prioritization Criterion 5:* The extent to which the results of the research are uniquely needed by NASA, as opposed to any other agencies (*needs unique to NASA exploration programs*).

The midterm committee also recognized the need to overlay the following additional and overlapping considerations that most directly affected the current feasibility or utility for exploration of the decadal survey recommendations:

- *Consideration A:* The extent to which the science is needed to support technology development for deep space exploration.
- *Consideration B:* The extent to which a long development lead time is required for the science to enable a needed technology.
- *Consideration C:* The extent to which the research question can only be adequately studied on the ISS or takes best advantage of remaining ISS time.
- *Consideration D:* The extent of urgency or criticality to a specific exploration goal, technology, or biomedical countermeasure.
- *Consideration E:* The extent to which the science may now be complete.

The midterm committee further recognized that it is not possible to quantify or objectively measure these considerations. Therefore the commmitee comprehensively discussed the science outputs from the discipline areas and consulted the literature and various NASA reports to apply these considerations in a deliberative process of additional prioritization. Some level of informed subjectivity was expected and even encouraged, to allow for specific science and program expertise to play a role in the prioritization within and across disciplines. A summary of the selection process and outcomes is presented in Figure 4.3.

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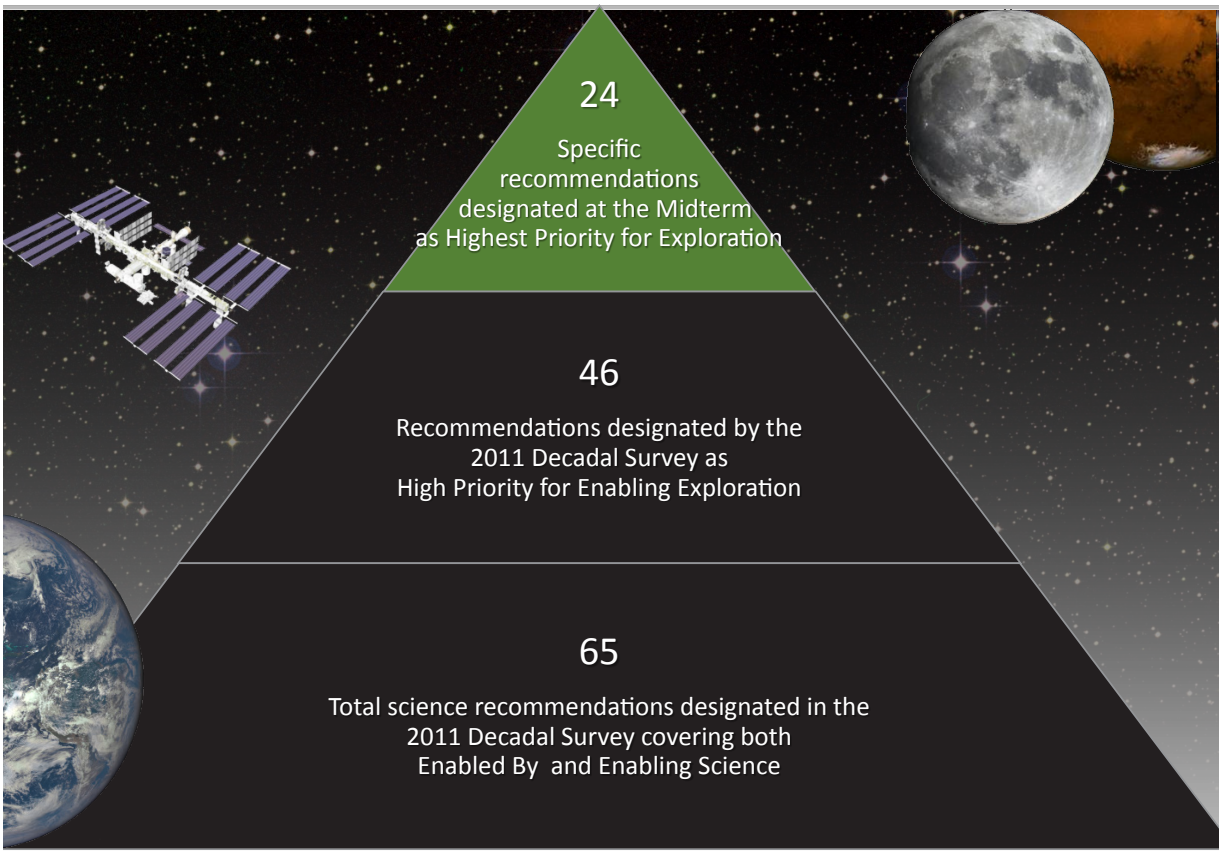

FIGURE 4.3 Selection of the highest-priority recommendations for science that best enables the exploration of space, as required by the statement of task for the midterm assessment committee. The 2011 decadal survey (National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C.) contained 65 recommendations, of those, 46 were identified as a high priority for exploration. The decadal survey used its Criteria 1, 2, 3, and 5 to arrive at those 46. Those recommendations, represented by the black areas of the pyramid, formed the basis of recommendations for further prioritization by the midterm assessment committee. Among those 46, 24 recommendations in the green area of the pyramid were identified by the midterm assessment committee as the highest priority for exploration for the remaining years of the decadal survey using the Considerations A through E, as presented in Chapter 4. Those recommendations are presented in Table 4.1.

4.3 HIGHEST-PRIORITY RECOMMENDATIONS FOR EXPLORATION

As directed by the statement of task, the committee drew strictly from recommendations that were listed in Table 13.1 the 2011 decadal survey. The committee used the decadal survey "Bounding Policy Option" of Mars Exploration to most directly address the statement of task, and to stay within the priorities identified in the decadal survey. All recommendations that were listed in Table 13.1 of the decadal survey report were further discussed in the context of their placements in Tables 13.2 and 13.3. Tables 13.1, 13.2, and 13.3 are reprinted in Appendix E. Then, based on the committee's decadal survey criteria and additional considerations as described above, the committee assigned a relative ranking of each of the recommendations, noting that these rankings considered the most important science needs in the pathway to exploration of deep space. The committee also considered whether it was necessary that the science of each recommendation be conducted in LEO or beyond LEO, and if the research could be conducted in ground-based or suborbital platforms. The results are summarized in Table 4.1, which lists

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TABLE 4.1 Ranking and Performance Environment of Research Priorities for Deep Space Exploration in the 2011 Decadal Survey[a]

Decadal Survey Recommendation Identifier	Summary Description of Recommendation[b]	Earth Analogs[c]	Low Earth Orbit	Beyond Earth Orbit[c,d]	Highest	Higher	High
Animal and Human Biology							
AH1	Efficacy of bisphosphonates	S	N	S			**X**
AH2	Bone strength and countermeasures	N/S	N/S	S			**X**
AH3	Bone loss in genetically altered mice	S	N	S			**X**
AH4	New osteoporosis drugs	S	N/S	S			**X**
AH5	Skeletal muscle protein balance	N/S	N	N/S		**X**	
AH6	Prototype exercise devices	S	N	N	**X**		
AH7	Levels and recruitment of flexor and extensor muscles	S	N	N/S		**X**	
AH8	Changes in Starling forces	S	N	N/S	**X**		
AH9	Effects of prolonged reduced gravity on task completion	S	S	N			**X**
AH10	Orthostatic intolerance	S	N/S	N		**X**	
AH11	Screening strategies for subclinical coronary heart disease	N	N	S		**X**	
AH16	Transmission of structural changes over generations	N	N	S			**X**
Applied Physics							
AP1	Reduced-gravity multiphase flows	S	N	S	**X**		
AP2	Interfacial flows	S	N	S	**X**		
AP3	Dynamic granular material behavior	N	S	N/S		**X**	
AP4	Methods for dust mitigation	N	N/S	S		**X**	
AP6	Flammability and fire suppression in space	S	N	S	**X**		
AP8	Numerical simulation of combustion	N	S	S	**X**		
AP10	Materials that enable operations in space environments	N	S	S	**X**		
AP11	Processing materials on extraterrestrial surfaces	N	N/S	S	**X**		
Behavior and Mental Health							
B1	Measures of mission relevant crew performance	N	N	N		**X**	
B2	Long-duration mission simulations	N	N	N	**X**		
B3	Underpinnings of individual differences	N	N/S	S	**X**		
B4	Multinational crews	N	S	S	**X**		

continued

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TABLE 4.1 Continued

Decadal Survey Recommendation Identifier	Summary Description of Recommendation[b]	Earth Analogs[c]	Low Earth Orbit	Beyond Earth Orbit[c,d]	Highest	Higher	High
Cross-Cutting Issues							
CC1	Post-landing vertigo and orthostatic intolerance	S	N	N			**X**
CC2	Artificial gravity as a multisystem countermeasure	S	N	N	**X**		
CC3	Low prevalence of decompression sickness	S	N	S			**X**
CC4	Optimal dietary strategies	N	N	N		**X**	
CC6	Food and energy intake	N	N	S			**X**
CC7	Cataract incidence, quality, and pathology	S	N	N		**X**	
CC8	Space radiation risks to humans	N	S	N	**X**		
Plant and Microbiology							
P1	Microbial observation	S	N	S		**X**	
P2	Plant and microbial responses and adaptations to spaceflight	S	N	N	**X**		
P3	Plant and microbial systems for life support	S	N	N	**X**		
Translation to Space Exploration[e]							
TSES1	Active two-phase flow relevant to thermal management	S	N	S	**X**		
TSES2	Zero-boiloff propellant storage	S	N	S	**X**		
TSES3	Surface mobility	N	S	N		**X**	
TSES4	Mitigate dust on extra-vehicular activity (EVA)	N	S	N	**X**		
TSES5	Thermal and impact mitigation on EVA	N	N	S	**X**		
TSES6	Heat and mass transfer in porous media	N	N	S	**X**		
TSES7	Thermoregulation on the lunar surface	N	S	N		**X**	
TSES10	Regenerative fuel cell technologies	N	N	S			**X**
TSES12	Fission surface power systems	N	S	S	**X**		
TSES13	Ascent and descent system technologies	N	N	N	**X**		
TSES14	Nuclear propulsion systems	N	N	S	**X**		
TSES16	In Situ Resource Utilization capability development and surface habitats	N	S	N	**X**		

[a] See National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C.

[b] These brief descriptions cannot adequately capture the sense of the decadal survey recommendation, and readers are referred to the more complete descriptions of the recommendations in the decadal survey and in Chapter 3 of this report.

[c] An S indicates that the location would be supporting for addressing some or all of the experiments needed to address the research priority. An N indicates that the location is likely to be necessary for completing some or all of the experiments that would be involved in addressing the priority.

[d] Beyond low Earth orbit includes both open space and the surfaces of other planetary bodies and asteroids.

[e] Translational priorities represent areas of physical sciences research identified in the decadal survey that support the TSES technology category. Thus TSES14 includes such research areas as liquid-metal cooling in reduced gravity.

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all of the high-priority recommendations of the 2011 decadal survey, along with the committee ranking identifying the higher- and highest-priority recommendations for exploration. In addition to this ranking, Table 4.1 summarizes the committee's assessment of the locations of the research activities to accomplish each of the science recommendations. The details of the committee's assessments within each individual discipline area, which informed the rankings in this table, can be found in the following sections of this chapter.

4.4 SCIENCE CONTEXT FOR THE RANKINGS

Each of the targeted priority rankings in Table 4.1 of this chapter arises from the context of current science accomplishments in areas that include, but reach beyond, NASA exploration. Each priority also stems directly from science needed to enable and enhance exploration of deep space. The committee therefore examined the progress in major discipline areas in order to establish a scientific context for the stated NASA plans for the ISS and the development of deep space exploration. This broader scientific context, further informed by the roles of the discipline in human exploration, serves as the basis for identifying and ranking the highest priorities for remaining years of the current decadal survey. This context also provides an overview of the progress of microgravity science compared to related fields outside of exploration-related science.

4.4.1 Plant Biology and Microbiology

Since the decadal survey was published in 2011, there has been rapid progress in several areas of plant and microbial biology, in particular great decreases in expense, and increases in power, of genomic and metagenomic sequencing, and in transcriptomic analysis by arrays and RNASeq for gene expression, genomic sequencing and assembly, and ChIP-seq for understanding regulatory processes. In the microbial area, this same set of methods has empowered metagenomic sequencing, which ties metabolic function to genomes and could be of great value in assessment of the bacterial and viral environment of the space environment, including the water, air, and surface growth of bacteria, and the microbiome of the astronauts, and of the plants grown for life support. One additional area that is under rapid development is genome modification (of bacteria, plants, and animals) using novel, bacterially based systems such as the CRISPR-Cas9 system. This allows large-scale and accurate engineering of genomes, a capability which could be of importance not only in experimental design and controls, but also in engineering organisms adapted to spaceflight conditions for bioregenerative and nutritional support of flight crews. An example of this would be the engineering of vegetables such as lettuce and tomatoes to reduce their response to environmental stress signals such as ethylene, a plant stress and fruit ripening/spoilage hormone that emanates from plastics and is generated by microorganisms present in spaceflight environments.

4.4.1.1 Role in Human Exploration

Plant and microbial biology are key components of the NASA exploration strategy due to the integral roles of microbes and plants within almost any human ecosystem. Plants provide the connection between humans and terrestrial nature, while also completing the life-support cycle by recycling air, water, and nutrients. And while NASA relies heavily on physio-chemical life support systems, plants continue to be envisioned as part of the regenerative life-support systems while potentially providing psychological support to the crew. A consistent body of research has demonstrated the positive psychological benefits of interacting with plants;[2] further, self-reports by astronauts have borne out the salutogenic, health-enhancing effects of the presence and maintenance of plants.[3]

Microbes inhabit all plant and human ecosystems, including exploration vehicles and habitats. The relevance of microbial life to long-term space missions such as those to Mars is two-fold: first, organisms that cause disease

[2] J. Stefan, N. Gueguen, and S. Meineri, 2015, Influence of the interior and outdoor plants on health: Synthesis of research, *Canadian Psychology* 56(4): 405-425.

[3] J. Bennington-Castro, 2014, "Up on the Farm? Five Reasons NASA Needs Space Greenhouses," *National Geographic*, January 21, http://news.nationalgeographic.com/news/2014/01/140121-space-greenhouses-plants-astronaut-mars.

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in plants or humans need to be monitored and controlled during the flight as a preventive contribution to astronaut health and for diagnosis of disease. Secondly, the nondisease-causing microbes that constitute important microbiomes of plants, humans, and potentially mutual microbiomes, are both contributors to, and indicators of, the state of the health of the system. Therefore these microbes will need to be monitored and understood in order to encourage beneficial microbiomes.

The decadal survey called out three major and specific areas for prioritized study applicable to the exploration goals of NASA: P1—multigenerational studies of microbial populations, P2—analysis in space of plant and microbial growth and of the physiological responses of plants and microbes to the spaceflight environment, and P3—a research program to demonstrate the roles of microbial-plant systems in long-term life support. As previously described in Chapter 3, NASA and its research partners have achieved much science in these three areas in the first 5 years since the decadal survey.

4.4.1.2 Plant and Microbial Biology Research Platforms and Approaches Relevant to Exploration

NASA exploration plans are reliant on transit vehicles without large-scale artificial gravity. The Gateway habitat, for example, will be in a continual microgravity state. Therefore, microgravity phenomena and the associated biological effects of microgravity remain key objectives for study even in the deep space era. Therefore, access to long-term LEO microgravity will be necessary for continued progress. Certain microgravity phenomena will have to be conducted in deep space, particularly any biological phenomena that are impacted by deep space radiation, making deep space research important for areas P2 and P3. In all cases, control experiments and testing of experimental protocols can be done on the ground, using Earth analogs that can inform the deep space experiments. The necessary facilities are available, and such needs are easily met with existing equipment and access to short-term microgravity in drop towers and suborbital flights.

Research associated with radiation and radiation effects on biological systems are expected to be a strong element of research conducted in space beyond LEO. The impacts of deep space radiation on biology have not been well explored in general, and with plant and microbial systems in particular. Deep space science would need to include best practice approaches to analyzing the genomic and physiological effects of deep space radiation on plants and microbes. For plants, the important areas would be the long-term viability and genomic stability of seeds that accompany deep space trips. For microbes, the important areas would include the effects of deep space radiation on the evolution of microbes and microbial systems. Long-term exposures on the Gateway, even during times without human habitation, would be an important part of the deep space biology portfolio.

Plant growth systems are envisioned as providing radiation protection to crew, particularly when the plant growth systems are arrayed on the vehicle walls. As part of the wall system, the plants and their water would provide shielding for the crew. However, the effects of such radiation on plant and plant/microbial interaction physiologies are largely unknown and would need priority assessment.

The decadal survey specifically called out the aggressive use of data-rich omics studies as part of this area of research. Much of the success and achievement in microbial and plant space biology has been through the directed application of omics technologies in alignment with the decadal survey recommendations. Examples are the GeneLab and Microbial Observatory initiatives within NASA and the ISS. The further advancement of omics technologies, processes, and instrumentation has driven plant and microbial space biology toward the capability of on-board omics analyses for rapid real-time evaluations on one hand, while sample storage technologies have also enabled the archiving of samples for detailed study once samples are returned from spaceflight.

A strong example of NASA commitment to supporting microgravity research resides in the facilities at its centers. For example, the NASA Kennedy Space Center (KSC) flight payload support group conducts integrated tests and performs near real time or post-flight ground-control experiments using ground unit hardware within the ISS environmental simulation (ISSES) chambers available at KSC. The ISSES chambers are configured to mimic those environmental conditions of temperature, relative humidity, light exposure, and CO_2 concentrations that the flight hardware experiences on the ISS. Additionally, KSC is in the process of establishing a suite of microgravity simulators that allow researchers the option of performing micro-, partial-, and hypergravity ground controls. The ISSES chambers and microgravity simulation facility are all part of a large and well-equipped infrastructure

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within the Space Station Processing Facility (SSPF) at KSC. The SSPF has payload integration support facilities, off line laboratories for principal investigators (PIs), office spaces, ISS experiment-monitoring areas, laboratory support and equipment.

4.4.1.3 Recent Advances Relevant to Plant and Microbial Microgravity Science

Among the DNA advances are development of rapid DNA sequencing methods that use equipment suited to the spaceflight environment. Next-generation sequencing has been developed to the point where sequence analysis in space is practical and recently demonstrated. This provides a path to diagnosis of infectious human and plant disease, and toward real-time, in-site environmental biological monitoring to identify plant and microbial adaptations, as well as future threats in the form of microbial evolution of host ranges or growth strategies. It also provides methods to monitor crew health through analysis of the gut and skin microbiome of the astronauts, the colonization of vehicle systems by microbes, and plant health within plant growth facilities designed provide bioregenerative life support.

Another advance has been in use of RNA sequencing methods to understand the physiological changes experienced by plants during short-term growth in space, which gives indications of the types of stresses that are relevant to long-term growth. These technologies provide data on plant and microbial responses to spaceflight and can be used to monitor plant health and production of plant products within exploration production systems.

One key action that can be taken to optimize the science value of the research program in context of enabling deep space exploration would be to develop robust methods, standards, and equipment for bacterial and virus monitoring in space. These methods are applicable to all decadal survey recommendations in this area, and are of direct use in space missions. The necessary actions consist of assessment of available sequencing technologies, miniaturization of the equipment, and testing equipment on multiple types of samples in spaceflight, with preserved samples returned to Earth for ground-based validation.

Also important for the further implementation of the decadal survey recommendations is the continued development of the facilities for long-term plant growth, with parallel development of microbial monitoring for plant pathogens and of the plant microbiome to assess (and correct) growth conditions for optimal growth. NASA has a long history of developing plant growth systems for spaceflight applications. Most of those systems have been research directed, in that they were designed for managing and controlling environmental conditions specific to science goals. The Veggie system currently on the ISS is one of the few spaceflight systems that has been designed for actual plant production on orbit. This seemingly subtle differentiation between a research plant growth unit and a production plant growth unit has driven some incredible, exploration support science. Within Veggie, plants have been grown for crew consumption, creating the procedures and methods for repeated plant production on orbit. In addition, these plant production experiments have revealed plant microbial pathologies on the ISS, exposing the need to more deeply understand the plant microbiome as it exists and evolves in an exploration habitat.

Finally, plants could provide a solution to the long-term storage of nutrients, such as vitamins and proteins, as called out in recommendation CC5 of the decadal survey, as growing plants from seeds could provide a freshly synthesized source through the term of a deep-space mission. However, plants have yet to be optimized for growth and production in microgravity spaceflight environments.

4.4.1.4 Highest-Priority Plant and Microbial Recommendations for Exploration

The results from that science have illuminated many space biology science effects, strongly suggesting that spaceflight has significant influence on biological systems and that biology in space is not the same as biology on the ground. Moreover, these results require that continued study of space plant and microbial biology be a priority during the era of deep-space exploration precisely because the results that have been obtained to date indicate that deeper understanding is required to truly make use of plants and microbes for supporting exploration.

The committee considered P2 (analysis in space of plant and microbial growth and of the physiological responses of plants and microbes to the spaceflight environment) and P3 (a research program to demonstrate the roles of microbial-plant systems in long-term life support) to be the highest-priority plant and microbial recom-

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mendations that are relevant to the exploration of deep space. P2 and P3 have the largest role in determining the biological adaptations to spaceflight and the understanding of how biological life support systems may function in support of exploration. P1, multigenerational studies of microbial populations, is still considered an important activity, yet this activity has been largely established and has entered a role of continued monitoring rather than of discovering principles needed for exploration. P2 and P3, the study of plant adaptations to spaceflight and the interactions of plant/microbial systems in spaceflight, is still largely in the mode of discovering principles that apply to exploration life support.

The patterns of gene regulation that respond to microgravity and the spaceflight environment, and the basic microbiology of plant-microbe interactions in the space environment, are rapidly being elucidated. However, long-term studies are lacking. And the research to apply this gene expression knowledge to improve plant growth for food and nutrition on a deep space mission—to mitigate the effects of weightlessness on plants and on their immediate environment—is in the future, and not clearly on course to be fully translated to life support production by 2024.

4.4.2 Behavior and Mental Health

Developments and progress in the wider nonspace research community related to behavior and mental health include the explosion of interest and knowledge in the neuroscience area to better understand normal as well as psychopathological processes. Omics advances, such as the identification of genes associated with types of depression and other forms of psychopathology and the study of epigenetic markers, are some of the areas in which recent progress has occurred. The research by Feinberg and colleagues[4] on epigenetic markers and the mechanisms of epigenetic modification in normal development and in mental disorders is particularly notable.

The development of the Research Domain Criteria (RDoC) by the National Institute of Mental Health (NIMH) as a new paradigm for understanding mental disorders focuses on dimensions of psychopathology (assessed by a combination of observable behavior and neurobiological and physiological processes) rather than categorical diagnostic types. This approach and the findings derived from studies using RDoC criteria have the potential to provide a more robust framework for understanding the complex and multifaceted nature of mental disorders and the commonalities across dimensions evident in different diagnostic groups.

There is ongoing research by the military and industrial organization/relations groups to identify individual traits of hardiness and the promotion of resilience, and also to better understand the dynamics of complex team systems. This area of research examines factors contributing to optimal single and multiteam performance in challenging situations/environments. In addition, advancements in virtual reality technologies based on games and other recreational pursuits have enhanced the fidelity of the virtual experience and the possibility to improve the effectiveness of virtual reality training procedures.

4.4.2.1 Role in Human Exploration and Future Implementation

Exploration missions require a small group of astronauts to live and work together for an extended period of time in a confined vehicle and habitat. The effectiveness of their individual and teamwork performance is crucial to the success of the mission. The decadal survey priorities designated for the behavior and mental health area cover both individual and team factors and are addressed below.

B1—*Develop sensitive, meaningful, and valid measures of mission-relevant performance for both astronauts and mission control personnel.*

Priority B1 was ranked relatively lower than priorities B2, B3, and B4 for the purposes of this study because of the overriding importance of identifying optimal individual and team characteristics to inform crew selection for long-duration exploration missions (LDEMs). Nonetheless, the B1 priority also requires additional research.

[4] A.P. Feinberg, M.A. Koldobskiy, and A. Göndör, Epigenetic modifiers and mediators in cancer aetiology and progression, *Nature Reviews Genetics* 17:284-299.

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Assessing the actual work performance of ground- and space-based personnel during a mission is a sensitive topic; however, it is critical to the eventual safe conduct of a deep-space exploration mission. Innovative strategies need to be developed to obtain this information without potential bias by the personnel involved. It should be possible to develop a global picture of the ideal astronaut based on debriefings and other materials collected on past and current astronauts. While specific technical skills are crucial, they are not a substitute for a broader conception in which aspects—such as cooperation, psychosocial skills, empathy, and excellence under stress—are taken into account. In addition, the monitoring measures currently used in space need to be enhanced. Improvement of both subjective and objective (including nonintrusive) measures of performance are needed. Analogs can provide relevant information, but there currently is no analog that can fully take the place of real-time performance of both astronauts and mission control personnel.

B2—*Conduct integrated translational research in which long-duration missions are simulated specifically for the purpose of studying the interrelationships among individual functioning, cognitive performance, sleep, and group dynamics.*

Current Human Research Program (HRP) funded projects conducting integrated translational research in the above area need to continue. Often this process consists of projects by different investigators coordinated at an analog site such as HI-SEAS (Hawai'i Space Exploration Analog and Simulation) or HERA (Human Exploration Research Analog). Inclusion of additional projects also is highly important. The challenge is to coordinate the activities and findings of the individual projects into a coherent whole through which overall conclusions can be reached.

B3—*Determine the genetic, physiological, and psychological underpinnings of individual differences in resilience to stressors during extended space missions, with development of an individualized medicine approach to sustaining astronauts during such missions.*

During the past decade, there has been a leap forward in research and knowledge in the neuroscience and omics areas. Improvement of various imaging technologies, including technical adaptations to enable the study of ongoing cognitive and emotional processes, is evident. These advances present an important opportunity to better understand the underpinnings of individual differences in adaptation to challenging and stressful environments, and possible changes in brain function and structure as a result of living and working in an isolated, confined, and extreme environment for an extended period. In addition, research on genetic and epigenetic factors, among other omics approaches related to resilience would be beneficial. Targeted research priorities integrating individual factors such as personality trait configuration, psychological and physiological markers of resilience following stressful events and sleep loss, and relationships to work performance are needed. Eventual understanding of the unique aspects of individual differences in the various combinations of genetic, physiological, and psychological underpinnings of resilience should provide the data for an individualized medicine approach.

B4—*Conduct research to enhance cohesiveness, team performance, and effectiveness of multinational crews, especially under conditions of extreme isolation and autonomy.*

The entire team performance area requires still greater attention, including additional research on multiteam systems. Research examining the dynamics of conflict resolution in small isolated and confined groups would be beneficial. Ongoing integrated analog studies of team performance are relevant to many aspects of crew functioning. A fine-grained analysis of the emergence of conflicts, effects on other team members, and the process of conflict resolution is important. Furthermore, mission control personnel are a vital part of the mission, and their inclusion in research is necessary. Another interrelated team issue is the different levels of autonomy the crew will experience. The crew will become more independent in functioning as they travel to deep space, and live and explore the Mars surface; however, the crew will need to deal with lesser autonomy as they return to Earth and progressively are under the greater command of mission control.

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4.4.2.2 Research Platforms and Approaches Relevant to Exploration

Both Earth analogs and flight experiments are important to furthering the research agendas during the remaining period of the current decadal survey. To ensure validity, as many projects as possible would need to be conducted in space. However, considerations of the total number of astronauts available on flights, as well as the need to obtain individual agreements to cooperate on a study, require that high-fidelity Earth analogs be used as well. The sample size of space flyers participating in research projects will be increased through greater cooperation with other space agencies, particularly Roscosmos. It may be useful to pursue this possibility more actively. Each analog will simulate different aspects of the vehicle or surface environment; increasing the number of analog sites available, including additional polar sites, will be helpful. Also, cooperative use of simulation facilities such as the Russian Mars habitat will add to the analog possibilities available.

A significant opportunity in the Behavior Medicine (BMed) area exists to make use of scientifically robust standardized data on astronaut personality traits (collected as part of the selection process) to predict performance in space and the reintegration back to Earth. These data may help answer a major risk for future Mars missions: the later development of psychopathology in initially psychologically healthy astronauts. Confidentiality could be ensured by having this effort funded and confined to the Johnson Space Center (JSC) operational group; a de-identified technical report could inform future research.

While the Behavioral Health and Performance (BHP) group has moved forward in funding analog projects, a significant challenge is the availability of enough relevant analogs to place funded studies in and to maintain these studies over consecutive years. A better understanding of the memorandum of understanding (MOU) between the NSF and NASA for access to Antarctic sites is needed. The current MOU is extremely general, referring only to "Space, Earth, and Biological Cooperative Activities" with no mention of access or collaboration at analog sites. Greater effort to obtain matching funding from other agencies such as the Defense Advanced Research Projects Agency (DARPA) and the National Institutes of Health (NIH) would be helpful; DARPA for studies of individual and team resilience in the face of intense challenges, NIH for studies of medication effects in extended duration microgravity and epigenetic factors.

4.4.2.3 Recent Advances Relevant to Microgravity Science

Developments and progress in the wider nonspace research community with relevance for space purposes include neuroscience research focused on better understanding normal as well as psychopathological processes. The research by Feinberg and colleagues on epigenetic markers and mechanisms of epigenetic modification[5] in normal development, mental disorders, and microgravity (the ISS Twins study[6])is highly relevant for LDEM. This knowledge has application for better understanding of the long-term physical impact of stressors on astronauts during LDEM, with possible effects on psychosocial functioning. These and other findings from the nonspace research community could potentially address aspects of the B3 decadal survey priority regarding genetic, physiological, and psychological underpinnings of individual differences in resilience to stressors.

As mentioned earlier, the development of the Research Domain Criteria (RDoC) by the NIMH as a new paradigm for understanding mental disorders focuses on dimensions of psychopathology rather than categorical diagnostic types. This approach and the findings derived from studies using RDoC criteria have the potential to inform the selection of astronauts for space missions based on genetics, epigenetics, neuroscience, and behavioral science criteria (B3).

The ongoing space-relevant research by the military and industrial organization/relations groups on the dynamics of complex team systems has application for identifying factors contributing to optimal single and multiteam performance in challenging situations/environments (B4). This area is highly relevant because it is possible that team performance may increase in risk factor status as the duration of missions increases. The mission challenge is the need for individual teams to function as part of multiteam systems, although the crew also must function with greater autonomy as the distance from Earth increases. In addition, advancements in virtual reality

[5] Ibid.

[6] See NASA, 2017, "Twins Study: The Research," last updated August 3, https://www.nasa.gov/twins-study/research.

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technologies can enhance the fidelity of the virtual reality programs that are being developed for space purposes to promote positive affect and stimulation and used as countermeasures to alleviate boredom and stress.

4.4.2.4 Further Considerations for Implementing Decadal Survey Recommendations

Relevant areas outside of NASA that influence this exploration research include NIH-sponsored projects on changes in brain structure and functioning under stressful conditions, and epigenetic studies assessing changes under different environmental conditions. Research sponsored by the military and in academic settings has provided relevant information on team functioning in challenging situations, as well as furthering the understanding of the personality characteristic of hardiness, and its influence on resilience under stress. NSF-funded studies in polar environments also have provided informative data on individual and team functioning in isolated, confined, and extreme environments.

4.4.2.5 Highest-Priority Recommendations for Exploration

It is likely that candidates currently being selected for the astronaut corps will be among the pool of astronauts selected for exploration missions. Therefore, it is important that the high-priority research designated below be carried out in an expedited manner both in analog and space environments to inform the necessary individual and team criteria optimal for long duration exploration missions.

Following the charge in the statement of task, the committee looked at the highest priorities from the decadal survey and ranked B2, B3, and B4 as most important in terms of relevance to exploration. These recommendations focus on the interrelationships among individual functioning, cognitive performance, sleep, and group dynamics; the genetic, physiological, and psychological underpinnings of individual differences in resilience to stressors during extended space missions; team performance; and effectiveness of multinational crews. While B1 was rated somewhat lower in terms of priority, clearly, the continued development of measures to assess work performance in space is also a significant priority. Considerable research will need to be continued following the 2024 end of the ISS to mitigate the risks designated by the NASA Human System Risk Board (HSRB). For 2024, HSRB categorized "Cognitive or Behavioral Conditions" as high risk and "Team Performance Decrements" as low risk, although not in the "optimized" category. (Risk factors that are not categorized as "optimized" require more research to bring countermeasure technologies to a level where they can be effectively applied during a Mars mission.)

4.4.3 Animal and Human Biology

Since the decadal survey in 2011, there has been rapid progress in several areas of human and animal biology and physiology and in particular, spectacular progress in genetic sequencing and analyses of gene expression for understanding developmental and regulatory processes. Another new area that is emphasized by NIH is research related to the microbiome and its role in health and disease. The microbiome may be adversely affected by degradation of micro-nutrients and food in general. Moreover, extreme confinement of crews during deep-space missions may also adversely affect the microbiome. Thus, NASA will benefit from past and future progress made in this field. Another area in which rapid progress is being made is drug development for prevention of muscle atrophy and bone loss. However, whether or not such drugs work in site-specific ways to counteract Type 1 muscle fiber atrophy and tibial and spinal bone losses needs further investigation. Finally, the rapidly growing field of virtual environment technology will greatly aid crew member adherence to exercise regimens needed to maintain crew health and well-being.

4.4.3.1 Role in Human Exploration

Animal research is necessary for assessing the effects of the space environment on humans in deep space, and for developing mitigating technologies for those effects that are adverse. Musculoskeletal loss including muscle atrophy and bone mass and strength losses (the subject of much successful past research) has now been largely

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mitigated by exercise and bone-building medications (bisphosphonates). However, the ARED (Advanced Resistive Exercise Device) resistance training device that has proven successful in mitigating muscle and bone loss is too heavy and large to fly on an exploration mission as presently configured. Moreover, there is evidence that some crew members experience back injuries with this hardware; they then are not allowed to exercise for variable periods of time after the injury. Also, some flight surgeons disagree about whether bisphosphonates are the best option for long-term treatment of bone loss, especially in healthy males. So even though bone loss is mitigated by ARED combined with bisphosphonates for LEO; better, smaller and more integrative countermeasures are needed for deep-space missions. Thus, there still is a need for development of pharmaceutical treatments to maintain muscle mass and protein balance, and multisystem countermeasures to maintain the cardiovascular system and to prevent the head-ward fluid shifts that probably contribute to vision problems. Human research is also necessary in these areas to demonstrate applicability of animal discoveries, and to assess the extent to which these issues affect astronaut performance and survival.

4.4.3.2 Research Platforms and Approaches Relevant to Exploration

The effects on humans of long-term space travel are largely responses to a microgravity environment and radiation (see below). Facilities for animal growth and culture, especially for rodents, are therefore central to future discovery. While simulated-microgravity environments are possible for plant research (for example, clinostats on the ground), there is no substitute for actual microgravity in animal research. Thus, experiments to determine the effects of microgravity on animals and humans would need to largely be done in LEO on the ISS. Ground-based controls are nonetheless necessary for all space-based experiments so that attribution of effects to the microgravity environment is soundly based. Ground-based controls, requiring duplication of space-based facilities, and careful control of nongravity parameters such as temperature, habitat size, nutrition, lighting, and so on, are therefore critical. With reference to one of the highest priorities (AH6), new prototype exercise devices are needed to optimize physical activity by employing multisystem countermeasures. Such multisystem countermeasure research can be supplemented in Earth analog studies, but progress will greatly depend on LEO and beyond Earth orbit platforms to evaluate new prototype exercise devices during actual microgravity. Also, with reference to the other highest priority (AH8) concerning Starling pressures, ground studies have already supplemented this research. However, actual microgravity is needed on LEO and beyond Earth orbit platforms to understand transcapillary fluid exchange and venous and lymphatic returns as well as possible impacts of loss of tissue weight that is only available during space flight.

The need for a flight centrifuge has been cited in National Academies studies reaching back to the 1980s.[7,8] The decadal survey frequently identified the loss of the 2.5-m-diameter animal and plant centrifuge facility on ISS as a severe impediment for providing an important one-gravity control condition for studying the effects of microgravity on animals and plants. Moreover, with the loss of this facility, other studies of partial-gravities (1/6 and 3/8G) and hyper-gravity are very limited. While there are three small centrifuges operational on ISS, none is sufficient to provide a useful 1-G control or to investigate partial- and hyper-gravities because of their small diameters and associated high-gravity gradients and Coriolis effects. These effects keep the small centrifuges from serving as useful controls for experiments with plants and animals. The absence of a large diameter centrifuge remains a severe limitation for fundamental biology on ISS. The absence of useful large-diameter centrifuges for flight experiments was an item that was raised several times in the community input workshop, and is a serious concern to the scientific users of ISS flight modules for animal and plant experiments. While the committee understands that it is not feasible to add a large diameter centrifuge to ISS at this juncture, the need for such a facility will remain unchanged as NASA looks toward development of LEO and beyond Earth orbit platforms. The life science community of researchers is very interested in partial gravity research in addition to using a large-diameter animal and plant centrifuge as a 1-G control.

[7] National Research Council (NRC), 1987, *A Strategy for Space Biology and Medical Science for the 1980s and 1990s*, National Academy Press, Washington, D.C.

[8] NRC, 1998, *A Strategy for Research in Space Biology and Medicine in the New Century*, National Academy Press, Washington, D.C.

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4.4.3.3 Recent Advances Relevant to Animal and Human Biology Microgravity Science

As in plant and microbial research (see above) new methods and applications of DNA and RNA sequencing have allowed major progress in methods to monitor crew health, including the state of the human microbiome and the diagnosis of human disease.

Since the decadal survey in 2011, there has been rapid progress in several areas of human and animal biology and physiology and outstanding progress in genetic sequencing and analyses of gene expression. Another new area is the microbiome and its role in health and disease. This is emphasized by NIH in its recent Earth-based research. NASA will benefit from past and future progress made in this field. Rapid progress is available for drug development for prevention of muscle atrophy and bone loss, but such drugs may not be site-specific to counteract Type 1 muscle fiber atrophy and bone loss. Also, the rapidly growing field of virtual environment technology will greatly aid crew member adherence to exercise regimens needed to maintain crew health and well-being.

4.4.3.4 Highest Priority Animal and Human Biology Recommendations for Exploration

Extended missions in microgravity continue to require a better understanding of relationships between exercise and microgravity effects. The effectiveness of crew members' individual and team work performance is crucial to the success of the mission. The decadal survey priorities focus on exercise devices and deterministic mechanisms that underlie countermeasures. New prototype exercise devices are required for deep-space missions. This is prioritized because of the need to optimize physical activity and develop a multisystem countermeasure that maintains cardiovascular health and musculoskeletal strength and prevents fluid shifts toward the head. These head-ward fluid shifts cause inflight congestion and elevated pressure in internal jugular veins,[9] a constant but subclinical elevation of intracranial pressure compared to that of upright posture on Earth,[10] increased post-flight, lateral ventricular volume,[11] and upward shifting of the brain,[12] all of which may cause or contribute to problems of vision. Determining the basic mechanism of adaptations and clinical significance of regional vascular/interstitial pressures, due to altered Starling pressures, continues to be important. This research is of high priority because an understanding of the potential causes and effects of head-ward fluid shifts on structural remodeling and loss of function of neuro-ocular systems is warranted for development of countermeasure strategies beyond those provided by current exercise devices. Lower-body negative pressure (LBNP) garments and LBNP exercise devices are also possible mitigation strategies whose study could be usefully included in this priority area.

Decadal survey priorities such as bone loss and possible countermeasures for bone loss such as drugs and exercise have already progressed well, so these priorities are ranked lower in importance. However, it is still unknown how the space flight environment affects bone fracture healing in-flight, so this risk is rated as a "red risk" for planetary exploration. Other decadal survey priorities such as muscle protein balance, recruitment of muscles, orthostatic intolerance, screening subclinical heart disease, and transmission of structural changes over generations were deemed less important for success of deep-space missions.

[9] M.B. Stenger, A.R. Hargens, S.A. Dulchavsky, P. Arbeille, R.W. Danielson, D.J. Ebert, K.M. Garcia, et al., 2017, "Fluid Shifts," presentation #17601 at A New Dawn: Enabling Human Space Exploration, NASA Human Research Program Investigators' Workshop on January 23-26, 2017, https://three.jsc.nasa.gov/iws/SRIW-Cvent-Program-2017.pdf.

[10] J.S. Lawley, L.G. Petersen, E.J. Howden, S. Sarma, W.K. Cornwell, R. Zhang, L.A. Whitworth, M.A. Williams, and B.D. Levine, 2017, Effect of gravity and microgravity on intracranial pressure, *J. Physiol.* 595(6):2115-2127.

[11] N. Alperin, A.M. Bagci, and S.H. Lee, 2017, Spaceflight-induced changes in white matter hyperintensity burden in astronauts, *Neurology* 89(21):2187-2191.

[12] D.R. Roberts, M.H. Albrecht, H.R. Collins, D. Asemani, A.R. Chatterjee, M.V. Spampinato, X. Zhu, M.I. Chimowitz, and M.U. Antonucci, 2017, Effects of spaceflight on astronaut brain structure as indicated on MRI, *N. Engl. J. Med.* 377(18):1746-1753.

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4.4.4 Cross-Cutting Issues for Humans in the Space Environment

4.4.4.1 Role in Human Exploration

Cross-cutting issues for humans to explore deep space are key components of the NASA exploration strategy to maintain the health and well-being of crew members. The decadal survey called out some major and specific areas for prioritized study applicable to the exploration goals of NASA: CC2 to determine whether continuous, artificial gravity (AG) by a long-arm centrifuge is needed as a multisystem countermeasure or if intermittent LBNP exercise or intermittent short-arm centrifugation is sufficient; and CC4 to determine optimal dietary strategies and food preservation for crews. The radiation components of cross-cutting decadal survey recommendations are described below in Section 4.4.5.

4.4.4.2 Research Platforms and Approaches Relevant to Exploration

Again, the human effects of long-term space exposures are largely responses to a microgravity environment and radiation (see Section 4.4.5 below). While a simulated microgravity environment such as head-down-tilt bed rest is possible for human research with and without artificial gravity, there is no substitute for actual microgravity and countermeasure development related to the highest risk category of vision impairment for deep-space missions. Thus, experiments to determine the effects of microgravity and other adverse environments on vision and development of optimized countermeasures for humans are necessary in LEO and beyond Earth orbit platforms. Although multisystem AG countermeasure research can be supplemented in Earth analog studies, progress will greatly depend on LEO and beyond Earth orbit platforms to evaluate these countermeasures effectively. Although not in the "highest-priority" category in Table 4.1, studies of food systems for over 12 months of storage are required for prolonged, deep-space flights. Food systems research is needed in ground-based analogs, during LEO on the ISS through 2024, as well as for beyond Earth orbit with higher radiation exposures.

The decadal survey indicated that NASA should explore ways to study the effects of partial gravity on the human body, not just microgravity. The biomedical community of researchers is very interested in partial gravity research that can be supplemented by Earth analog studies, but further progress will depend on beyond Earth orbit platforms to evaluate the effects of partial gravity on the human body.

4.4.4.3 Recent Advances Relevant to Cross-Cutting Issues for Humans in the Space Environment

As described in preceding research sections, new methods and applications of DNA and RNA sequencing have allowed significant progress in methods to monitor crew health, including the state of the human health and the diagnosis of human disease. Since the decadal survey in 2011, there has been rapid progress in areas of human biology and omics research and great strides in genetic sequencing and analyses of gene expression for understanding adaptations to extreme environments. A new area that is emphasized by NIH is research related to the microbiome and its role in health and disease. As noted previously, the microbiome may be adversely affected by degradation of micro-nutrients and food in general, as well as by extreme confinement of crews during deep-space missions. Thus, NASA will benefit from past and future progress made in this field. Finally, the rapidly growing field of virtual environment technology will greatly aid multisystem, AG countermeasure research.

4.4.4.4 Highest-Priority Cross-Cutting Recommendations for Exploration

CC2: Artificial gravity as a multisystem countermeasure is needed for deep-space missions. Research is still needed to develop and test integrated AG countermeasures, including exercise, for periods exceeding 180 days in microgravity with and without artificial gravity by centrifugation and LBNP. It is important for integrated countermeasures in space to reproduce normal musculoskeletal loads and fluid shifts that occur on Earth and to which

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the body is adapted.[13,14] Such integration of countermeasures may maintain not only the structure and function of muscle and bone, but also cardiovascular and cerebral health. Merely miniaturizing exercise hardware presently on ISS for a smaller deep-space habitat is unlikely to maintain health and well-being of crew members. For example, it is known that head-ward fluid shifts in space have undesirable effects on cerebral venous circulations,[15] lymphatic function[16] and near-sighted vision.[17] This syndrome is now referred to as Spaceflight-Associated Neuro-ocular Syndrome (SANS). Moreover, it is likely that all crew members will be affected by these ocular changes as deep-space missions are lengthened. Presently, there is no hardware in space, except lower body negative pressure and perhaps thigh cuffs, to counteract head-ward fluid shifts and attempt to reproduce daily cycles of fluid shifts common to postural transitions on Earth. An alternate hypothesis is that some crew members may be genetically predisposed to endothelial dysfunction induced by oxidative stress.[18] Separately, hardware is needed to preferentially activate and maintain muscles that provide low-level but prolonged contractions common to upright posture on Earth. Current exercise hardware such as the Advanced Resistive Exercise Device and ISS treadmill do not preferentially maintain these postural, core muscle groups.

The addition of virtual environment (VE) technology is recommended for multisystem exercise hardware to improve the exercise experience for crew members and to foster adherence to exercise regimens. VE technology may also contribute to crew behavioral health and performance in small, confined habitats. These objectives will need to be met in integrative and coordinated research schemes, combining hardware development and protocol optimization. This is of utmost priority and is needed soon on ISS to achieve a reasonable sample of human subjects. Ground-based (e.g., bed rest) studies are an important initial step in developing an integrated, multisystem countermeasure combining exercise and foot-ward fluid shift. However, ground-based studies are not always adequate, especially with respect to simulating head-ward fluid shifts and their role in vision impairment. National and international cooperative studies utilizing existing translational networks and NASA's new Translational Research Institute may promote greater resource sharing and accelerate research. Such studies are necessary in LEO and beyond Earth orbit.

Decadal survey priorities identified in Table 4.1 include some other research. The decadal survey priority of optimizing dietary strategies and food systems was categorized as "higher," but was determined to be less important for success of deep-space missions than multisystem, AG countermeasure research. Other research related to post-landing vertigo, decompression sickness, and food and energy intake has already progressed well, so these priorities are ranked lower in importance.

4.4.5 Radiation Component of Cross-Cutting Decadal Survey Recommendations

Recent work with whole genome sequencing, especially in cancer research on the varied genomic effects of mutagenesis, has led to increased throughput and capacity for genome reconstruction approaches not only for sequencing of whole organisms but also for sequencing of single cells. At the time that the decadal survey review was being carried out, these techniques were cumbersome and it was difficult to realize their application to everyday science approaches. With the advent of new sequencing approaches, utility for space radiation studies has become feasible. Thus it could be useful for NASA to examine new approaches to consideration of mechanisms of DNA repair, and to direct measurement of genomic damage. This research could be particularly important, as mechanisms

[13] L.G. Petersen, J.C. Petersen, M. Andresen, N.H. Secher, and M. Juhler, 2016, Postural influence on intracranial and cerebral perfusion pressure in ambulatory neurosurgical patients. Am. J. Physiol. Regul. Integr. Comp. Physiol. 310(1):R100-R104.

[14] J.S. Lawley, L.G. Petersen, E.J. Howden, S. Sarma, W.K. Cornwell, R. Zhang, L.A. Whitworth, M.A. Williams, and B.D. Levine, 2017, Effect of gravity and microgravity on intracranial pressure, *J. Physiol*. 595(6):2115-2127.

[15] M.B. Stenger, A.R. Hargens, S.A. Dulchavsky, P. Arbeille, R.W. Danielson, D.J. Ebert, K.M. Garcia, et al., 2017, "Fluid Shifts," presentation #17601 at A New Dawn: Enabling Human Space Exploration, NASA Human Research Program Investigators' Workshop on January 23-26, 2017, https://three.jsc.nasa.gov/iws/SRIW-Cvent-Program-2017.pdf.

[16] A.A. Gashev, M.D. Delp, and D.C. Zawieja, 2006, Inhibition of active lymph pump by simulated microgravity in rats. *Am. J. Physiol. Heart Circ. Physiol*. 290(6):H2295-H2308.

[17] A.G. Lee, T.H. Mader, C.R. Gibson, and W. Tarver, 2017, Space flight-associated neuro-ocular syndrome, *JAMA Ophthalmol*. 135(9):992-994.

[18] S.R. Zwart, C.R. Gibson, J.F. Gregory, T.H. Mader, P.J. Stover, S.H. Zeisel, and S.M. Smith, 2017, Astronaut ophthalmic syndrome, *FASEB J*. 31(9):3746-3756.

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of repair of the complex DNA lesions that are more common following HZE (high Z (atomic number) and energy) radiation than low LET (linear energy transfer) radiations may be of relevance to radiation mitigation strategies.

4.4.5.1 Role in Human Exploration

It is well understood that radiation is a very high risk for exploration missions, particularly deep-space missions that are planned for Mars and beyond. Risks from exposure to the HZE radiation that will be encountered in space are poorly understood because most of Earth's environment is shielded from such particle irradiation by the Van Allen belts. Studies of the effects of these qualities of radiation are largely limited to those done in recent years by the space radiation community.

4.4.5.2 Highest-Priority Recommendations for Exploration

Space radiation risks to humans (CC8) highlights a major research topic that is ranked "highest priority" in Table 4.1, a priority that includes a deeper understanding of deep space radiation effects on biological and physical systems. The decadal survey study has detailed some information about risks from such radiation exposures, but recent work by NASA has listed four predominant risks associated with space radiation exposure in the Human Research Program Roadmap: risk of acute (in-flight) and late central nervous system effects from radiation exposure, risks of acute radiation syndromes due to solar particle events (SPEs), risk of cardiovascular disease and other degenerative tissue effects from radiation exposure, and risk from carcinogenesis. Of these four risks, most radiation biologists argue that the risk of acute toxicities is probably not substantial enough to warrant additional studies in this area. Much is known about acute toxicities and mitigation approaches for doses up to 10 Gy have been developed predominantly for cytokine and growth factor stimulation of bone marrow function and bone marrow transplantation. Expected doses on a round-trip journey to Mars are expected to be a maximum of 1.2 Gy, far below the usual threshold for bone marrow syndrome at 2 Gy and therefore of little relevance to Mars space travel.

Other risks however remain significant and worthy of study. Probably the most important is the risk of CNS (central nervous system) effects since acute toxicities (if they truly exist) could limit the functioning of the crew during the mission. Evidence suggests that there may be some acute CNS toxicity (such as cognitive dysfunction) in animals that have been exposed to HZE radiation even at low doses; this work clearly warrants verification and additional study particularly from a mechanisms perspective to assess whether similar effects might occur in humans. This is the highest-priority work because of the possible threat to the mission and because such effects could be "show stoppers." The extrapolation from animals to humans for this work may be especially difficult because cognitive effects in animals are so difficult to compare to those in humans and because neurological function of different species is quite different. Unraveling the effects in animals and then extrapolating to humans may be very challenging.

Among the other risks that are listed for radiation exposure are late-effect toxicities that may affect astronauts after their return to Earth. Like acute CNS damage at low HZE doses, late CNS endpoints are also poorly understood, and while they are not likely to impact the mission, understanding them may help in defining some of the mechanisms underlying early tissue toxicities (as is the case for other late tissue effects with low-LET radiation). Thus, late CNS effects would be an important priority after the mission-threatening toxicities. In addition, the impact on astronauts following their service to NASA should also be considered as important; a full awareness of the risks of space travel (as much as is possible) would be important decision considerations on whether space flight is worth its risks both for humanity as a whole and also for individual astronauts. Considerations of such late effects will need to play a role in decision-making. Finally (and perhaps most importantly) knowing that a risk exists and understanding the mechanism by which tissue damage is induced might provide approaches to mitigation.

Risks for cardiovascular disease (CVD) also appear to be (at present) mostly late tissue toxicities of radiation exposure. Even low doses of low LET radiation have been associated with late life induction of CVD in recent studies of the atomic bomb survivors' data sets. While the radiation qualities are different from those encountered in space, space HZE radiations are more damaging in most cases than low LET radiation and thus could pose an even greater risk than what was experienced by the atomic bomb survivor population. These risks are most likely not life-threatening and thus are similar to the CNS late effects noted above. Nevertheless, as in the CNS effects,

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for CVD effects there are no real mechanisms that have been identified and the exact risks are not fully understood. CVD risks from HZE radiation are even less well understood than those for CNS toxicities and thus warrant study for the same reason as other late effects.

The risk for cancer induction is perhaps one of the most well-studied endpoints in the radiation field, although there are still some unknowns about HZE radiation. In particular, many of the studies that have been done revolve around acute radiation exposures given in a few minutes or a few hours, but space radiation encounters are likely to be of a much lower dose rate and longer duration. For low LET radiation, a lower dose-rate usually implies a lower risk of carcinogenesis, but most of the data with high-LET radiation (such as neutrons or alpha particles, for example) suggest that there is no dose-rate effect or even possibly an inverse dose-rate effect making a long duration exposure equivalent to the same exposure given in a few moments. This aspect of HZE radiation exposure is not well characterized and could pose a significant danger from radiation exposure in space, although again this will be a late effect. The latent period for most solid cancers induced by low LET radiations is 15+ years, and data in animal models suggests that this may be similar for high-LET radiation. As such, radiation-induced cancer incidence is also not a risk that is likely to occur during the course of the mission but years after the astronauts have returned. As such, this risk needs to be understood in more detail particularly with regard to dose-rate effects and it will be important to initiate more work on countermeasures. This work continues to be important but NASA has continued to work in carcinogenesis at a somewhat reduced level while elevating the importance of studies related to CNS risks because (1) cancer is better understood and (2) CNS problems during the mission could compromise completion.

4.4.5.3 Research Platforms and Approaches Relevant to Exploration

Most of the space radiation work is currently done at the Brookhaven National Laboratory's Space Radiation Laboratory. This lab was designed specifically to provide exposure to HZE radiations on Earth while mimicking the types of exposures that might be encountered in space. Single ion and now mixed field types of exposures (similar to what would be encountered in space) are possible in the facility. Most exposures in LEO or other similar platforms will not generate sufficient exposure to HZE radiation because (1) LEO is under the Van Allen belts and thus is protected from the effects of HZE radiation to a large extent and (2) generation of a sufficient dose equivalent to a Mars mission is very difficult in that environment. For these reasons, NASA has put significant effort into developing and maintaining the Brookhaven facility. One current limitation to the facility is the difficulty of doing large animal work there; as CNS effects become more important, use of nonhuman primates and larger animal species with a CNS structure more similar to humans may be necessary. Brookhaven currently has no capacity to handle these large animal studies. It will be important to conduct some aspects of radiation research beyond Earth orbit in order to expose microbes, plants, animals, and people to cosmic radiation similar to that for exploration missions.

4.4.5.4 Recent Advances Relevant to Microgravity Science

Research at DOE and NIAID (National Institute of Allergy and Infectious Diseases) has some overlap with NASA's radiation research so collaborations may be an option for NASA to advance this field further. It is clear from carcinogenesis studies that use of countermeasures to prevent or reduce mutation induction in space might be of value, but other countermeasures such as those for CNS effects and others could reduce space flight radiation risks. NIAID has a strong countermeasures program in part aimed at identifying new protective compounds. Partnerships between NIAID and NASA and some drug companies working to discover and provide these compounds could be important for risk mitigation. Such collaborations could potentially be important in the NASA portfolio.

4.4.6 Fundamental Physical Sciences in Space

The research on complex fluids and soft matter has progressed considerably in recent years. The researchers know now how to prepare colloids with functionalized interfaces in order to control the forces between them and

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to succeed in the elaboration of metamaterials.[19] Elaborate numerical simulations of crystallization/gelation with different types of colloids is throwing light on the protein aggregation process that frequently involve more than two species (cataract for instance).[20] Microgravity experiments will now proceed on excellent grounds.

The study of protein crystallization became easier with available powerful synchrotrons that allow using small, easier to grow crystals. Furthermore, it is now possible to visualize the conformation changes during protein function using nuclear magnetic resonance (NMR). Microgravity studies on proteins therefore no longer focus on growing large crystals, but rather on issues such as the role of aggregation prior to crystallization. These studies should allow researchers to better master the crystallization process and possibly improve the crystal quality.

Foams studies have recently clarified the issue of drainage in the case of small liquid fractions ("dry" foams),[21] but the behavior of "wet" foams remains elusive. Foam coarsening is not well understood. Despite the still limited fundamental knowledge, numerous foam applications have been developed during recent years. Two applications are of interest for space exploration: water purification using flotation and solid foams for light materials.

Quantum physics research gave rise to groundbreaking achievements in recent years. This research was distinguished by a number of Nobel prizes, exemplified by the 1997 cold atoms, 2001 Bose-Einstein condensates, and 2012 quantum particle measurement. Important applications in the fields of computers and telecommunications are foreseen in the coming years

4.4.6.1 Role in Human Exploration

Fundamental physics, by its nature, does not directly "enable" space exploration in the near-term. However, results from the fundamental physics discipline will provide the knowledge and discoveries that will likely "enable" new capabilities (i.e. quantum computing) for space exploration over a longer time horizon.

As described in Section 3.5, in the 2011 decadal survey, the Fundamental Physical Sciences Panel identified four research areas that were especially important in the fundamental physical sciences. The panel prioritized them in order as (FP1) Research on Complex Fluids and Soft Matter, (FP2) Research that Tests and Expands Understanding of the Fundamental Forces and Symmetries of Nature, (FP3) Physics and Applications of Quantum Gases, and (FP4) Investigations of Matter in the Vicinity of Critical Points.

4.4.6.2 Highest-Priority Recommendations for Exploration

The committee, in keeping with the criteria discussed later in the report in Section 5.2, did not identify any of the decadal survey established priorities in Fundamental Physics as a high priority for space exploration. While none of the four research areas apply to the statement of task request to rank exploration relevant research, the midterm committee feels strongly that these FP program elements are quite relevant to the broad interests of NASA, and a discussion of these elements is appropriate for this study. In this context, the program elements are discussed in the priority order FP1, FP3, FP2 and FP4.

For the Soft Matter (FP1) projects, most NASA-financed projects concern crystallization/gelation in colloids. These projects are important, because not only can they shed light on the glass transition but they will help to find ways to produce high-quality metamaterials.

For Expanding Understanding of Fundamental Forces and Symmetries of Nature (FP2), the need is for improved understanding of some of the most fundamental questions in science, many of which are addressed through the lens of precision measurements, both ground- and space-based. While much of the work is ground-based, there is a crucial component of a space-based clock as a component of the ACES (Atomic Clock Ensemble in Space) network of synchronized clocks. The tests of the most fundamental properties of nature almost invariably lead to surprises and critical new understandings of our world. It seems completely appropriate that NASA should

[19] S. Sacanna, W.T.M. Irvine, P.M. Chaikin, and D.J. Pine, 2010, Lock and key colloids, *Nature* 464:575-578.

[20] A. Stradner, G. Foffi, N. Dorsaz, G. Thurston, and P. Schurtenberger, 2007, New insight into cataract formation: Enhanced stability through mutual attraction, *Physical Review Letters* 99.

[21] S.A. Koehler, S. Hilgenfeldt, and H.A. Stone, 2000, A generalized view of foam drainage: Experiment and theory, *Langmuir* 16:6327-6341.

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be a player in this field. It is also likely, as mentioned in FP4, that new developments in this work may have an important effect in quantum information and quantum computing.

For Physics and Applications of Quantum Gases (FP3), it is clear that microgravity studies offer the possibility of unique, groundbreaking scientific results. The pico-Kelvin Bose-Einstein Condensate experiments offer the exciting opportunity to explore a new regime, accessible only in microgravity environment. Development of Cold Atom Laboratory (CAL) and these experiments has been expensive, and will come on-line soon with the expected deployment of CAL. It should be noted that the costly design, construction, evaluation and launch phases of this project have been completed and the CAL payload is scheduled to launch in the Fall of 2017. Research on orbit will not impose demands on the limited ISS crew time.

The priority, Matter in the Vicinity of Critical Points (FP4), is fundamental in nature; it was classified by the committee as "enabled by space exploration" only. The committee noted that the research on critical fluids led to the discovery of the "piston" effect, allowing heat transfer in microgravity through pressure waves (on Earth, heath is transported much more efficiently via convection). This effect allows a much faster heat transfer than diffusion and brings a solution to otherwise difficult technical problems.

4.4.6.3 Research Platforms and Approaches Relevant to Exploration

It is apparent that the ISS will be necessary in order to address the unique microgravity and partial gravity environment posed by space exploration and prior to eventual human flights to Mars or to the Moon. Much of the cold atom science will be carried out in the CAL and specifically the pico Kelvin BEC experiments require the extended microgravity period available on the ISS. By its nature, research in Fundamental Physical Sciences, considered "enabled by" the microgravity environment, cannot be "complete" by the 2024 ISS transition.

Earth facilities such as drop towers, parabolic flights, and sounding rockets are extremely useful for conducting low gravity research. These platforms are accessible and do not demand lengthy and costly development of dedicated infrastructure. The drawback is that the microgravity duration is not long enough to obtain answers for many of the projects directed at the decadal survey priorities. For instance, colloid and protein crystallization/gelation are slow processes, liquid crystal domains and foam bubbles growth take hours, and critical fluids require similarly long temperature equilibration times.

New options may be available through the use of CubeSats[22] and other free-flyers. However, the available volume with respect to experiments within CubeSats is limited with respect to the experiments in this area of science. As a consequence, only very small and simple experiments can be performed. Suborbital systems such as Virgin Galactic and Blue Origin have developed initial infrastructure in support of this area of research. Suborbital opportunities (e.g. sounding rockets, commercial) can offer more experiment volume than free-flyers and CubeSats, and provide attractive options for those experiments that can be accomplished within a few minutes.

Most of the types of projects associated with the CAL depend on extensive testing and evaluation if they are to meet their research goals (as opposed to a demonstration of principles). The earliest transition timeline away from ISS in 2024 will not allow completion of the program necessary to finalize the fundamental physics projects that can be conducted on CAL. If, however, the ISS can be continued until 2028, it will be important to make this decision rapidly in order to avoid waste of opportunities. The committee notes also that the CAL may well be continued through European developments, in the form of a CAL2 instrument.

4.4.6.4 Recent Advances Relevant to Microgravity Science

The 2015 NASA technology roadmaps document[23] focuses on development activities that "expand knowledge and capabilities in aeronautics, science and space." While it is correct to say that fundamental physics in space

[22] National Academies of Sciences, Engineering, and Medicine, 2016, *Achieving Science with CubeSats: Thinking Inside the Box*, The National Academies Press, Washington, D.C.

[23] See NASA Office of the Chief Technologist, "2015 NASA Technology Roadmaps," https://www.nasa.gov/offices/oct/home/roadmaps/index.html, accessed October 5, 2017.

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contributes to this technology development, the prime focus of fundamental physics in space remains science. As stated in the roadmap introduction, "The roadmaps focus on applied research and development activities." Still, there is clear overlap between fundamental physical science and roadmaps TA 5 (Communications, Navigation, and Orbital Debris Tracking and Characterization Systems), TA 6 (Human Health, Life Support, and Habitation Systems), TA 8 (Science Instruments, Observatories, and Sensor Systems), and TA 10 Nanotechnology).

In the future, an emerging opportunity presents itself in the form of space-based, global, fully secure quantum telecommunications and, more generally, quantum information and quantum computing. The Chinese have already launched a satellite to test quantum communication technology.[24] Within NASA, funding for such science appears quite limited, but the potential for both terrestrial and space benefits are very significant.

4.4.7 Applied Physical Sciences in Space

Applied physical sciences are central to many key exploration technologies. Reliable, efficient, and safe life support systems require the scientific knowledge of fluid physics, combustion and material science in confined microgravity environments with limited resources. Many of the design challenges of new exploration technology systems were listed in the decadal survey report. The general progress made in these fields and the need for further progress are discussed below under each discipline.

4.4.7.1 Fluid Physics

4.4.7.1.1 Role in Human Exploration and Future Implementation

The manipulation of fluids is essential to our everyday terrestrial life. From our bodies to our agriculture to our engineered world, gravity plays a usually helpful, ever-present, ever-passive role to collect heavier liquids, particles, and gases below lighter ones. Liquid fuels provide a pertinent example along with power cycle working fluids, coolants, and myriad operations for processing water. The control of such phenomena broadly referred to here as "multiphase fluids transport." Multiphase fluids transport is similarly ubiquitous in the fluid systems in space, but pose critical challenges to the designers of nearly all fluid systems for spacecraft due to the effective near absence of gravity. Such systems include liquid propellants, fuels, cryogens, refrigerants, plant and animal habitats, equipment for water recycling and wastes, and others. Many current systems employed aboard spacecraft (i.e., ISS life support systems) are maintained by regular crew maintenance and routine resupply, but systems that are far more robust are essential and mission-enabling for the further human exploration of space. As discussed in Chapter 3, NASA has made progress to address critical Fluid Physics decadal survey items such as forced multiphase flows and transport (AP1), capillary flows and phenomena (AP2), complex fluids (AP5) and continued awareness of granular materials (AP3) and flow phenomena relevant to dust mitigation (AP4).

4.4.7.1.2 Research Platforms and Approaches Relevant to Exploration

Reduced-gravity fluids research is often pursued at small size scales in terrestrial laboratories, but access to the low-*g* environment is required for thorough investigations at appropriate scales for database collection, benchmarking, and establishment of increased technology readiness level. U.S. and international low-*g* platforms include government, commercial and academic drop towers, low-*g* aircraft, commercial sub-orbitals, and potentially other free flyers, with the International Space Station providing the most routine, most versatile option with a highly relevant low-*g* environment. Fluids experiments developed for ISS usually carry the highest resource commitments, but total cost per data point for certain (i.e., fast-to-flight, automated) ISS experiments can compete with ground-based experimental facilities. It is important to be aware of the capabilities, costs, and availability of the various reduced-gravity platforms for both soliciting and evaluating low-*g* fluid physics research.

[24] See D. Castelvecchi, 2017, China's quantum satellite clears major hurdle on way to ultrasecure communications, *Nature News*, June 15, https://www.nature.com/news/china-s-quantum-satellite-clears-major-hurdle-on-way-to-ultrasecure-communications-1.22142.

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4.4.7.1.3 Recent Advances Relevant to Microgravity Science

Terrestrial academic and commercial advances in experimental and numerical methods continue to ease the burden of developing space experiments by providing off-the-shelf tools for more rapid, lower-cost, experiment design toward flight certification. Unfortunately, the low-*g* environment can also lead to unique fluid behavior requiring designs that require an additional level of research prior to the experiment design process. In order to expand the impact of the experimental and theoretical work to complex system interactions, NASA would need to require that numerical analyses accompany most if not all fluids investigations. Such bench-marked algorithms would serve as foundational design tools for extrapolation to advanced full-scale system development for exploration. Commercial software and specific low-*g* investigator open-software tools may be exploited for the potentially "big data" archive management.

4.4.7.1.4 Application of Fluid Physics Research

The importance of fluid physics to applications in space exploration is reflected by the large number of translational highest-priority listings in Table 13.2 of the decadal survey (TSES1, TSES2, TSES4-S6, TSES12-16). In a cross-cutting manner, due to long term NASA support in this area of research, low-*g* Fluid Physics results are beginning to mature to the point of direct application to next generation spacecraft fluid systems design as highlighted in Section 3.6. For example, for the highest Fluid Physics Fields (decadal survey identifier topics AP1 and AP2), specific applications being addressed concern advanced cryogenic storage analysis and design, multiphase heat transfer by way of pool and flow boiling and condensation, fluid phase flow separators and devices, plant and animal habitats, water processing systems including urine collection and drying, condensing heat exchanger design, advanced passive CO_2 scrubbing methods, and others.

4.4.7.1.5 Highest-Priority Recommendations for Exploration

As indicated in Table 4.1, it remains clear that forced multiphase flows and transport (AP1) and capillary flows and phenomena (AP2) are the most practical and thus highest-priority fluids research areas for NASA's immediate future. These topics might in turn be combined into the single subject of "low-*g* fluid interfacial transport." This is due to the fact that either by design or accident (i.e., local boiling, degassification, aerobic activity, etc.) most fluid systems aboard spacecraft in turn become multiphase systems that are strongly and often surprisingly impacted by the absence of a strong gravitational acceleration. In such common situations, the locations of the gases and/or liquids are uncertain, requiring special knowledge and accommodation. These areas directly impact progress in the development of robust life support systems as well as other cross-cutting fields cited specifically in Table 4.1 (i.e., TSES1, 2, 6, 12-16). But it is also clear that the lack of understanding of low-*g* fluid interfacial transport adds risks to the ability to study nearly all fluids operations relating to most subjects of this report; i.e., Plant and Microbiology, Animal and Human Biology, Fundamental Sciences, and all of the Applied Physical Sciences. It is the committee's view that NASA has done well to maintain flight experiments with this focus despite limited resources (see previous Section 3.6.2 Fluid Physics for discussion of prior work).

The most significant progress expected for applied science research within the fluid mechanics discipline targeting design needs for exploration will most likely come by way of practical stability limits, regime limits, onsets, transitions, and low-*g* benchmarked numerical methods. An historic example concerns phase change heat transport processes of sufficient inertia to assure *g*-independence—in other words, quantification of the limits above which terrestrial and low-*g* flow processes are essentially identical. It is reasonable to expect significant "component-level" progress along these lines by the 2024 time frame. However, it is not certain whether such phenomena may be adequately studied at the "system level" (i.e., entire systems including evaporators, conduits, condensers, filters, valves, manifolds, parallel paths, reservoirs, etc.) and plans do not appear to be in place to pursue such research in the near term. System level stability and interactions are critical to every fluid system aboard spacecraft with special concerns during start-up, shut-down, safing, and transient response to excursions and off-nominal events. Benchmarked methods to model such processes would have to be developed to reduce risks of advanced system performance at an accelerated pace if NASA hopes to deliver significant design products by 2024. The development of such system level fluid physics experiments will likely require more resources than currently available for this physical sciences subdiscipline.

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The behavior of granular materials (AP3) and flow phenomena relevant to dust mitigation (AP4) remain "higher" priorities, but not primarily for low-*g* environments. Barring crewed-asteroid missions, such knowledge is more critical to In Situ Resource Utilization (ISRU) in lunar and Martian environments. Although Complex Fluids (AP5) continues as an active and fruitful "enabled by" field of research on ISS, it is outside of the central thrust of the low-*g* fluid interfacial transport fields (AP1 and 2) and is winding down on ISS. Complex Fluids research is not currently considered high-priority research for exploration.

4.4.7.2 Combustion

4.4.7.2.1 Role in Human Exploration and Future Implementation

Combustion processes are central to chemical propulsion which will power space exploration under the current plan. In addition, fire in microgravity is a serious safety concern. Fire behaves very differently without gravity. In human spacecraft the existence of a low and purely-forced flow, and the possibility of using Normoxic atmosphere (higher oxygen percentage and lower total pressure than air), can alter the material flammability responses from what is known on Earth. Since there is no escape from a spacecraft far from Earth in a fire, it is essential that fire behavior in such an environment is understood.

4.4.7.2.2 Highest-Priority Recommendations for Exploration

Overall, the combustion program directed by NASA has done well, as described in Section 3.6. Progress has been made in all the three high-priority areas listed in the decadal survey, but they are far from completion. These areas will continue to be the highest-priority items for space exploration research during the next phase.

AP-6, material flammability and fire suppression in space, is directly related to spacecraft safety. By using a small flow tunnel in glovebox leftover from a previous project, the BASS (Burning and Suppression of Solids) project aboard ISS for solid combustion has been a great success. Compared to other projects in combustion, the turn-around time was short. The interaction between ground-based scientists and astronauts performing the tests was superb. The experiments produced valuable results on solid burning, flame spread, and extinction that are unique in a long-duration microgravity environment. Many of the findings have already been published in top journals. NASA is encouraged to continue the use of glovebox facility when appropriate despite its limitations (e.g., atmospheric pressure and limited diagnostics).

The next project for solid combustion on ISS is Sofie (Solid Fuel Ignition and Extinction) to be carried out in the Combustion Integrated Rack (CIR) that is better instrumented. The test atmosphere is expanded to include different oxygen percentage and pressure levels that have large influence on combustion behavior. The five PIs will share the same flow tunnel and other instruments to save resources. Each PI's project has different goals, sample configurations and in some cases different solid materials. The current launch date for the Sofie insert is 2020. In order to maintain good progress towards meeting decadal survey priorities that support exploration, it will be important for NASA to meet this date or even to accelerate the schedule if possible.

Saffire (AES Spacecraft Fire Safety Demonstration), a large-scale fire experiment, is not on ISS but related to ISS as Cygnus is a supply vehicle to ISS. As mentioned previously, this is an innovative use of resources and the creation of a new microgravity test platform. Three Saffire tests have been completed. One of the most important results is the experimental verification that flame over solid will reach a limiting size in purely forced flow in microgravity. This is in contrast to fire on Earth where an upward spreading flame normally accelerates. Several Saffire flights are in the planning stage and NASA is encouraged to continue this experimental series. Both Sofie and Saffire will include experiments using the Normoxic atmosphere (same partial oxygen pressure as in air but with higher oxygen percentage and lower total pressure), which is a condition being considered for future space exploration. Related to this is the need to test barrier materials such as Nomex fabric, which has been used in space to prevent ignition and burning. The effectiveness of the barrier material in Normoxic atmosphere need to be determined.

The use of carbon dioxide as a flame suppressant has been studied extensively in FLEX (Flame Extinguishment Experiments) using liquid droplets as the fuel. There is a need to investigate other suppression methods such as water mist on other fire configurations including solid burning in microgravity.

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Decadal survey priority AP-7 refers to combustion processes in microgravity, and while it did not receive the highest possible ranking against exploration criteria in the decadal survey, it plays a broad supporting role that is worth citing here. Study of combustion fundamentals is crucial for understanding the effect of gravity on flames and the validation of combustion and kinetic models. Microgravity also can enable deeper insight into terrestrial combustion processes. There have been plenty of unexpected findings in microgravity experiments conducted since 2011. One example is the appearance of cool flames after the radiation quenching of the hot flame in the FLEX experiments. These cool flames provide excellent data to study the low-temperature chemistry of liquid fuels that is crucial to knock control in internal combustion engines and the determination of flammability limits.

ACME (Advanced Combustion via Microgravity Experiments) is scheduled to be in CIR aboard ISS in 2017. The primary goal is to improve efficiency and reduce pollutant emissions in terrestrial combustion. The secondary goal is fire prevention. ACME has five independent experiments to investigate the burning rates and soot formation in laminar, gaseous, non-premixed flames with and without an electric field.

Decadal survey priority AP-8 refers to the numerical model simulation of microgravity flames and fire in spacecraft. Numerical simulation has become a powerful research tool in physical sciences including combustion. It is indispensable in engineering design, including fire spread prediction in buildings. In microgravity combustion research, two notable examples are the predictions of flame quenching by radiation and the reach of limiting length in spreading flames over solids. Theoretical models often provide clue for doing novel experiments, and experiments can provide data to improve the model. Given the limited flight opportunities and the high expense to perform microgravity experiments, modeling is not only helpful but necessary. There are a few public and commercial computational codes that can be used for routine engineering tasks. But for more specialized cases in microgravity combustion, special efforts are necessary to establish the proper model and to perform the experimental comparison.

The recent initiated NASA-PSI (Physical Sciences Informatics) program that fund studies by utilizing data from finished space experiments is an important way to expand the investment on the space experiments. This could be continued and expanded.

The committee does see the potential for collaboration with other government agencies and private industries in this area of research. The Center for the Advancement of Science in Space (CASIS)-initiated collaboration with the National Science Foundation (NSF) on fluid and combustion research that resulted in new funding to new or existing PIs is to be congratulated. Opportunities for collaboration with other federal agencies including the Department of Defense (DoD) could be exploited. The work with Orbital ATK on the Saffire project is one good example of collaboration with private industry.

As in many areas of microgravity research, international collaboration continues to be important for combustion research. In addition to collaboration with international colleagues on an individual basis through the PIs grant, systematic international team collaboration exists in the FLARE project led by the Japan Aerospace Exploration Agency and in the Saffire project between NASA, the European Space Agency, and scientists from Europe, Russia, Japan, and Australia. Such collaboration will continue to be important in addressing decadal survey priorities.

4.4.7.2.3 Research Platforms for Combustion and Approaches Relevant to Exploration

Access to the long-duration microgravity environment, such as on ISS or supply spacecraft, is needed to carry out solid material combustion and fire tests. Preparatory ground-based tests are normally performed for combustion projects before flight experiments. Tests in normal gravity and partial gravity, if appropriate, are often used to highlight the impact of gravity on flames. The use of drop-tower (and sometime airplanes) for short duration (~10 s) of microgravity tests have been useful in providing transient data and in identifying shortcomings of the design hardware. Numerical simulations of combustion and fire are indispensable in combustion and spacecraft fire safety research. In order to support Mars exploration needs, it will be important to have a long duration, high-quality partial gravity experimental platform for combustion and fire safety studies. Because of the increased human flight time for the Mars mission, the complexity of the combustion phenomena, the diversity of materials employed in space applications and the limited opportunity to perform tests in microgravity, sufficient scientific understanding of the gravitational effects on material flammability is unlikely to be achieved in the next several years. Continued scientific research into the post-2024 era will be required to minimize the risk of fire hazard in space exploration.

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4.4.7.3 Material Science

4.4.7.3.1 Role in Human Exploration, Recent Advances, and Future Implementation

It should be noted that the decadal survey report indicates that "NASA has relied on existing materials," and that "very little fundamental research has been conducted on advanced materials for space exploration." This lack of long-term investment in materials science and technology, which has continued since the decadal survey report was written, may limit progress in future exploration. In addition, there are numerous processes and technologies have been developed or proposed for the support of exploration that require strong focus and emphasis. These include 3D printing and additive manufacturing, smart materials (sensitive to pressure, light temperature, and other external stimuli), smart coatings (nanostructured, super-hydrophilic or hydrophobic), flexible plastic electronics, and self-healing materials. Because these materials have broad applications across many industries, NASA has been able to take advantage of the advances being made by other groups and it will benefit them to continue to do so. At the same time, additional investment by NASA is necessary.

As described in Chapter 3, considering the budget limitations, here has been a strong response by NASA to the materials science priorities in the decadal survey, and several projects have been funded. The findings from this work, combined with the earlier NASA support for microgravity materials studies conducted primarily on space shuttle missions and on the ISS, have established strong foundations on understanding materials behavior and processing in reduced gravity. For example, it has been shown that basic materials processes such as nucleation and growth, phase formation kinetics, interfacial stability and microstructure pattern formation, melt-substrate wetting, phase sedimentation, gas bubble formation, particle pushing, mixing and diffusion, and buoyancy-induced convection, are all significantly altered under reduced gravity conditions. Unfortunately, these findings are still very limited and further emphasis and funding will be necessary to address additional materials processing and behavior in complex systems.

Considerable progress has been also made in modeling materials in recent years. Indeed, materials scientists and engineers have been able to utilize computational methods and numerical techniques to predict materials behavior and/or design new materials. Through these research activities, which have been funded by NASA and other agencies, it has also become obvious that predicting materials behavior even in simple systems is much more complicated compared to their counterparts in structural, mechanical, electrical, and nuclear systems. The modeling and calculation requirements become even more complicated when trying to predict complex materials systems' behavior under unusual environments encountered in extraterrestrial environments. Eventually, however, sheer computational power and reduced computational costs will enable materials practitioners to address the enormously difficult task of designing new materials.

4.4.7.3.2 Highest-Priority Recommendations for Exploration and Research Platforms

Both AP10 (materials that enable operations in space environments) and AP11 (processing materials on extraterrestrial surfaces) were ranked very high against exploration criteria in the decadal survey. This report also ranked them as having the highest importance to exploration because they can provide critically needed reliable materials performance in harsh and demanding environments, such as better crew shielding or protective coating of surfaces to minimize bacterial or algae growth during long occupancy in crew compartments, or extracting and producing materials, return fuel as well as emergency metallurgical repairs such as welding, joining, melting, or casting.

It is obvious that understanding materials behavior during exploration requires systematic and strategic investigations using ground-based facilities, as well as long-term experiments aboard the ISS.

4.4.7.4 Implementing Decadal Survey Recommendations in Applied Physics

Looking toward the remaining years of the current decadal survey, the committee offers the following guidance for the science implementation within NASA's Low-*g* Applied Physical Science Program. This guidance is presented in the context that recognizes that NASA has done well to deliver an applied low-*g* fluid science program despite a clearly limiting funding level. The committee also acknowledges the quality and relevancy of the current suite of conducted and to-be-conducted flight experiments.

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Like most of the science disciplines in the decadal survey, low-*g* Applied Physical Sciences benefit from both "enabling" and "enabled by" research in order to fully exploit the finite duration of the ideally suited ISS, as well as to continue to draw new investigators and investigations into the field. Applied physical sciences also benefits from connections (intra-agency, government, academia, industry, international and even nonspace-faring nations) to identify priority studies. Fundamental and Applied Physical Science includes projects that are either new or were not necessarily emphasized by the 2011 decadal survey report, such as modern heat pipe research and low-*g* water transport phenomena relevant to robust long-term plant watering methods and systems. Applied physical sciences also benefits from scientists incentivized to reach outside of the Division of Space Life and Physical Sciences Research and Applications (SLPSRA) to other NASA groups to speed maturation of science and engineering knowledge to the level of applications (i.e., advance ZBOT knowledge to the technology demonstration stage with impacts to current applications as well as advance systems design).

Fluid physics is a prime example of how government, commercial, and academic platforms (U.S./international, terrestrial, ground-based, drop tower, aircraft, commercial suborbitals, ISS, and other free flyers) can be exploited for moving science forward through the fostering of connected, multiuser studies. These studies would be performed using the following approaches: with and without hardware adaption, re-runs, data mining to establish "depth" of science return, broadened research base, maximized investment, and pre-during-post experiment theory using numerical analysis.

4.4.8 Translation to Space Exploration Systems

Looking at the SLPSRA projects identified by NASA as relating to the translational decadal survey recommendations, there has been good progress in the areas of fire safety, cryogenic fluid management, and energy conversion research.

In relation to the technology roadmaps of NASA, new materials such as photonic materials are mentioned. These materials will clearly be an output of some of the NASA projects on colloids. More generally, the research on colloids is important for the elaboration of nanomaterials, nano-enhanced composites (TA10 1), nanostructured (TA10 2), or nano-sized materials (TA10 1.1). Clearly underrepresented is the research on granular materials (AP3), including, addressing dust problems (TA07 2). There are seemingly no projects in plasma research in the works, although this topic could be quite relevant to space exploration.

Other interesting topics are smart materials and materials that are sensitive to light, pressure, etc., or multifunctional (TA10-1) and self-repairing materials (TA7 4.2), which are frequently processed starting from soft materials. Omnipresent is the requirement for lightweight materials. In Europe, there is an ESA project on polymers and metallic foam in microgravity. Indeed, foams might be interesting to produce structures for space and even fabricated in situ in space.

Despite this success, it is important that more projects determine their translational research path to exploration systems as this enabling research receives more attention with NASA's increased focus on exploration. Thus, in the future it would be useful for NASA to identify far more projects that have a reference to this portion of the decadal survey (anecdotally, the committee knew of some projects in other NASA programs that fit this description) and to make sure these research projects have clear traceability to NASA's technologies and missions.

As identified in the decadal survey, it is important that NASA continue to focus efforts on transitioning basic research into technology to meet mission needs. Driving toward an improved partnership between scientists, engineers, and program managers who all have important roles in the process is critical. More attention will need to be given to understanding how to accomplish this transition within NASA and systematically track this process. Navigating the gap between research studies and application has been commonly referred to as the "valley of death." NASA must determine ways to better bridge this valley of death. This is not an uncommon problem, and the committee is aware that other industries are struggling with the same challenges, but having some success. In the medical field, there are several examples of connecting patients to the research as end users and researching and charting that path. NASA could attempt to do the same between mission program managers and research scientists, so that connections are clear and places where the process breaks down is well understood. Simply identifying these pathways by improved traceability would be a good first step.

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4.4.8.1 Role in Human Exploration

The 16 TSES priorities are all critical to future human exploration, by definition. But since the NASA Technology Roadmaps work that has been performed since the publication of this decadal survey, it is probably merited that this list of 16 be revisited and tied more closely to the technology prioritization performed by the National Academies and NASA itself.

4.5 SUMMARY OF SCIENCE AND TOP PRIORITIES

In this chapter, the decadal survey science recommendations were examined by the committee within the dual contexts of the 2011 decadal survey and the current NASA exploration strategies. The specific highest-priority recommendations of the committee, as identified in the decadal survey and compared to NASA exploration needs, are presented earlier in this chapter in Table 4.1. After reviewing various issues relevant to the recommendations for each of the discipline areas of space life and physical sciences, the committee assessment produced two additional findings.

Finding 4-2: The current funding levels are insufficient to fully address the significant unknowns and risks of human exploration beyond LEO. Fundamental understanding of human health and behavior risks in microgravity, combined with fundamental microgravity physics and materials, in an integrated manner, is essential to extending the human neighborhood beyond LEO. Significant risks remain, particularly in understanding the radiation environment and its effects, environmental control and life support, human behavior, and protecting long-term crew health with integrated countermeasures. These risks are best addressed in the respective disciplines and in an increasingly integrated fashion.

Finding 4-3: NASA has pursued means to improve internal cross-organizational efforts across the research and technology development landscape, based on the linkages with the Technology Roadmaps.

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5

Recommendations for Implementing the Decadal Portfolio Over Remaining Years Within Constraints

In this chapter, the committee provides a series of recommendations based on its assessment of the National Aeronautics and Space Administration's (NASA's) implementation of programs in space life and physical sciences pertaining to the 2011 decadal survey. In keeping with the statement of task, the committee presents this chapter to address recommendations for implementing the recommended portfolio during the remaining years of the decadal survey and in preparation for the next decadal survey, while recognizing resource and program constraints. The committee believes that these recommendations will be useful not only to NASA, but to the broader explorations community that now includes other agencies as well as commercial and international partners. These recommendations may also guide future advisory committees that will engage in the next decadal survey. However, in large part based on this assessment, the committee fully understands that the remaining years of the present decadal survey will continue to be years of rapid evolution as NASA better defines its exploration goals and the ecosystem of interests in low Earth orbit (LEO) matures—making evaluation of these recommendations a very important activity.

As in any effort as complex and important as space exploration, especially in the current context of rapid evolution, there is ample opportunity for continued assessment and resultant improvement. That said, the committee bases these recommendations fundamentally upon the positive responses by NASA to the decadal survey. In fact, Findings 1 and 2 in Chapter 2 specifically recognize the positive changes enacted by NASA. The committee believes, based on the remaining findings of the assessment presented in Chapters 2 through 5, that addressing these recommendations will provide the best possible exploration research within the existing strategic, programmatic, and funding constraints.

The recommendations are organized in three phases: (1) recommendations to address the current status and challenges, (2) recommendations to complete the science effort toward the 2011 decadal survey, and (3) recommendations addressing the pathways to exploration and the International Space Station (ISS) beyond 2024. Each recommendation is meant to stand on its own, be addressed individually, and just as importantly be addressed as part of the integrated set of recommendations.

Specific recommendations are provided to address the findings documented previously in this report, as well as other considerations needed for the portfolio implementation.

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5.1 CURRENT STATUS AND CHALLENGES THAT CONSTRAIN RESOURCES

Programmatic challenges arise within the Division of Space Life and Physical Sciences Research and Applications (SLPSRA) largely due to budgetary restrictions. Besides the obvious need for greater research funding overall, toward something similar to the 2004 levels (Chapter 2), the decadal survey offered programmatic recommendations that bear attention in the current assessment.

The major operational constraints to progress in space life and physical sciences inevitably focused on the ISS. The committee recognizes that the decadal survey calls out multiple platforms capable of delivering science goals of the portfolio. However the funding lines for SLPSRA and the intense focus on maximizing use of the ISS drive much of the research emphasis toward the ISS. This drive creates a series of limitations that beg solutions. Given the current funding, legislative and organizational constraints, and increased research pressures from the Center for the Advancement of Science in Space (CASIS) and other government agencies, careful attention is needed to maximize the opportunity exploration related potential research results.

5.1.1 Create Regular Requests for Research Proposals in Order to Enhance Science Capability

Recommendation 5-1: NASA should recognize the need for regular requests for research proposals, in order to keep an active external research community available to do exploration-related space life and physical sciences research.

Regular requests for proposals foster a vibrant research enterprise that can respond to NASA needs. Regular requests are the only way that university science can be expected to grow the next generation of scientists. Low and inconsistent funding leads researchers to pursue avenues of research outside of space exploration.

5.1.2 Maximize Use of Crew Time for Decadal Survey Research

Recommendation 5-2: NASA should continue and increase its efforts to maximize International Space Station (ISS) resource synergy across the ISS National Laboratory, international partners, and the Division of Space Life and Physical Sciences Research and Applications, particularly with regards to crew time availability and research priority on the ISS. Continued efforts to increase cargo and crew transport to and from the ISS should be expedited as much as possible.

The limitations driven by the available crew time, upmass, and downmass can be mitigated with synergistic approaches to the detailed research implementation. Significant progress and effort has been made by SLPSRA, CASIS, and other government agencies recognizing these needs and the limited resources. The Committee encourages this continued effort.

5.1.3 Significantly Improve Research Result Reporting and Integration

Continued efforts to expand and improve research reporting and management of open databases for experiment data is essential for program planning in the remaining years of the ISS and beyond.

Recommendation 5-3: NASA should consider decadal survey priority tracking integration within Agency elements and utilize existing, commercially available, well-known research reporting and open-science database tools that are in use across the academic research spectrum for accurate, timely, and sustainable information. NASA should also make a determined effort to build on the significant improvements in the International Space Station program for communicating the value of the investigations.

As assessed in Chapter 2, NASA recognizes the value in tracking research projects as they apply to decadal survey recommendations. However, current reporting within the agency varies among the divisions and makes it

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difficult to provide high-resolution mapping of projects and results as they relate to decadal survey goals. This creates challenges as NASA develops programs to procure science and works to build advocacy for the research needed.

5.1.4 Strengthen Relationships with Other Agencies and International Partners

Recommendation 5-4: Relationships with the National Institutes of Health, the National Institute of Standards and Technology, the National Science Foundation, the Department of Defense, the Department of Energy, and other agencies should be strengthened to better address the decadal survey and midterm review identified research priorities, especially exploration priorities. NASA should consider negotiation with the Center for the Advancement of Science in Space regarding International Space Station research allocations to better address NASA's exploration priorities.

The unique international construct of the ISS, together with the rise of the ISS National Laboratory, creates certain opportunities for stakeholders in parallel with its operational constraints. In particular, the ISS National Laboratory and its CASIS management, as discussed in Chapter 2, create an opportunity for other governmental agencies and private industry to experience the value of microgravity research, and that research inevitably feeds progress toward exploration. Research synergies with these other entities, as well as the established ISS international partners will positively affect exploration research outcomes.

A structured approach is needed to identify areas of potential partnerships, potential synergistic approaches and communication of the results. The NASA Authorization Act of 2010 (P.L. 111-267) enables NASA to negotiate with CASIS if NASA needs greater than 50 percent for exploration research. The committee recognizes that expectations of substantial funding support from other agencies are unrealistic given the highly constrained current and future science funding environment. Efforts to maintain and grow international partnerships with the European Space Agency, Japan, Russia, and other countries will further enhance research opportunities and capabilities.

5.2 MAXIMUM SCIENCE PROGRESS FOR THE NEXT DECADAL SURVEY

Again referring to the dynamic evolution of space utilization that is presently occurring, and recognizing the need to maximize research returns before the transitions of ISS potentially beginning in 2024, the committee has the following recommendations to maximize the progress prior to the next decadal survey. These recommendations are aimed primarily at improved coordination and use of available research platforms, while assuring that exploration priorities still consider the fundamental sciences. In particular, the committee recognizes the strong need to properly communicate and advocate within the federal budget process for microgravity research. The committee believes that these recommendations, if adequately addressed, will greatly aid the budget advocacy process and bring in more stakeholders to benefit from the research. The use of the various research platforms will help address the research needs as expeditiously as possible, and provide impetus for additional players or stakeholders in the execution and use of the research results.

The programmatic findings of Chapter 2 are brought together with the science findings of Chapters 3 and 4 to guide these recommendations that span science and program areas.

5.2.1 Traceability of Exploration Strategy to Research Priorities

Recommendation 5-5: NASA should establish and document traceability of the research priorities to its Technology Roadmaps, Design Reference Missions, and exploration strategy.

This increased traceability would lead to clear selection criteria for research investigations in support of exploration. These criteria, selection decisions, and outcomes would then have clear, direct linkage to the exploration and research needs, and inform key decision points in the overall strategic approach. Based on the negative impacts of frequent but irregular changes in exploration strategy, NASA's periodic refining of the Design Reference Missions concept based on a regular review process would enhance this traceability.

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5.2.2 Cross-Organizational Efforts

Recommendation 5-6: NASA should further balance communication and reporting efforts across the organization.

The disparate budget reporting, investigation and research reporting, and the unclear reporting authority of the Human Research Program relative to SLPSRA highlight the need for improved organizational integration of communication and reporting activities. Improving the organizational integration will enhance NASA's ability to advocate and justify the investments in microgravity research.

5.2.3 Direct Larger Balance of Budget to Science Research

Recommendation 5-7: NASA should direct an increasingly higher priority toward the conduct of science within existing International Space Station (ISS) hardware and research capabilities. Utilization of existing, including privately developed, ISS facilities should be maximized in recognition of the current funding limits, the ISS transition timeline, and the need for high-priority microgravity research.

5.2.4 Proactively Use Full Range of Platforms

Recommendation 5-8: In order to maximize the implementation of decadal survey priorities within its constrained resources, NASA should continue to be mindful of the full range of platforms (including drop towers, aircraft, balloons, suborbital vehicles, and free-flyers) and terrestrial analogs and ground-based laboratories available for decadal survey research.

5.3 PATH TO EXPLORATION AND BEYOND 2024

Human space exploration currently depends on the ISS for research and for developing long-term space exploration experience. Within the human exploration endeavor, there are two overarching issues affecting the overall research prioritization and funding. The first issue is NASA's exploration strategy has experienced significant changes since the 2004-2005 timeframe. This includes changes from Constellation to SLS/Orion, the Asteroid Redirect Mission, the Journey to Mars, and the current Deep Space Gateway. These variations create difficulty in mapping science to specific mission needs, yet the findings and the science in Chapters 3 and 4 suggest that any of these missions and mechanisms require significant space life and physical science research for long-term survival of humans in space. However, the finer prioritization required to develop specific science in support of these potential missions respond better in the presence of a clearer exploration roadmap.

The second major open issue is the strategy for the ISS and the U.S. National Laboratory in the post-2024 timeframe. The International partners have all committed to funding their ISS partnerships through 2024, however, there is currently no strategy for ISS in the post-2024 timeframe. It is imperative that NASA develop this strategy for the ISS or other orbital platforms for research as soon as possible in order to provide a basis for planning and prioritization. The strategy must be both realistic and executable.

The following recommendations address the need for an exploration strategy and concomitant need for an LEO post-2024 strategy. These recommendations should be addressed in an integrated fashion.

5.3.1 Provide Additional Funding to Address Significant Risks and Unknowns for Human Exploration

Recommendation 5-9: In light of the resource constraints, NASA should raise the priority of Space Life and Physical Sciences Research and Applications Division research within the International Space Station (ISS) to address the risks and unknowns of human space exploration, particularly given the value of microgravity research for exploration and the urgency resulting from the potential transition of the ISS. These priorities should be directly traceable to the space exploration strategy, linked research priorities, and related technologies.

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5.3.2 Expeditiously Develop LEO Post 2024 Strategy

Recommendation 5-10: It is essential that NASA as quickly as possible develop an International Space Station post-2024 strategy. This development factors strongly in the overall exploration strategy, space life and physical sciences research priorities, and resource allocation in terms of crew time, cargo delivery, and funding. This post-2024 strategy should address clear cost allocation among the various research activities and partners.

While NASA has stated they plan to address this issue by the end of 2017, it is essential that this be completed as soon as possible as it drives overall strategic direction, key programmatic decisions, key decision criteria, and funding requests and allocation. In the context of this strategy, realistic expectations for transition to commercial operations will need to be considered.

5.3.3 Develop the Identified High-Priority Science Research Areas

Recommendation 5-11: **NASA should aggressively lead in the 46 research priorities for deep space exploration identified in Table 4.1 of this midterm report to provide as much "pull" as possible for exploration enhancement using space life and physical sciences. NASA should, for example, lead in the development of microgravity-adapted biological and physical systems, making maximum use of all available platforms, including the International Space Station, specifically for the science behind the design and implementation of microgravity-optimized operation.**

5.4 SUMMARY: ADDRESSING EXPLORATION AND BASIC SCIENCE

The recommendations presented here were developed in direct response to the statement of task, taking each specific task element as a concept for gathering data to inform an assessment. Each of those element assessments interrelates with other element assessments to inform the recommendations. These recommendations are offered to NASA to meet the exploration needs and to maximize the overall research benefit. The committee emphasizes the need to address these recommendations in an integrated manner. The interrelationships are extremely important and must be addressed.

In the larger, more overarching appreciation for how science progresses, NASA has done a credible job of including fundamental research in the funding portfolio. Given the budget and research constraints, it would be easy to focus only on those investigations viewed as directly "enabling" human exploration. Fundamental physics and research that is labeled "enabled by" needs to be carefully considered and included where appropriate, as evidenced by fundamental physics investigations such as the Cold Atom Lab (CAL) and the Alpha Magnetic Spectrometer (AMS). These experiments are fully dependent on microgravity and the space environment, and can lead to game changing potential for science and as yet unappreciated future applications. Also, the "enabled by" research has been carefully considered, as this research can become the bedrock for healthier, more efficient, and less costly exploration approaches in the future.

The committee recognizes an underlying structural conundrum, one that seemingly places NASA exploration goals against the basic science of the space and life sciences portfolio. On one hand, the committee recognizes the need for NASA to focus on exploration, and the statement of task specifically limits the assessment to those priorities that enable deep space exploration. On the other hand, the committee also recognizes the large expense that NASA has put into the fundamental sciences such as CAL, decidedly nonexploration parts of the portfolio. The conundrum arises when balancing the costs of such basic research against the exploration goals in the current NASA emphasis. Yet with a true NASA focus on deep space exploration NASA could prioritize only projects that enable that deep space exploration, at the expense of pure basic research.

In summary, NASA, along with its partners, has done a commendable job of addressing the decadal survey priorities within the imposed constraints. It will be important for NASA to continue to build the key partnerships, continue to evolve and refine the exploration strategy, and maintain the focus on the "enabling" research with

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careful consideration and implementation of fundamental physics and "enabled by" research needed for long-term exploration.

The committee applauds NASA's efforts to use space and the ISS for both science research aligned with science for exploration needs and science conducted for the sake of fundamental science research. However, the committee believes that a judicious approach is needed when shifting NASA research more toward those types of experiments necessary for deep space exploration, so as to reap the benefits of important basic experiments already started, that may in the long term have game-changing effects on space science. Fundamental science research answers key questions for hardware designers and developers and provides the knowledge that can identify disruptive and innovative technologies that can dramatically increase exploration capabilities or reduce costs.

Recommendation 5-12: The committee recommends that a cautious approach be used when shifting the NASA research portfolio more toward those types of experiments necessary for deep space exploration, so as to maintain the benefits of important basic experiments, especially those uniquely enabled by International Space Station microgravity and already in progress, which may in the long term have the potential for major impacts in fundamental physical science.

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Appendixes

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A

Statement of Task

The National Academies of Sciences, Engineering, and Medicine shall convene an ad hoc committee to review the progress of NASA's Space Life and Physical Sciences Research and Application program in addressing the strategies, goals, and priorities outlined in the 2011 NRC decadal survey report, *Recapturing a Future for Space Exploration, Life and Physical Sciences Research for a New Era*.

1. The committee's assessment will describe:
   a. NASA's approach to and progress on realizing the strategies, goals, and priorities outlined in the 2011 decadal survey and other relevant Academies reports;
   b. Existing and emerging challenges to and opportunities for the implementation of decadal strategies, goals, and priorities, including:
      - Plans for the International Space Station and the long-term future for research infrastructure in low Earth orbit (LEO)—including the provision of privately-owned and -operated infrastructure;
      - Resource and program constraints,
      - The opportunities and challenges of a multisponsor (NASA, Center for the Advancement of Science in Space, other government agencies, commercial) science procurement approach using Reference Experiments and Open Databases; and
      - Developments in the research community and its engagement with NASA, CASIS, and other research sponsors.
   c. While considering current and forecasted resources, actions that could be taken to optimize the science value of the program in the context of enabling the expansion of human exploration into deep space. In this regard:
      - The committee will identify and rank from among the "highest priority recommendations" in the decadal survey a set of targeted research priorities which are critical for NASA to conduct in order to enable the expansion of human exploration into deep space.
      - To the greatest extent possible, the committee will also identify the platform environment where the priority research should take place in order to be successful—such as: Earth analogs, LEO (on ISS through 2024), LEO (on other platforms), beyond Earth orbit (such as, a cislunar habitat).
      - In identifying and ranking its targeted research priorities, the committee will consider the full set of "highest priority recommendations" in the decadal survey in conjunction with relevant exploration

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technology development priorities identified in the NRC report *NASA Space Technology Roadmaps and Priorities: Restoring NASA's Technological Edge and Paving the Way for a New Era in Space*. The committee will also consider NASA's deep space exploration goals and priorities as presented by the agency.

- Using the remaining "highest priority recommendations" from the decadal survey, the committee will identify where possible those that are aligned with ongoing activities in other government agencies and, where feasible, the commercial/private sector.

2. The committee will provide guidance about implementation of the recommended portfolio for the remaining years of the current decadal survey, recognizing likely resource and program constraints.

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B

Meeting Agendas

MEETING 1: FEBRUARY 7-9, 2017
NATIONAL ACADEMY OF SCIENCES, WASHINGTON, D.C.

February 7, 2017

Executive Session (8:00 a.m. to 12:00 p.m.)

Open Session

12:00 p.m.	Working Lunch	
1:00	Division of Space Life and Physical Sciences Research and Applications (SLPSRA) Program Overview and Directions —SLPSRA organizational overview —SLPSRA and the decadal—Past, present, and future directions —Midterm task discussion	*Craig Kundrot, NASA*
2:00	Physical Sciences Research and Accomplishments —Program review —Q&A	*Francis Chiaramonte, NASA*
3:00	Break	
3:15	Biological Sciences Research and Accomplishments —Program review —Q&A	*David Tomko, NASA*
4:15	Human Research and Accomplishments —Program review —Q&A	*Steve Davison, NASA*

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5:15	Fundamental Physics Research and Accomplishments —Program review —Q&A	*Mark Lee, NASA*

February 8, 2017

Open Session

9:00 a.m.	Perspective on Role of SLPSRA in NASA —Organizational connections —Support and science roles —Potential future roles —Q&A	*Gale Allen, NASA*
10:00	ISS Research: Past, Present, and Future —Research highlights —Understanding progress —Capabilities —Q&A	*Julie Robinson, NASA*
11:00	Break	
11:30	CASIS Role in Microgravity Research —History and development (legislation, role, and relationship with NASA microgravity and ISS programs) —Research portfolio —Relationship to decadal —Q&A	*Michael Roberts, CASIS*
12:30 p.m.	Working Lunch	

Executive Session (1:30 to 5:00 p.m.)

February 9, 2017

Executive Session (8:00 a.m. to 12:00 p.m.)

MEETING 2 AND COMMUNITY INPUT COLLOQUIUM
APRIL 18-20, 2017
NATIONAL ACADEMY OF SCIENCES, WASHINGTON, D.C.

April 18, 2017

Executive Session (7:30 to 10:30 a.m.)

Open Session

10:45 a.m.	Presentation on NASA Data Request #1	*Craig Kundrot, NASA*

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11:30	Microgravity Synergies with Non-NASA Agencies *Kris Kandarpa, NIH* *Richard E. Ricker, NIST* *Vyacheslav Lukin, NSF*	
12:30 p.m.	Working Lunch	

Executive Session (1:30 to 5:30 p.m.)

April 19, 2017

Microgravity Research at NASA: Community Input Colloquium

Open Session

8:30 a.m.	Integrated Exploration Architecture and Science	*Gale Allen, NASA*
9:00	Session 1: Radiation Panelists: *Francis A. Cucinotta, University of Nevada, Las Vegas* *Kathryn Held, Massachusetts General Hospital/Harvard Medical School* *Charles Limoli, University of California*	Moderator *Gayle Woloschak*
10:00	Break	
10:30	Session 2: Biological Adaptation to Spaceflight Panelists: *Charles Fuller, University of California, Davis* *Cheryl Nickerson, Arizona State University* *Anna-Lisa Paul, University of Florida*	Moderator *Elliot Meyerowitz*
11:30	Session 3: Materials and Combustion Panelists: *Martin Glicksman, Florida Institute of Technology* *Carlos Fernandez-Pello, University of California, Berkeley* *David Urban, NASA*	Moderator *Reza Abbaschian*
12:30 p.m.	Working Lunch	
1:30	Session 4: Fluid Physics and Complex Fluids Panelists: *Molly Anderson, NASA* *Mark J. McCready, University of Notre Dame* *Gretar Tryggvason, University of Notre Dame*	Moderator *Mark Weislogel*
2:30	Break	

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3:00	Session 5: Fundamental Physics Panelists: *Brian DeMarco, University of Illinois* *Eric Cornell, University of Colorado, Boulder* *Ronald Walsworth, Harvard University*	Moderator *Carl Lineberger*
4:00	Session 6: Behavior and Countermeasures Panelists: *Andrew Feinberg, Johns Hopkins University* *Stephen Zaccaro, George Mason University* *Peter Norsk, Baylor College of Medicine*	Moderator *Gloria Leon*
5:00	Wrap-up Discussion and Remarks from Chairs	

April 20, 2017

Executive Session

MEETING 3: JUNE 20-22, 2017
J. ERIK JONNSON CONFERENCE CENTER, WOODS HOLE, MASSACHUSETTS

June 20, 2017

Executive Session (9:00 a.m. to 2:30 p.m.)

Open Session

2:30 p.m.	ISS Experiments Data Request Presentation	*Julie Robinson, NASA* Via WebEx
3:30	Break	
3:45	Developing an ISS Strategy for Beyond 2024	*Robyn Gatens, NASA* Via WebEx

Executive Session (4:30 to 6:00 p.m.)

June 21-22, 2017

Executive Sessions

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C

Biographies of Committee Members, Consultant, and Staff

DANIEL L. DUMBACHER, *Co-Chair,* is a professor of engineering practice in the School of Aeronautics and Astronautics at Purdue University. At Purdue, Mr. Dumbacher specializes in program and project management, complex, state-of-the-art systems engineering, and propulsion and power systems engineering. He has been instrumental in the development of the Purdue Systems Collaboratory, an effort to integrate better the engineering disciplines with business, political science, communications, and philosophy. Mr. Dumbacher began his career at NASA in 1981 where he served in numerous engineering and management roles for a wide variety of missions. Prior to Purdue, Mr. Dumbacher served as deputy associate administrator in the Exploration Systems Development Division for the Human Exploration and Operations Mission Directorate at NASA Headquarters. Prior to this, he served as director of the Engineering Directorate at NASA's Marshall Space Flight Center, leading 1,400 civil service and 1,200 contractor employees supporting the Space Shuttle propulsion elements, Ares I and Ares V launch vehicle development, Earth science payloads, planetary science payloads, lunar science payloads, International Space Station (ISS) global science operations, and ISS environmental and life support activities. He has been awarded NASA's Silver Snoopy Award, the NASA Distinguished Service Medal, and the Presidential Rank Award for Meritorious service, and Purdue's W.A. Gustafson Outstanding Teacher Award, among others. He earned his M.B.A. from the University of Alabama and completed the Senior Managers in Government study program at Harvard University. Mr. Dumbacher has authored several publications and given numerous presentations, in addition to offering congressional testimony.

ROBERT J. FERL, *Co-Chair,* is a professor and director of the Interdisciplinary Center for Biotechnology Research at the University of Florida (UF). Dr. Ferl's research agenda includes analysis of the fundamental biological processes involved in plant adaptations to environments, with an emphasis on the particular environments and opportunities presented by the space exploration life sciences. He is an expert in the area of plant gene responses and adaptations to environmental stresses and the signal transduction processes that control environmental responses. The fundamental issues driving his research program include the recognition of environmental stress, the signal-transduction mechanisms that convert the recognition of stress into biochemical activity, and the gene activation that ultimately leads to response and adaptation to environmental stress. Most recently, these studies have led to the examination of protein interactions as fundamental mechanisms for metabolic regulation of plant biochemistry. His work has also looked at the application of basic science to the questions of advanced life support for NASA's exploration initiative. Dr. Ferl's work has been funded by NASA, the National Science Foundation (NSF), and

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the U.S. Department of Agriculture (USDA), and he has frequently been asked to serve as a scientific adviser and reviewer for national agencies, including NASA, NSF, and USDA. He serves as the director of the virtual center for Exploration Life Sciences, a joint academic research and education venture between UF/Institute of Food and Agricultural Sciences (IFAS) and NASA Kennedy Space Center (KSC). He has also served on the Science Council of the Division of Space Life Sciences of the Universities Space Research Association. In 2016, Dr. Ferl was presented the American Institute of Aeronautics and Astronautics (AIAA) 2016 Jeffries Aerospace Medicine and Life Sciences Research Award and the NASA Exceptional Scientific Achievement Medal. He received his Ph.D. in biology from Indiana University. For the National Academies of Sciences, Engineering, and Medicine, he has served on the Plant and Microbial Biology Panel of the 2011 Decadal Survey on Biological and Physical Sciences in Space and currently serves as the co-chair of the Committee on Biological and Physical Sciences in Space, a standing committee of the Space Studies Board (SSB) and the Aeronautics and Space Engineering Board (ASEB).

REZA ABBASCHIAN is the William R. Johnson, Jr. Family Professor and Distinguished Professor of Engineering at the University of California, Riverside (UCR). He has served as dean of the Bourns College of Engineering from 2005 to 2016. The emphasis of his research is the fundamental understanding of the role of interfaces on the processing and/or properties of material. Current projects involve investigations in solidification, high-pressure, high-temperature growth of diamond crystals, and electromagnetic levitation processing of alloys. Prior to his appointment at UCR, Dr. Abbaschian was the Vladimir A. Grodsky Professor of Materials Science and Engineering Department at the University of Florida, where he also served as chair of the department for 16 years. His awards include the TMS (The Minerals, Metals, & Materials Society) Educator Award, the Structural Material Division's Distinguished Scientist/Engineer Award, the TMS Leadership Award, the American Society for Engineering Education (ASEE) Donald E. Marlowe Award, the Davis Productivity Award of the State of Florida, the ASM Albert Sauveur Achievement Award, and AIME honorary member. He is a fellow of TMS, ASM, and AAAS. Dr. Abbaschian received his Ph.D. in materials science and engineering from the University of California, Berkeley.

ALAN R. HARGENS is a professor of orthopaedic surgery and the director of the Clinical Physiology Laboratory at the University of California, San Diego (UCSD) Medical Center. His recent research concerns gravity effects on the cardiovascular and musculoskeletal systems of humans and animals. He also investigates exercise devices to maintain astronaut health and performance in microgravity and partial gravities. This research is translated to aid post-surgical treatment and rehabilitation of orthopaedic patients and to improve performance of athletes. Dr. Hargens has edited eight books and published more than 300 peer-reviewed articles and 56 chapters. He also holds eight patents. Prior to joining UCSD, Dr. Hargens was the senior research physiologist (1996-2000) and a space station project scientist (1988-1997) at NASA Ames Research Center. He has received numerous academic awards and professional honors, including the 2008 Recognition Award from the American Physiology Society History Group, the Elizabeth Winston Lanier Award from the American Academy of Orthopaedic Surgeons and Orthopaedic Research Society, the Outstanding Performance Award from NASA, and the Nello Pace Award for Outstanding Leadership in and Contributions to the Field of Gravitational Physiology. He received his Ph.D. from the Scripps Institution of Oceanography at UCSD. He served on the Integrative and Translational Research for the Human System Panel of the 2011 Decadal Survey on Biological and Physical Sciences in Space of the National Academies.

YIGUANG JU is the Robert Porter Patterson Professor in Mechanical and Aerospace Engineering and director of the Sustainable Energy Program at Princeton University. Dr. Ju's research interests include energy, combustion, and propulsion in the areas of near limit combustion, microscale combustion, plasma assisted propulsion, alternative fuels, chemical kinetics, multiscale modeling, and functional nano-materials. Previously, Dr. Ju was the Changjiang Professor and director of the Thermophysics Institute at Tsinghua University. He has received a number of awards, including the Distinguished Paper Award from the International Symposium on Combustion (2011, 2015), the NASA Director's Certificate of Appreciation award (2011), and the Friedrich Wilhelm Bessel Research Award by the Alexander von Humboldt Foundation (2011). He has published more than 180 refereed journal articles. He is the president elect of the Eastern State Section of Combustion Institute (International), an

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American Society of Mechanical Engineers (ASME) fellow, and an associate editor for *AIAA Journal*, *Proceedings of the Combustion Institute*, and *Frontiers in Energy*. He earned his Ph.D. in mechanical and aerospace engineering from Tohoku University in 1994.

DOMINIQUE LANGEVIN is the research director at the Laboratoire de Physique des Solides of the University of Paris-Sud. Dr. Langevin works in the field of soft matter, with contribution to systems such as liquid crystals, polymer solutions, colloidal dispersions, microemulsions, emulsions and foams. She is familiar with a variety of experimental techniques, among them scattering of radiation (light, X ray, neutrons), microscopy (optical, atomic force microscopy) and rheology. Previously, she was director of Centre de Recherche Paul Pascal in Bordeaux, one of the largest laboratories in colloid science, where she created a surface of fluids group. Dr. Langevin is a member of Academia Europea and has received various awards, among them the CNRS (National Center for Scientific Research) silver medal, the L'Oreal Unesco award for Women in Science, and the Overbeek gold medal for interface and colloidal science. She is the chair of the microgravity panel of the European Space Sciences Committee, the sister committee of the SSB in Europe. She received her Ph.D. from the École Normale Supérieure in Paris.

GLORIA R. LEON is professor emeritus of psychology at the University of Minnesota. Dr. Leon served for 10 years as director of the Clinical Psychology graduate program at the University of Minnesota and 7 prior years as assistant/associate director. She continues to carry out research and serve on advisory committees in both space and disaster-related areas. She has conducted extensive research on personality, behavioral functioning, and team processes of different polar expedition groups, studying teams composed of single gender, mixed gender, and cross-national members. Dr. Leon recently completed longitudinal research projects with Danish military groups in Greenland as an analog for Mars missions. Over a 13-year period, she was co-principal investigator (PI) on NASA-funded research on the development and testing of cooling garments for space purposes, with implications for better monitoring of astronauts during extended extravehicular activities. She has twice been awarded the NASA Certificate of Recognition for the creative development of a technical innovation and is the recipient of the NASA Johnson Space Center Group Achievement Award to the Sex and Gender Report Team She has also served on NASA peer review panels and was a member of the external advisory committee (EAC) of the National Space Biomedical Research Institute (NSBRI), advising the Neurobehavioral and Psychosocial team. Dr. Leon has served a 4-year term as chair of the NASA Human Research Program Standing Review Panel in the Behavioral Health and Performance area. She received her Ph.D. in mental health psychology from the University of Maryland, College Park. Dr. Leon has served as a member of several National Academies committees, including the Committee on NASA's Research on Human Health Risks, the Planning Committee for the Workshop on Analog Environments for Research on Health Impacts of Space Travel, the Committee on Aerospace Medicine and Medicine of Extreme Environments (CAMMEE), and the Human Behavior and Mental Health panel of the 2011 Decadal Survey on Biological and Physical Sciences in Space.

CARL LINEBERGER is the E.U. Condon Distinguished Professor of Chemistry at the University of Colorado, Boulder and a fellow of JILA, a joint University of Colorado-NIST facility. His research interests include the structure and stability of ions and free radicals, photoelectron spectroscopy of anions, and photophysics and dynamics of cluster ions. His work is primarily experimental, using a wide variety of laser-based techniques to study structure and reactivity of gas phase ions. Prior to joining the faculty at the University of Colorado, Dr. Lineberger held various positions including that of research physicist at the U.S. Army Ballistic Research Laboratory and chair of JILA. Dr. Lineberger is a member of the National Academy of Sciences (NAS), the American Academy of Arts and Sciences, and a fellow of both the American Physical Society and the American Association for the Advancement of Science. He has received many awards, including the Peter Debye Award in Physical Chemistry, the William Lloyd Evans Award, and the Sierra Nevada Section of American Chemical Society Distinguished Chemist Award. Dr. Lineberger received his B.E.E., M.S.E.E., and Ph.D. degrees from the Georgia Institute of Technology. He has served on several National Academies committees, including the Committee on Responsible Science, the Laboratory Assessments Board, and the Joint Advisory Group on Becoming the Online Resource Center for Ethics Education in Engineering and Science.

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ELLIOT MEYEROWITZ is a HHMI-GBMF Plant Science Investigator and George W. Beadle Professor of Biology in the Division of Biology and Biological Engineering at the California Institute of Technology. He has made outstanding contributions toward the understanding of development in plants and animals. Dr. Meyerowitz's work was crucial in establishing Arabidopsis thaliana as a model plant for molecular-genetic studies. He has identified Arabidopsis genes that regulate flower development and genes that determine responsiveness to the plant hormone ethylene. Dr. Meyerowitz is a member of the NAS, the American Academy of Arts and Sciences, and the American Philosophical Society, and is a foreign member of the French Académie des Sciences and the U.K.'s Royal Society. He is a recipient of the Ross Harrison Prize, the Wilbur Cross Medal, and the Richard Lounsbery Award. He earned his M.Phil. and Ph.D. degrees in biology from Yale University. Dr. Meyerowitz has served on several National Academies committees, including the Committee on Space Biology and Medicine, the Committee on a New Biology for the 21st Century, and the Commission on Life Sciences. More recently, he has served on the 2016 NAS Nominating Committee and the Council of the National Academy of Sciences, and he is currently a member of both the International Temporary Nominating Group for Class II: Biological Sciences and the 2017 NAS Class II Membership Committee.

TODD J. MOSHER is the vice president of engineering for Syncroness, where he leads the Syncroness product development engineering organization in developing medical, aerospace, and other commercial products. Dr. Mosher has 25 years of experience as an engineering professional working in industry and serving as a professor at two universities. He has directed the design of both human spaceflight and robotic spacecraft projects. Previously, Dr. Mosher was the senior director of strategic opportunities for Sierra Nevada Corporation's (SNC's) Space Exploration Systems business area within the Space Systems Group. In that role he led the formation of strategic partnerships with Lockheed Martin, United Launch Alliance, Draper Laboratory, Aerojet Rocketdyne, the Walt Disney Corporation, and Lucasfilm. He directed the proposal efforts for the next phase of the NASA Commercial Crew Program and NASA's next Commercial Resupply Services contracts. Prior to that role, Dr. Mosher was the director of design and development for the Dream Chaser program managing the design team for all of the major subsystems and a staff of over 100 SNC engineers and contractors while keeping design and development milestones on schedule and within budget. He has been recognized as one of *The Denver Post* Colorado Top Thinkers (2012) and the University of Colorado Kalpana Chawla Outstanding Recent Alumni award (2012). At SNC, he was awarded the Explorer's Cup Management Team Award (2012), the SNC Director of the Year (2011), and the STAR Award for Technical Excellence (2010). Dr. Mosher has a Ph.D. and M.S. in aerospace engineering from the University of Colorado, an M.S. in systems engineering from the University of Alabama, Huntsville, and a B.S. in aerospace engineering from San Diego State University. He has served on multiple National Academies studies, including co-chairing the Committee on NASA Technology Roadmaps and serving as a member of the Committee on Assessment of NASA Laboratory Capabilities and the Committee to Review NASA's Exploration Technology Development Programs.

ELAINE ORAN is the Glenn L. Martin Institute Professor of Engineering in the A. James Clark School of Engineering at the University of Maryland (UMD). She also is an emeritus scientist at the U.S. Naval Research Laboratory, adjunct professor at the University of Michigan, and a visiting professor at Leeds University in the United Kingdom. Her research interests span computational fluid dynamics and its application to reactive and complex flows, deflagrations, detonations and multiphase flows, rarefied gas flow, high-performance computing and parallel architectures, turbulence in reacting and nonreacting flows, astrophysical phenomena, shocks and shock interactions in gases and condensed phases, nonequilibrium molecular dynamics applied to physical and biological systems, and numerical methods for noncompressible flow. Previously, Dr. Oran was the senior scientist for Reactive Flow Physics at the U.S. Naval Research Laboratory. She received her Ph.D. in physics from Yale University. Dr. Oran is a member of the National Academy of Engineering (NAE) and serves on the National Academies Committee on Human Rights, the NAE's Awards Committee (as the Search Committee Executive of the Aerospace Engineering Search Committee), as a member of the ASEB. She also served on the Applied Physical Sciences Panel for the 2011 Decadal Survey on Biological and Physical Sciences in Space.

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JAMES A. PAWELCZYK is an associate professor of physiology and kinesiology at Pennsylvania State University. Dr. Pawelczyk served as a payload specialist on STS-90 Neurolab. During the 16-day Spacelab flight, the seven-person crew aboard NASA Space Shuttle *Columbia* served as both experiment subjects and operators for 26 individual life sciences experiments focusing on the effects of microgravity on the brain and nervous system. Dr. Pawelczyk's primary research interests include the neural control of circulation, particularly skeletal muscle blood flow, as it is affected by exercise or spaceflight. He is a member of the American Physiological Society and the American College of Sports Medicine. He has won numerous awards, including the Young Investigator Award from the Life Sciences Project Division of the NASA Office of Life and Microgravity Science Applications (1994) and the NASA Space Flight Medal (1998). He has served on NASA's Human Exploration Operations Research Subcommittee. He earned his B.A. in biology and psychology from the University of Rochester, an M.S. in physiology from Pennsylvania State University, and a Ph.D. in biology (physiology) from the University of North Texas. Dr. Pawelczyk has previously served as a member of the National Academies Review of NASA Strategic Roadmaps: Space Station Panel, the Committee on NASA's Bioastronautics Critical Path Roadmap, the Committee to Review NASA's Space Flight Standards, the Planning Committee for the Issues in Space Science and Technology Workshop Series, the Committee on NASA's Research on Human Health Risks, the Committee on Aerospace Medicine and Medicine of Extreme Environments, and the SSB. He also served as chair of the Integrative and Translational Research for the Human Systems Panel of the 2011 Decadal Survey on Biological and Physical Sciences in Space. He currently serves as a member of the National Academies Committee to Review NASA's Evidence Reports on Human Health Risks and the Committee on Biological and Physical Sciences in Space.

JAMES T'IEN is the Leonard Case Jr. Professor of Engineering in the Department of Mechanical and Aerospace Engineering at Case Western Reserve University. Dr. T'ien's research interests are in the areas of combustion, propulsion, fire research, and chemically reacting flows. He has received numerous awards, including a Daniel and Florence Guggenheim Fellowship in Jet Propulsion, a Public Service Medal from NASA, and the Space Processing Award from the AIAA. In addition, he is a fellow of the Combustion Institute, an honorable member of the Combustion Institute's Chinese Taipei Section, and a member of the American Society of Mechanical Engineers. Dr. T'ien earned his Ph.D. in aerospace and mechanical sciences from Princeton University. For the National Academies, he served on the Applied Physical Sciences Panel of the 2011 Decadal Survey on Biological and Physical Sciences in Space and on the Committee to Identify Innovative Research Needs to Foster Improved Fire Safety in the United States.

MARK M. WEISLOGEL is a professor in the Thermal and Fluid Sciences Group in the Maseeh College of Engineering and Computer Science at Portland State University (PSU). Dr. Weislogel has research experience from government and private institutions. While employed by NASA, he proposed and conducted experiments relating to microgravity fluid mechanics. He continues to make extensive use of NASA ground-based low-gravity facilities such as drop towers and low-gravity aircraft and has completed experiments on the Space Shuttle, the Russian Mir space station, and the ISS. While in the private sector, Dr. Weislogel served as a PI for applied research projects concerning high-performance heat transport systems, micrometeorite-safe space-based radiators, microscale cooling systems, emergency oxygen supply systems, and astronaut sleep stations. His research at PSU includes passive noncapillary cooling cycles for satellite thermal control and capillary fluidics at both micro- and macro-length scales. His teaching interests include heat transfer, fluid mechanics, and applied mathematics as it relates to these subjects. He earned a Ph.D. from Northwestern University. Dr. Weislogel served as a member of the Applied Physical Sciences Panel for the 2011 Decadal Survey on Biological and Physical Sciences in Space of the National Academies.

GAYLE E. WOLOSCHAK is a professor of radiation oncology and radiology in the Departments of Radiation Oncology, Radiology, and Cell and Molecular Biology at Northwestern University's Feinberg School of Medicine in Chicago. Her research is focused on nanocomposites and genes controlling radiosensitivity and motor neuron dysfunction. Dr. Woloschak's work is oriented toward function use of nanocomposites for intracellular manipulation, imaging, and gene silencing. Her work on motor neuron disease is designed to understand the molecular basis

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for the combined abnormalities from a molecular-cellular perspective. Prior to joining the Northwestern faculty in 2002, Dr. Woloschak was the senior molecular biologist in the Biosciences Division at Argonne National Laboratory. Dr. Woloschak has served as a member of the National Institute of Health's radiation study section and as chair of NASA's peer review radiation biology committee. She received her Ph.D. in medical sciences (microbiology) from the Medical College of Ohio. Dr. Woloschak previously served as a member of the steering committee for the 2011 National Academies Decadal Survey on Biological and Physical Sciences in Space as well as a member of the survey's Integrative and Translational Research for the Human System Panel. She has served on numerous other National Academies committees as well, including the Committee to Review NASA's Evidence Reports on Human Health Risks, the Committee on the Analysis of Cancer Risks in Populations Near Nuclear Facilities, and the Committee on Evaluation of Radiation Shielding. She is currently a member of the standing Committee on Aerospace Medicine and Medicine of Extreme Environments.

CONSULTANT

LISELOTTE J. SCHIOLER is president of Schioler Consulting. She retired in early 2016 from the National Institute of Aerospace (NIA) where she was responsible for Federal Aviation Administration and non-NASA Langley Research Center government agency programs. She has over 30 years of experience in fundamental research, as well as program and proposal development, proposal consulting, and program management. Prior to her employment at NIA, she worked for the federal government as a researcher in high-temperature structural ceramics (U.S. Army) and as a program manager for ceramics/high-temperature materials (U.S. Air Force Office of Scientific Research and the National Science Foundation), as well as at a large aerospace company, a small high-tech business, and her own consulting company. Dr. Schioler is a fellow of the American Ceramic Society and a recipient of its James I. Mueller Memorial Award, Engineering Ceramics Division. She holds a Sc.D. in ceramic science from the Massachusetts Institute of Technology. She has served on several advisory committees for the Department of Energy and NASA. At the National Academies, she has served as a member of the 2012 Committee on NASA Technology Roadmaps and as the co-chair of the 2016 Committee on NASA Technology Roadmaps.

STAFF

SANDRA J. GRAHAM, *Study Director*, has been a senior program officer at the National Research Council's Space Studies Board (SSB) since 1994. During that time Dr. Graham has directed a large number of major studies, many of them focused on space research in biological and physical sciences and technology. High-profile studies in other areas have included an assessment of servicing options for the Hubble Space Telescope, a study of the societal impacts of severe space weather, and the 2014 *Pathways to Exploration* report reviewing the U.S. human space program. As a precursor to this midterm review, she directed the work of the committee and seven panels to develop the comprehensive decadal survey report *Recapturing a Future for Space Exploration—Life and Microgravity Sciences Research for a New Era*. Prior to joining the SSB, Dr. Graham held the position of senior scientist at the Bionetics Corporation, where she provided technical and science management support for NASA's Microgravity Science and Applications Division. She received her Ph.D. in inorganic chemistry from Duke University, where her research focused primarily on topics in bioinorganic chemistry, including rate modeling and reaction chemistry of biological metal complexes and their analogs.

MARCHEL HOLLE joined the SSB as a research associate in 2016. He received his B.A. in government with minors in physics and history from Hamilton College in 2016. While in school, Mr. Holle gained a wide breadth of experience in legislative and government affairs working with the Satellite Industry Association, the Aerospace Industries Association, SpaceX, the Space Foundation, and the Grocery Manufacturers Association.

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DIONNA J. WILLIAMS is a program coordinator with the SSB, having previously worked for the National Academies' Division of Behavioral and Social Sciences and Education for 5 years. Ms. Williams has a long career in office administration, having worked as a supervisor in a number of capacities and fields. Ms. Williams attended the University of Colorado, Colorado Springs, and majored in psychology.

MICHAEL MOLONEY was the director for space and aeronautics at the SSB and the Aeronautics and Space Engineering Board (ASEB) of the National Academies (though March 2018). Since joining the ASEB/SSB, Dr. Moloney has overseen the production of more than 40 reports, including four decadal surveys—in astronomy and astrophysics, planetary science, life and microgravity science, and solar and space physics—a review of the goals and direction of the U.S. human exploration program, a prioritization of NASA space technology roadmaps, as well as reports on issues such as NASA's Strategic Direction, orbital debris, the future of NASA's astronaut corps, and NASA's flight research program. Before joining the SSB and ASEB in 2010, Dr. Moloney was associate director of the Board on Physics and Astronomy and study director for the decadal survey for astronomy and astrophysics (Astro2010). Since joining the Academies in 2001, Dr. Moloney has served as a study director at the National Materials Advisory Board, the Board on Physics and Astronomy, the Board on Manufacturing and Engineering Design, and the Center for Economic, Governance, and International Studies. Dr. Moloney has served as study director or senior staff for a series of reports on subject matters as varied as quantum physics, nanotechnology, cosmology, the operation of the nation's helium reserve, new anti-counterfeiting technologies for currency, corrosion science, and nuclear fusion. In addition to his professional experience at the National Academies, Dr. Moloney has more than 7 years' experience as a foreign-service officer for the Irish government—including serving at the Irish Embassy in Washington and the Irish Mission to the United Nations in New York. A physicist, Dr. Moloney did his Ph.D. work at Trinity College Dublin in Ireland. He received his undergraduate degree in experimental physics at University College Dublin, where he was awarded the Nevin Medal for Physics.

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Acronyms

ACES	Atomic Clock Ensemble in Space
ACME	Advanced Combustion via Microgravity Experiments
AES	Advanced Exploration Systems
AG	artificial gravity
AMS	Alpha Magnetic Spectrometer
ARED	Advanced Resistive Exercise
ARL	Army Research Laboratory
ARS	acute radiation syndromes
ASI	Italian Space Agency
BASS	Burning and Suppression of Solids
BEC	Bose-Einstein condensate
BHP	Behavioral Health and Performance
BMed	Behavioral Medicine
BPS	Biological and Physical Sciences Program
CAL	Cold Atom Laboratory
CASIS	Center for the Advancement of Science in Space
CBPSS	Committee on Biological and Physical Sciences in Space
CIR	Combustion Integrated Rack
CNES	Centre national d'études spatiales (space agency of France)
CNS	central nervous system
COTS	Commercial Orbital Transportation Services
CSA	Canadian Space Agency
CVD	cardiovascular disease
DARPA	Defense Advanced Research Projects Agency
DLR	German Aerospace Center
DoD	U.S. Department of Defense

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DOE	U.S. Department of Energy
DRM	design reference mission
ESA	European Space Agency
EVA	extra-vehicular activity
EXPRESS	Expedite the Processing of Experiments for Space Station
FLEX	Flame Extinguishment Experiments
FLEX-ICE-GA	FLEX-Italian Combustion Experiment for Green Air
FY	fiscal year
GCR	galactic cosmic rays
HEOMD	Human Exploration and Operations Mission Directorate
HERA	Human Exploration Research Analog
HFBP	Human Factors Behavioral Performance
HHC	Human health and Countermeasures
HI-SEAS	Hawaii Space Exploration Analog and Simulation
HRP	Human Research Program
HSRB	Human System Risk Board
HZE	high Z (atomic number) and energy
iRED	interim resistive exercise device
ISRU	in situ resource utilization
ISS	International Space Station
ISSES	ISS environmental simulation
JAXA	Japan Aerospace Exploration Agency
JSC	Johnson Space Center
KSC	Kennedy Space Center
LBNP	lower-body negative pressure
LDEM	long-duration exploration mission
LEO	low Earth orbit
LET	linear energy transfer
MoBE	Microbiology of the Built Environment
MOU	memoranda of understanding
NASA	National Aeronautics and Space Administration
NCI	National Cancer Institute
NIAID	National Institute of Allergy and Infectious Diseases
NIH	National Institutes of Health
NIMH	National Institute of Mental Health
NIST	National Institute of Standards and Technology
NMR	Nuclear Magnetic Resonance
NRA	NASA Research Announcement
NRC	National Research Council
NSBRI	National Space Biomedical Research Institute

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NSCOR	NASA Specialized Center of Research
NSF	National Science Foundation
OLMSA	Office of Life and Microgravity Science and Applications
PI	principal investigator
PSI	Physical Sciences Informatics [data repository]
RDoC	Research Domain Criteria
SAA	Space Act Agreement
SANS	Spaceflight-Associated Neuro-ocular Syndrome
SBIR	Small Business Innovative Research
SLPSRA	Space Life and Physical Sciences Research and Applications (Division of)
SLS	Space Launch System
Sofie	Solid Fuel Ignition and Extinction Experiment
SPE	solar particle event
SSPF	Space Station Processing Facility
STMD	Space Technology Mission Directorate
STTR	Small Business Technology Transfer
TRI	Translation Research Institute
TSES	Translation to Space Exploration Systems
USOS	U.S. Orbital Segment
VE	virtual environment

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E

Criteria and Table Reprinted from the 2011 Decadal Survey

Box 13.2 and Tables 13.1 to 13.3 from the 2011 decadal survey[1] are reprinted below.

[1] National Research Council, 2011, *Recapturing a Future for Space Exploration: Life and Physical Sciences Research for a New Era*, The National Academies Press, Washington, D.C.

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BOX 13.2
Criteria Used for Categorization of Research Recommendations

In its categorization of research, whether basic, applied, or translational, the committee used the following eight prioritization criteria developed to capture the potential value of the results of research (information, engineered systems, publications, or new concepts).

- Prioritization Criterion 1: The extent to which the results of the research will reduce uncertainty about both the benefits and the risks of space exploration (*Positive Impact on Exploration Efforts, Improved Access to Data or to Samples, Risk Reduction*)
- Prioritization Criterion 2: The extent to which the results of the research will reduce the costs of space exploration (*Potential to Enhance Mission Options or to Reduce Mission Costs*)
- Prioritization Criterion 3: The extent to which the results of the research may lead to entirely new options for exploration missions (*Positive Impact on Exploration Efforts, Improved Access to Data or to Samples*)
- Prioritization Criterion 4: The extent to which the results of the research will provide full or partial answers to grand science challenges that the space environment provides a unique means to address (*Relative Impact Within Research Field*)
- Prioritization Criterion 5: The extent to which the results of the research are uniquely needed by NASA, as opposed to any other agencies (*Needs Unique to NASA Exploration Programs*)
- Prioritization Criterion 6: The extent to which the results of the research can be synergistic with other agencies' needs (*Research Programs That Could Be Dual-Use*)
- Prioritization Criterion 7: The extent to which the research must use the space environment to achieve useful knowledge (*Research Value of Using Reduced-Gravity Environment*)
- Prioritization Criterion 8: The extent to which the results of the research could lead to either faster or better solutions to terrestrial problems or to terrestrial economic benefit (*Ability to Translate Results to Terrestrial Needs*)

The committee did not weight these criteria, a step that would require assumptions about policy decisions not yet made, or subject to change in the future. The criteria and priorities outlined in this chapter, based on clear metrics, provide a basis for a complete, transparent, and robust research program, which the committee believes is required to fully address NASA's future needs for the life and physical sciences research essential to successful space exploration.

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TABLE 13.1 Summary of Highest-Priority Recommendations Made in Chapters 4 Through 10

Recommendation Identifier[a]	Recommendation	Enabled by (EB) and/or Enabling (E) Space Exploration
Plant and Microbial Biology (Chapter 4)		
P1	Establish a microbial observatory program on the ISS to conduct long-term, multigenerational studies of microbial population dynamics.	EB
P2	Establish a robust spaceflight program of research analyzing plant and microbial growth and physiological responses to the multiple stimuli encountered in spaceflight environments.	EB
P3	Develop a research program aimed at demonstrating the roles of microbial-plant systems in long-term life support systems.	EB/E
Behavior and Mental Health (Chapter 5)		
B1	Develop sensitive, meaningful, and valid measures of mission-relevant performance for both astronauts and mission control personnel.	E
B2	Conduct integrated translational research in which long-duration missions are simulated specifically for the purpose of studying the interrelationships among individual functioning, cognitive performance, sleep, and group dynamics.	E
B3	Determine the genetic, physiological, and psychological underpinnings of individual differences in resilience to stressors during extended space missions, with development of an individualized medicine approach to sustaining astronauts during such missions.	E
B4	Conduct research to enhance cohesiveness, team performance, and effectiveness of multinational crews, especially under conditions of extreme isolation and autonomy.	EB/E
Animal and Human Biology (Chapter 6)		
AH1	The efficacy of bisphosphonates should be tested in an adequate population of astronauts on the ISS during a 6-month mission.	EB/E
AH2	The preservation/reversibility of bone structure/strength should be evaluated when assessing countermeasures.	EB/E
AH3	Bone loss studies of genetically altered mice exposed to weightlessness are strongly recommended.	EB
AH4	New osteoporosis drugs under clinical development should be tested in animal models of weightlessness.	EB
AH5	Conduct studies to identify underlying mechanisms regulating net skeletal muscle protein balance and protein turnover during states of unloading and recovery.	EB/E
AH6	Conduct studies to develop and test new prototype exercise devices and to optimize physical activity paradigms/prescriptions targeting multisystem countermeasures.	EB/E
AH7	Determine the daily levels and pattern of recruitment of flexor and extensor muscles of the neck, trunk, arms, and legs at 1 *g* and after being in a novel gravitational environment for up to 6 months.	EB
AH8	Determine the basic mechanisms, adaptations, and clinical significance of changes in regional vascular/interstitial pressures (Starling forces) during long-duration space missions.	EB/E
AH9	Investigate the effects of prolonged periods of microgravity and partial gravity (3/8 or 1/6 *g*) on the determinants of task-specific, enabling levels of work capacity.	EB/E
AH10	Determine the integrative mechanisms of orthostatic intolerance after restoration of gravitational gradients (both 1 *g* and 3/8 *g*).	EB/E

continued

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TABLE 13.1 Continued

Recommendation Identifier[a]	Recommendation	Enabled by (EB) and/or Enabling (E) Space Exploration
AH11	Collaborative studies among flight medicine and cardiovascular epidemiologists are recommended to determine the best screening strategies to avoid flying astronauts with subclinical coronary heart disease that could become manifest during a long-duration exploration-class mission (3 years).	EB/E
AH12	Determine the amount and site of the deposition of aerosols of different sizes in the lungs of humans and animals in microgravity.	EB/E
AH13	Multiple parameters of T cell activation in cells should be obtained from astronauts before and after re-entry to establish which parameters are altered during flight.	EB
AH14	Both to address the mechanism(s) of the changes in the immune system and to develop measures to limit the changes, data from multiple organ/system-based studies need to be integrated.	EB/E
AH15	Perform mouse studies of immunization and challenge on the ISS, using immune samples acquired both prior to and immediately upon re-entry, to establish the biological relevance of the changes observed in the immune system. Parameters examined need to be aligned with those in humans influenced by flight.	EB
AH16	Studies should be conducted on transmission across generations of structural and functional changes induced by exposure to space during development. Ground-based studies should be conducted to develop specialized habitats to support reproducing and developing rodents in space.	EB
Crosscutting Issues for Humans in the Space Environment (Chapter 7)		
CC1	To ensure the safety of future commercial orbital and exploration crews, quantify post-landing vertigo and orthostatic intolerance in a sufficiently large sample of returning ISS crews, as part of the immediate post-flight medical exam.	EB/E
CC2	Determine whether artificial gravity (AG) is needed as a multisystem countermeasure and whether continuous large-radius AG is needed or intermittent exercise within lower-body negative pressure or short-radius AG is sufficient. Human studies in ground laboratories are essential to establish dose-response relationships, and what gravity level, gradient, rotations per minute, duration, and frequency are adequate.	E
CC3	Conduct studies on humans to determine whether there is an effect of gravity on micronucleation and/or intrapulmonary shunting or whether the unexpectedly low prevalence of decompression sickness on the space shuttle/ISS is due to underreporting. Conduct studies to determine operationally acceptable low suit pressure and hypobaric hypoxia limits.	E
CC4	Determine optimal dietary strategies for crews and food preservation strategies that will maintain bioavailability for 12 or more months.	E
CC5	Initiate a robust food science program focused on preserving nutrient stability for 3 or more years.	E
CC6	Include food and energy intake as an outcome variable in dietary intervention trials in humans.	EB/E
CC7	Conduct longitudinal studies of astronauts for cataract incidence, quality, and pathology related to radiation exposures to understand both cataract risk and radiation-induced late tissue toxicities in humans.	E
CC8	Expand the use of animal studies to assess space radiation risks to humans from cancer, cataracts, cardiovascular disease, neurologic dysfunction, degenerative diseases, and acute toxicities such as fever, nausea, bone marrow suppression, and others.	E

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TABLE 13.1 Continued

Recommendation Identifier[a]	Recommendation	Enabled by (EB) and/or Enabling (E) Space Exploration
CC9	Continue ground-based cellular studies to develop end points and markers for acute and late radiation toxicities, using radiation facilities that are able to mimic space radiation exposures.	E
CC10	Expand understanding of gender differences in adaptation to the spaceflight environment through flight- and ground-based research, particularly potential differences in bone, muscle, and cardiovascular function and long-term radiation risks.	EB/E
CC11	Investigate the biophysical principles of thermal balance to determine whether microgravity reduces the threshold for thermal intolerance.	EB/E
Fundamental Physical Sciences in Space (Chapter 8)		
FP1	Research on complex fluids and soft matter. Microgravity provides a unique opportunity to study structures and forces important to the properties of these materials without the interference caused by Earth-strength gravity.	EB/E
FP2	Understanding of the fundamental forces and symmetries of nature. High-precision measurements in space can test relativistic gravity, fundamental high-energy physics, and related symmetries in ways that are not practical on Earth. Novel effects predicted by new theoretical approaches provide distinct signatures for precision experimental searches that are often best carried out in space.	EB
FP3	Research related to the physics and applications of quantum gases. The space environment enables many investigations, not feasible on Earth, of the remarkably unusual properties of quantum gases and degenerate Fermi gases.	EB/E
FP4	Investigations of matter near a critical phase transition. Experiments that have already been designed and brought to a level of flight readiness can elucidate how materials behave in the vicinity of thermodynamically determined critical points. These experiments, which require a microgravity environment, will provide insights into new effects observable when such systems are driven away from equilibrium conditions.	EB
Applied Physical Sciences in Space (Chapter 9)		
AP1	Reduced-gravity multiphase flows, cryogenics and heat transfer database and modeling, including phase separation and distribution (i.e., flow regimes), phase-change heat transfer, pressure drop, and multiphase system stability.	EB/E
AP2	Interfacial flows and phenomena (including induced and spontaneous multiphase flows with or without phase change) relevant to storage and handling systems for cryogens and other liquids, life support systems, power generation, thermal control systems, and other important multiphase systems.	EB/E
AP3	Dynamic granular material behavior and subsurface geotechnics to improve predictions and site-specific models of lunar and martian soil behavior.	E
AP4	Development of fundamentals-based strategies and methods for dust mitigation during advanced human and robotic exploration of planetary bodies.	E
AP5	Experiments on the ISS to understand complex fluid physics in microgravity, including fluid behavior of granular materials, colloids and foams, biofluids, non-Newtonian and critical point fluids, etc.	EB
AP6	Fire safety research to improve methods for screening materials for flammability and fire suppression in space environments.	E
AP7	Combustion processes research, including reduced-gravity experiments with longer durations, larger scales, new fuels, and practical aerospace materials relevant to future missions.	EB/E

continued

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TABLE 13.1 Continued

Recommendation Identifier[a]	Recommendation	Enabled by (EB) and/or Enabling (E) Space Exploration
AP8	Research on numerical simulation of combustion to develop and validate detailed single phase and multiphase combustion models for interpreting and facilitating combustion experiments and tests.	E
AP9	Reduced-gravity research on materials synthesis and processing and control of microstructure and properties, to improve the properties of existing and new materials on the ground.	EB/E
AP10	Development of new and advanced materials that enable operations in harsh space environments and reduce the cost of human space exploration.	E
AP11	Fundamental and applied research to develop technologies that facilitate extraction, synthesis, and processing of minerals, metals, and other materials available on extraterrestrial surfaces.	EB/E
Translation to Space Exploration Systems (Chapter 10)		
TSES1	Conduct research to address issues for active two-phase flow relevant to thermal management (T1).	E
TSES2	To support zero-boiloff propellant storage and cryogenic fluid management technologies, conduct research on advanced insulation materials research, active cooling, multiphase flows, and capillary effectiveness (T2), as well as active and passive storage, fluid transfer, gauging, pressurization, pressure control, leak detection, and mixing destratification (T3).	E
TSES3	NASA should enhance surface mobility; relevant research includes suited astronaut computational modeling, biomechanics analysis for partial gravity, robot-human testing of advanced spacesuit joints and full body suits, and musculoskeletal modeling and suited range-of-motion studies (T4), plus studies of human-robot interaction (including teleoperations) for the construction and operation of planetary surface habitats (T26).	E
TSES4	NASA should develop and demonstrate technologies to mitigate the effects of dust on extravehicular activity (EVA) systems and suits, life support systems, and surface construction systems. Supporting research includes impact mechanics of particulates, design of outer-layer dust garments, advanced material and design concepts for micrometeoroid mitigation, magnetic repulsive technologies, and the quantification of plasma electrodynamic interactions with EVA systems (T5); dynamics of electrostatic field coupling with dust (T23); and regolith mechanics and gravity-dependent soil models (T27).	E
TSES5	NASA should define requirements for thermal control, micrometeoroid and orbital debris impact and protection, and radiation protection for EVA systems, rovers, and habitats and develop a plan for radiation shelters (T19).	E
TSES6	NASA should conduct research for the development and demonstration of closed-loop life support systems and supporting technologies. Fundamental research includes heat and mass transfer in porous media under partial gravity and microgravity conditions (T6) and understanding the effect of variable gravity on multiphase flow systems (T21, T22).	E
TSES7	NASA should develop and demonstrate technologies to support thermoregulation of habitats, rovers, and spacesuits on the lunar surface (T20).	E
TSES8	NASA should perform critical fire safety research to develop new standards to qualify materials for flight and to improve fire and particle detectors. Supporting research is necessary in materials qualification for ignition, flame spread, and generation of toxic and/or corrosive gases (T7) and in characterizing particle sizes from smoldering and flaming fires under reduced gravity (T8).	E

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TABLE 13.1 Continued

Recommendation Identifier[a]	Recommendation	Enabled by (EB) and/or Enabling (E) Space Exploration
TSES9	NASA should develop a standard methodology for qualifying fire suppression systems in relevant atmospheres and gravity levels and would benefit from strategies for safe post-fire recovery. Specific research is needed to characterize the effectiveness of fire suppression agents and systems under reduced gravity (T9) and to assess the toxicity of various fire products (T10).	E
TSES10	Research should be conducted to allow regenerative fuel cell technologies to be demonstrated in reduced-gravity environments (T11).	E
TSES11	To support the development of new energy conversion technologies, research should be done on high-temperature energy conversion cycles, device coupling to essential working fluids, heat rejection systems, materials, etc. (T12). Research is also required on more efficient surface-base primary power and on the technologies to enable solar electric propulsion as an option to transfer large masses of propellant and cargo to distant locations (T18).	E
TSES12	To make fission surface power systems a viable option, research is needed on high-temperature, low-weight materials for power conversion and radiators and on other supporting technologies (T13).	E
TSES13	Development and demonstration of ascent and descent system technologies are needed, including ascent/descent propulsion technologies, inflatable aerodynamic decelerators, and supersonic retro propulsion systems. The required research includes propellant ignition, flame stability, and active thermal control (T14); lightweight flexible materials (T15); and rocket plume aerothermodynamics and vehicle dynamics and control (T16).	E
TSES14	Research is required to support the development and demonstration of space nuclear propulsion systems, including liquid-metal cooling under reduced gravity, thawing under reduced gravity, and system dynamics (T17).	E
TSES15	Research is needed to identify and adapt excavation, extraction, preparation, handling, and processing techniques for a lunar water/oxygen extraction system (T24).	E
TSES16	NASA should establish plans for surface operations, particularly ISRU capability development and surface habitats. Research is needed to characterize resources available at lunar and martian surface destinations (T25) and to define surface habitability systems design requirements (T28).	E

[a]Identifiers correspond to the identifiers given to the highest-priority recommendations listed at the ends of Chapters 4 through 10, which provide context and clarifying discussion.

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TABLE 13.2 Highest-Priority Recommendations That Provide **High** Support in Meeting Each of Eight Specific Prioritization Criteria

	<----------Prioritization Criteria---------->							
	(1) Positive Impact on Exploration Efforts, Improved Access to Data or to Samples, Risk Reduction	**(2)** Potential to Enhance Mission Options or to Reduce Mission Costs	**(3)** Positive Impact on Exploration Efforts, Improved Access to Data or to Samples	**(4)** Relative Impact Within Research Field	**(5)** Needs Unique to NASA Exploration Programs	**(6)** Research Programs That Could Be Dual-Use	**(7)** Research Value of Using Reduced-Gravity Environment	**(8)** Ability to Translate Results to Terrestrial Needs
Life Sciences	P2, P3, B1, B2, B3, B4, AH1, AH2, AH3, AH5, AH6, AH7, AH8, AH9, AH10, AH11	P3, B1, B2, B3, B4, AH6, AH9, AH10, AH11	P3, B4, AH1, AH2, AH3, AH5, AH6, AH7, AH8, AH9, AH10, AH11	P1, P2, B3, B4, AH9, AH10, AH11, AH16	P1, P2, P3, AH1, AH2, AH3, AH4, AH5, AH6, AH7, AH8, AH9, AH10, AH11, AH16	B1, B2, B3, B4, AH1, AH2, AH3, AH4, AH5, AH6, AH7, AH9, AH10	P1, B1, B4, AH12, AH16	B1, B2, B3, B4, AH1, AH2, AH3, AH4, AH5, AH6, AH7
Translational Life Sciences	CCH2, CCH4, CCH7	CCH2, CCH4, CCH6, CCH7	CCH2, CCH4, CCH6, CCH7, CCH8	CCH2, CCH6	CCH1, CCH2, CHH3, CCH6, CCH7, CCH8		CCH1, CHH2, CHH3, CCH7, CCH11	
Physical Sciences	AP1, AP4, AP6, AP8, AP11	AP1, AP2, AP10, AP11	AP1, AP2, AP3, AP10, AP11	FP1, FP2, FP3, AP5, AP7, AP8, AP9	AP1, AP2, AP3, AP4, AP6, AP11	AP7, AP8, AP9, AP10	FP1, FP2, FP3, FP4, AP1, AP2, AP5, AP6, AP7, AP9	AP1, AP2, AP7, AP8, AP9
Translational Physical Sciences	TSES1, TSES2, TSES3, TSES14	TSES1, TSES3, TSES5, TSES10	TSES14		TSES2, TSES3, TSES4, TSES5, TSES6, TSES7, TSES12, TSES13, TSES14, TSES 16	TSES10, TSES11, TSES12	TSES1, TSES2, TSES3, TSES4, TSES5, TSES12, TSES13, TSES14, TSES15, TSES16	TSES10

NOTE: Identifiers are as listed in Table 13.1 and correspond with the recommendations listed there and also presented with clarifying discussion in Chapters 4 through 10.

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TABLE 13.3 Level of Support Provided by High-Priority Recommendations for Each of Eight Prioritization Criteria

Recommendation Identifier[a] Within Suggested Program Elements	**(1)** Positive Impact on Exploration Efforts, Improved Access to Data or to Samples, Risk Reduction	**(2)** Potential to Enhance Mission Options or to Reduce Mission Costs	**(3)** Positive Impact on Exploration Efforts, Improved Access to Data or to Samples	**(4)** Relative Impact Within Research Field	**(5)** Needs Unique to NASA Exploration Programs	**(6)** Research Programs That Could Be Dual-Use	**(7)** Research Value of Using Reduced-Gravity Environment	**(8)** Ability to Translate Results to Terrestrial Needs
	<----------Prioritization Criteria---------->							
Plant and Microbial Biology Research								
P1	Medium	Low	Low	**High**	**High**	Medium	**High**	Medium
P2	**High**	Medium	Medium	**High**	**High**	Medium	Medium	Medium
P3	**High**	**High**	**High**	Low	**High**	Medium	Medium	Medium
Human Behavior and Mental Health Research								
B1	**High**	**High**	Low	Medium	Low	**High**	**High**	**High**
B2	**High**	**High**	Low	Medium	Low	**High**	Low	**High**
B3	**High**	**High**	Medium	**High**	Low	**High**	Low	**High**
B4	**High**	**High**	**High**	**High**	Medium	**High**	**High**	**High**
Animal and Human Biological Research								
AH1	**High**	Medium	**High**	Medium	**High**	**High**	Medium	**High**
AH2	**High**	Medium	**High**	Medium	**High**	**High**	Medium	**High**
AH3	**High**	Medium	**High**	Medium	**High**	**High**	Medium	**High**
AH4	Medium	Medium	Medium	Medium	**High**	**High**	Medium	**High**
AH5	**High**	Medium	**High**	Medium	**High**	**High**	Medium	**High**
AH6	**High**	**High**	**High**	Medium	**High**	**High**	Medium	**High**
AH7	**High**	Medium	**High**	Medium	**High**	**High**	Medium	**High**
AH8	**High**	Medium	**High**	Medium	**High**	Medium	Medium	Medium
AH9	**High**	**High**	**High**	**High**	**High**	**High**	Medium	Medium
AH10	**High**	**High**	**High**	**High**	**High**	**High**	Medium	Medium
AH11	**High**	**High**	**High**	**High**	**High**	Medium	Medium	Medium
AH12	Medium	Medium	Medium	Medium	Medium	Low	**High**	Medium
AH13	Medium	Low	Medium	Medium	Medium	Medium	Medium	Medium
AH14	Medium	Low	Medium	Medium	Medium	Medium	Medium	Medium
AH15	Medium/Low	Low	Medium	Medium	Medium	Medium	Medium	Medium
AH16	Medium/Low	Medium/Low	Medium/Low	**High**	**High**	Low	**High**	Medium

continued

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TABLE 13.3 Continued

Recommendation Identifier[a] Within Suggested Program Elements	<----------Prioritization Criteria---------->							
	(1) Positive Impact on Exploration Efforts, Improved Access to Data or to Samples, Risk Reduction	**(2)** Potential to Enhance Mission Options or to Reduce Mission Costs	**(3)** Positive Impact on Exploration Efforts, Improved Access to Data or to Samples	**(4)** Relative Impact Within Research Field	**(5)** Needs Unique to NASA Exploration Programs	**(6)** Research Programs That Could Be Dual-Use	**(7)** Research Value of Using Reduced-Gravity Environment	**(8)** Ability to Translate Results to Terrestrial Needs
Crosscutting Research for the Human System								
CC1	Medium	Low	Low	**Low**	**High**	Low	**High**	Medium
CC2	**High**	**High**	**High**	**High**	**High**	Low	**High**	Low
CC3	Medium	Medium	Medium	Low	**High**	Low	**High**	Low
CC4	**High**	**High**	**High**	Medium	Medium	Medium	**Medium**	Medium
CC5	Medium	Medium	Medium	Medium	Medium	Medium	Medium	Medium
CC6	Medium	**High**	**High**	**High**	**High**	Medium	Low	Medium
CC7	**High**	**High**	**High**	Low	**High**	Low	**High**	Low
CC8	Medium	Medium	**High**	Low	**High**	Low	Low	Low
CC9	Medium	Low	Low	Low	Medium	Low	Low	Low
CC10	Medium	Medium/Low	Medium	Low	Medium	Medium	Low	Medium
CC11	Medium	Medium/Low	Medium	Low	Medium	Medium/Low	High/Medium	Medium
Fundamental Physical Sciences Research								
FP1	Low	Low	Medium	**High**	Low	Medium	**High**	Medium
FP2	Low	Low	Low	**High**	Low	Medium	**High**	Medium
FP3	Low	Low	Medium	**High**	Low	Medium	**High**	Medium
FP4	Low	Low	Low	**Medium**	Low	Medium	**High**	Medium

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Applied Physical Sciences Research								
AP1	**High**	**High**	**High**	Medium	**High**	Low	**High**	**High**
AP2	Medium	**High**	**High**	Medium	**High**	Medium	**High**	**High**
AP3	Medium	Medium	**High**	Low	**High**	N/A	Low	Low
AP4	**High**	Medium	Medium	Low	**High**	N/A	Medium	Low
AP5	Low	Low	Medium	**High**	Low	Medium	**High**	Medium
AP6	**High**	Medium	Low	Low	**High**	Low	**High**	Medium
AP7	Medium	N/A	N/A	**High**	Medium	**High**	**High**	**High**
AP8	**High**	Medium	Low	**High**	Medium	**High**	N/A	**High**
AP9	N/A	N/A	Low	**High**	Low	**High**	**High**	**High**
AP10	Low	**High**	**High**	Medium	Medium	**High**	Low	Medium
AP11	**High**	**High**	**High**	Low	**High**	N/A	Medium	N/A
Translation to Space Exploration Systems Research								
TSES1	**High**	**High**	Low	Low	Medium	Medium	**High**	Low
TSES2	**High**	High	Medium	Low	**High**	Medium	**High**	Medium
TSES3	**High**	**High**	**High**	Low	**High**	Medium	**High**	Medium
TSES4	Medium	Medium	**High**	Low	**High**	Low	**High**	Low
TSES5	Medium	Medium	**High**	Low	**High**	Low	Medium	Low
TSES6	Medium	Medium	Medium	Low	**High**	Low	Medium	Low
TSES7	Medium	Medium	**High**	Low	**High**	Medium	Medium	Medium
TSES8	Low	Low	Low	Low	Medium	Medium	**High**	Medium
TSES9	Low	Low	Low	Low	Medium	Medium	**High**	Medium
TSES10	Medium	**High**	Low	Low	Medium	**High**	Medium	Medium
TSES11	Medium	Medium	Low	Low	Medium	**High**	Low	Medium
TSES12	Medium	Medium	Low	Low	**High**	**High**	**High**	Medium
TSES13	Medium	Medium	Low	Low	**High**	Medium	**High**	Medium
TSES14	**High**	Medium	**High**	Medium	**High**	Medium	**High**	Medium
TSES15	Medium	Medium	Low	Low	Medium	Low	**High**	Low
TSES16	Medium	Medium	Low	Low	**High**	Low	**High**	Low

[a]Identifiers are listed in Table 13.1 and correspond with the recommendations listed there and also presented the ends of Chapters 4 through 10, which provide context and clarifying discussion.

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F

Commercial Spaceflight Federation Listing of Microgravity Experiments Since 2011 that Have Flown on Balloons, Parabolic, or Suborbital Missions

TABLE F.1 Commercial Spaceflight Federation Listing of Microgravity Experiments Since 2011 that Have Flown on Balloons, Parabolic, or Suborbital Missions

					Commercial Spaceflight Federation List of Microgravity Experiments Since 2011 that Have Flown on Balloons, Parabolic, or Suborbital Missions		
Title #	Status	Year Launched	Type of Flight (P/S/B)	Origin	Payload	Affiliation	Commercial Company
T0001-PS	Active	2011	2/2/-	DIR.	Suborbital Flight Environment Monitor (SFEM)	NASA Ames Research Center	
T0003-PS	Active	2011	2/-/-	AFO1	On-Orbit Propellant Storage Stability	Embry-Riddle Aeronautical University	
T0004-PS	Active	2011	4/-/-	AFO1	Printing the Space Future	Made in Space, Inc.	Made in Space
T0008-P	Active	2011	4	AFO1	Indexing Media Filtration System	Aerfil, LLC	
T0015-S	Active	2011		AFO1	Electromagnetic Field Measurements on sRLV	Johns Hopkins University Applied Physics Laboratory	
T0020-PS	Active	2011	3/-/-	AFO2	Microgravity Multi-Phase Flow Experiment for Suborbital Testing (MFEST)	NASA Johnson Space Center	Zero Gravity Corporation
T0021-S	Active	2011	2	AFO2	Application of controlled vibrations to multiphase systems	Universitat Politècnica de Catalunya/University of Alabama, Huntsville	
T0022-S	Active	2011		AFO2	Environment monitoring suite on sRLV	Johns Hopkins University Applied Physics Laboratory	

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Title #	Status	Year Launched	Type of Flight (P/S/B)	Origin	Payload	Affiliation	Commercial Company
T0023-S	Pending	2011		AFO2	Measurement of the Atmospheric Background in the Mesosphere	Silicon Valley Space Center	
T0024-S	Active	2011		AFO2	RF Gauging of the Liquid Oxygen Tank on sRLV	NASA Glenn Research Center	
T0027-P*	Completed	2011	2	AFO2	Autonomous Cell Culture Apparatus for Growing 3-Dimensional Tissues in Microgravity	NASA Johnson Space Center	Zero Gravity Corporation
T0034-S	Active	2012		AFO3	Terrain Relative Navigation Descent Imager (TRNDI)	Draper Laboratory	
T0035-PS	Active	2012	1/-/-	AFO3	Near-Zero Gravity Cryogenic Line Chilldown Experiment in a Suborbital Reusable Launch Vehicle	University of Florida	
T0036-S	Active	2012		AFO3	Collisions Into Dust Experiment on a Commercial Suborbital Vehicle	University of Central Florida	Blue Origin
T0039-P	Pending	2012		AFO3	Fuel Mass Gauging Based on Electrical Capacitance Volumetric Tomography Techniques	NASA Goddard Space Flight Center	
T0043-P	Pending	2012	1	AFO3	Parabolic Flight: Validation of Electro- Hydrodynamic Gas-Liquid Phase Separation in Microgravity	New Jersey Institute of Technology	
T0044-P	Active	2012	1	AFO3	Sintering of Composite Materials under Reduced Gravity Conditions	University of Central Florida	
T0045-P	Active	2012	1	AFO3	Evaporative Heat Transfer Mechanisms within a Heat Melt Compactor (EHeM HMC)	Advanced Technical Institute I.T.I.S "E. Fermi," Italy	
T0046-S	Pending	2012		AFO3	Polar Mesospheric Cloud Imaging and Tomography Experiment	Space Science Institute	
T0049-P	Active	2012	3	AFO3	Parabolic Flight Evaluation of a Hermetic Surgery System for Reduced Gravity Experiment	NASA Glenn Research Center	
T0050-B	Active	2012		AFO3	Flight Demonstration of an Integrated Camera and Solid-State Fine Steering System	Southwest Research Institute	
T0052-S	Active	2012		AFO3	Collection of Regolith Experiment (CORE) on a Commercial Suborbital Vehicle	University of Louisville	

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Title #	Status	Year Launched	Type of Flight (P/S/B)	Origin	Payload	Affiliation	Commercial Company
T0053-PS	Active	2012	1/-/-	AFO4	Validating Telemetric Imaging Hardware for Crew Assisted and Crew-Autonomous Biological Imaging in Suborbital Applications	University of Central Florida	
T0054-B	Active	2012	1	AFO4	Stratospheric Parabolic Flight Technology	Purdue University	
T0056-P	Pending	2013		AFO5	UAH ChargerSat-2 Parabolic Flight Testing	University of Alabama, Huntsville	
T0057-P*	Active	2013	2	AFO5	Hybrid Ultimate Lifting Kit (HULK) Device Demonstration during Parabolic Flight	University of Florida	
T0058-P	Pending	2013		AFO5	Assessing Otolith-Organ Function with Vestibular Evoked Myogenic Potentials	NASA Glenn Research Center	
T0059-P	Active	2013	1	AFO5	On the performance of a nanocatalyst-based direct ammonia alkaline fuel cell (VEMPs) in Parabolic Flight	Johns Hopkins University School of Medicine	
T0061-SB	Active	2013	-/1/1	AFO5	Flight Testing of a UAT ADS-B Transmitter Prototype for Commercial Space Transportation Using sRLV	Embry-Riddle Aeronautical University	
T0062-P	Pending	2013		AFO5	In-Flight Lab Analysis Technology Demonstration in Reduced Gravity (DAAFC) under microgravity conditions	University of Puerto Rico, Rio Pedras	
T0064-B	Active	2013		AFO5	Deployable Rigid Adjustable Guided Final Landing Approach Pinions (DRAG FLAPs)	Masten Space Systems Inc.	Masten Space Systems
T0066-B*	Completed	2013	3	AFO5	Guided Parafoil High Altitude Research	Airborne Systems North America of CA, Inc.	
T0073-S*	Completed	2012	1	DIR.	Radial Core Heat Spreader	NASA Glenn Research Center	
T0075-S	Active	2012	1	DIR.	Exo-Atmosperic Aerobrake	NASA Ames Research Center	
T0076-S	Pending	2012		NRA1	Demonstration of Vertically Aligned Carbon Nano-tubes for Earth Climate Remote Sensing	Johns Hopkins University Applied Physics Laboratory	
T0077-S*	Active	2012	2	NRA1	Facility for Microgravity Research and Submicroradian Stabilization using sRLVs	Controlled Dynamics Inc	
T0078-S	Pending	2012		NRA1	Enhanced Thermal Switch	Johns Hopkins University Applied Physics Laboratory	

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Title #	Status	Year Launched	Type of Flight (P/S/B)	Origin	Payload	Affiliation	Commercial Company
T0079-S	Active	2012		NRA1	Autonomous Flight Manager for Human-in-the-Loop Immersive Simulation and	NASA Glenn Research Center	
T0080-S	Completed	2012	1	NRA1	Advanced Micro Sun Sensor	NASA/Jet Propulsion Laboratory	
T0081-PS	Active	2012	1/-/-	NRA1	Demonstration of Variable Radiator	Texas A&M University	
T0082-S	Pending	2012		NRA1	Dynamic Microscopy System Flight Test of Terrestrial Rockets	Draper Laboratory	
T0083-PS	Active	2012	1/-/-	NRA1	Design and Development of a Micro Satellite Attitude Control System	State University of New York, Buffalo	
T0084-S	Active	2012	1	NRA1	Suborbital Test of a Robotics-Based Method for In Orbit Identification of Spacecraft Inertia Properties	Inertia Properties New Mexico State University	
T0085-S	Active	2012		NRA1	SwRI Solar Instrument Pointing Platform	Southwest Research Institute	
T0086-S	Active	2012		NRA1	Saturated Fluid Pistonless Pump Technology Demonstrator	University of Colorado	
T0087-S	Pending	2012		NRA1	Electric-hydrodynamic Control of Two-Phase Heat Transfer in Microgravity	New Jersey Institute of Technology	
T0088-S*	Active	2012	1	NRA1	An FPGA-based, Radiation Tolerant, Reconfigurable Computer System with Real Time Fault Detection, Avoidance, and Repair	Montana State University	
T0092-S*	Completed	2013	1	AFO6	Precision Formation Flying Sensor	University of Colorado	
T0095-SB	Active	2013	-/1/1	AFO6	Test of Satellite Communications Systems on-board Suborbital Platforms to Provide Low-Cost Data Communications	Solstar Communications	
T0097-B*	Pending	2013	2	AFO6	Planetary Atmosphere Minor Species Sensor	University of Central Florida	
T0098-S	Active	2013		AFO6	Navigation Doppler Lidar Sensor Demonstration for Precision Landing on Solar System Bodies	NASA Langley Research Center	
T0100-P	Completed	2013	2	AFO6	Creation of Titanium-Based Nanofoams in Reduced Gravity for Dye-Sensitized Solar	Techshot, Inc.	

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Title #	Status	Year Launched	Type of Flight (P/S/B)	Origin	Payload	Affiliation	Commercial Company
T0102-P	Pending	2013	1	AFO6	Demonstration of Adjustable Fluidic Lens in Microgravity	University of Arizona	
T0104-PS	Pending	2013		AFO6	Real Time Conformational Analysis of Rhodopsin using Plasmon Waveguide Resonance Spectroscopy Cell Production	Northwestern University	
T0106-B	Active	2013		AFO6	Low-Cost Suborbital Reusable Launch Vehicle (sRLV)	Surrogate GSSL Inc.	
T0109-P	Active	2013		AFO6	Advanced Optical Mass Gauge	Mass Dynamix, Inc.	
T0110-P	Active	2013	1	DIR.	Wet Lab	NASA Ames Research Center	
T0111-S	Pending	2013		NRA2	Rocket Flight of a Delta-Doped CCD Focal Plane Array for CHESS	Arizona State University	
T0112-S	Active	2013		NRA2	Spacecraft Disturbance Isolation and Rejection Platform (SDIRP)	NASA Jet Propulsion Laboratory	
T0113-B	Pending	2013		NRA2	Focal Plane Actuation to Achieve Ultra-High Resolution on Suborbital Balloon Payloads	Arizona State University	
T0114-S	Pending	2013		NRA2	Technology Demonstration of Graphene Ion Membranes for Earth and Space Applications	Johns Hopkins University Applied Physics Laboratory	
T0115-S	Active	2013		NRA2	EDL Technology Development for the Maraia Earth Return Capsule	NASA Johnson Space Center	
T0116-S	Active	2013		NRA2	Operational Demonstration of the MPS-120 CubeSat High-impulse Adaptable Modular Propulsion System	Aerojet General Corporation	
T0117-S	Pending	2013		NRA2	1U CubeSat Green Propulsion System with Post- Launch Pressurization	Busek Co. Inc.	
T0118-S	Pending	2013		NRA2	Iodine RF Ion Thruster Development	Busek Co. Inc.	
T0119-S	Active	2013		NRA2	Inductively Coupled Electromagnetic (ICE) Thruster System Development for Small Spacecraft Propulsion	MSNW LLC	
T0120-S	Pending	2013		NRA2	Advanced Hybrid Rocket Motor Propulsion Unit for CubeSats (PUC)r	The Aerospace Corporation	
T0121-S*	Completed	2013	1	USIP	Flyover Mapping and Modeling of Terrain Features	Carnegie Mellon University	Masten Space Systems

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Title #	Status	Year Launched	Type of Flight (P/S/B)	Origin	Payload	Affiliation	Commercial Company
T0123-P	Completed	2013	2	USIP	Microgravity Propellant Gauging Using Modal Analysis	Carthage College	
T0124-B*	Completed	2013	1	USIP	Gannon University's Cosmic-Ray Calorimeter (GU-CRC)	Gannon University	
T0126-P*	Pending	2014	1	AFO8	Validating Microgravity Mobility Models for Hopping/Tumbling Robots	NASA/Jet Propulsion Laboratory	
T0128-S	Active	2014		AFO8	Zero-gravity Green Propellant Management Technology	Purdue University	
T0129-P	Active	2014	1	AFO8	Testing of a Microgravity Rock Coring Drill Using Microspines	NASA/Jet Propulsion Laboratory	
T0131-P	Active	2014	1	AFO8	Enhanced Dynamic Load Sensors for ISS (EDLS-ISS) Operational Feasibility for ARED	Aurora Flight Sciences	
T0132-P*	Completed	2014	2	AFO8	Effects of Microgravity on Intracranial Pressure	University of Texas Southwestern Medical Center	
T0133-P	Active	2014	1	AFO8	Payload Separation Performance of a New 6U CubeSat Canisterized Satellite Dispenser	Air Force Research Laboratory	
T0135-P*	Completed	2014	2	AFO8	Testing ON-OFF Gecko Adhesive Grippers in Microgravity	NASA/Jet Propulsion Laboratory	
T0137-S	Active	2014	2	AFO8	Fuel Optimal and Accurate Landing System Test Flights	NASA/Jet Propulsion Laboratory	
T0138-P	Completed	2014	2	AFO8	Reduced Gravity Flight Demonstration of SPHERES INSPECT	Massachusetts Institute of Technology	
T0139-S	Active	2014		DIR.	ADEPT	NASA Ames Research Center	
T0140-P	Active	2014	1	DIR.	Lunar Plants	NASA Ames Research Center	
T0141-B*	Completed	2014	1	NRA2	Demonstration of Enabling Communications Technologies for Future Low-Cost Small Earth Return Vehicles	Terminal Velocity Aerospace	
T0142-S	Active	2014		DIR.	Affordable Vehicle Avionics (AVA)	NASA Ames Research Center	
T0143-B	Active	2015		DIR.	ORS Bi-Static RF Imager	Air Force Research Laboratory	
T0144-S	Active	2015		DIR.	PULSAR	NASA Marshall Space Flight Center	

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Title #	Status	Year Launched	Type of Flight (P/S/B)	Origin	Payload	Affiliation	Commercial Company
T0145-P	Active	2015		REDDI14	Low-Gravity Flow Boiling on Modern Textured Surfaces	Purdue University	
T0146-P	Active	2015		REDDI14	Droplet Pinning in Microgravity	Rensselaer Polytechnic Institute	
T0147-P	Active	2015		REDDI14	Modal Propellant Gauging in Microgravity	Carthage College	
T0148-S	Pending	2015		REDDI14	Demonstration of One-Fault Tolerant Precision Navigation for Autolanding Airless	Bodies: Flight Two Astrobotic Technology, Inc.	
T0149-P	Active	2015		REDDI14	Microgravity Fabrication of Freeze-cast Titanium Foams	Northwestern University	
T0150-P	Active	2015		REDDI14	Advancing Diaphragm Modeling Technology for Propellant Management	Purdue University	
T0151-B	Active	2015		DIR.	PRANDTL-M	NASA Armstrong Flight Research Center	
T0152-P	Active	2015		DIR.	ISS Life Science Glovebox (LSG) Crew Restraint	NASA Ames Resarch Center	
T0153-B	Active	2015		DIR.	Mars Electric Reusable Flyer	NASA Langley Research Center	

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